Lecture 1

Monday, 16 January 2023 7:22 AM

→ Electrochem Engineering → Translating fundamental principles to applied/industrial contexts
↳ (of electrochem transformations)

→ Why?
↦ CO_2-free (green e^-) → Decarbonization
↦ Driven by e^- rather than heat

→ How?
↦ Concepts: Thermo, kinetics, Transport
Charge, voltage, conservation of charge } Fundamentals

↦ Current distributions, Rotating electrodes
Electroanalytical methods } Techniques, modelling to test and understand e-chem systems

↦ Porous electrodes/batteries,
Electrocatalysis, decarbonization } Applied systems

→ Goal?
↦ Technical working knowledge of e-chem phenomena for e^- driven processes

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Electrochemical reaction: A change in the molecular identity of a chemical species in which one or more e^- are reactant or product

product

eg: $2\,Cl^-_{(aq)} \rightarrow Cl_{2\,(g)} + 2e^-$

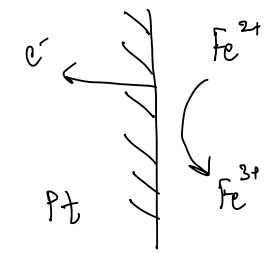
(rxn does not occur unless e^- transfer happens)

↳ e^- transfer to/from electrode to a different phase

↓ ↓

have to be free e^- — conducting/semi-conducting

↳ echem reactions are inherently heterogeneous.



- Driving force: Thermodynamics (e-chem potential)
- Proximity: Physics
- Transport: interfacial region

↳ Charge neutrality: takes an enormous amount of energy to separate charge. Therefore, we can assume charge neutrality everywhere except over very short distances.

↓

This means that there must be another electrode

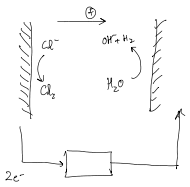
↳ Charge transfer @ each electrode ("half reaction")

↳ Half reactions are chemically independent.

Lecture 2.

Monday, 16 January 2023 8:12 AM

Electroneutrality!



Electrolyte: phase thru which ions can move
1) Can be liquid (aq or organic)
2) Polymeric (Nafion)
3) Solid (ceramic w/ oxide vacancies or solid-oxide electrolyte)

Ex: "Chlor-Alkali Process" OIL RIG

$2\,Cl^-_{(aq)} \rightarrow Cl_2(g) + 2e^-$ (Oxidation)

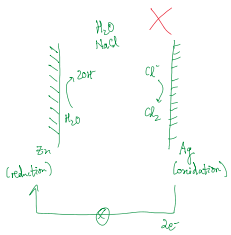
$2H_2O + 2e^- \rightarrow 2OH^- + H_2(g)$ (Reduction)

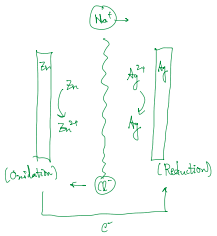
$2NaCl_{(aq)} + 2H_2O \rightarrow Cl_2(g) + 2NaOH + H_2(g)$

Shorthand: $Pt \mid Cl^-_{(aq)} \parallel NaOH_{(aq)} \mid H_2, Cl_2(g) \mid Pt_{(s)}$

↑ Phase boundary ↑ transition to different electrolyte composition

Ex: Alessandro Volta: first battery in 1790's





Wind + Solar → enable electrochemical processes in manufacturing to take place

$AgO + H_2O + 2e^- \rightarrow Ag + 2OH^-$

$Zn + 2OH^- \rightarrow ZnO + H_2O + 2e^-$

Note: Valency $2+ \rightarrow 3+$ (Lost an e^-) } Oxidation

∴ Number increasing → oxidation

e^- being released (right hand side) → Oxidation

Lecture 3

Tuesday, 17 January 2023 1:25 PM

HW1 (due Thursday)

HW0, part 1 → Course evaluation (due Friday)

↳ Heterogenous reaction (e^- transfer between electrodes)

→ written as half-reactions (oxidation/reduction)

→ short hand notation : e^- chemical cell

ex: Chlor-Alkali

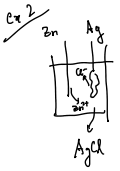
$2Cl^-_{(aq)} \rightarrow Cl_2(g) + 2e^-$

$2H_2O + 2e^- \rightarrow 2OH^- + H_2(g)$

$Pt \mid Cl_2(g) \mid Cl^-, Na^+ \,||\, Na^+, OH^-, H_2O \mid H_2(g) \mid Pt$

(|| denotes change in composition (although in the same phase) (separator))

Ex 2



$Zn(s) \mid Zn^{2+}, Cl^- \mid AgCl(s) \mid Ag$

Electrochemical Concepts and Conventions

+ At least 2 electrodes

↳ Anode → Oxidation reaction (An-Ox)

$R \rightarrow O + e^-$

(R: reductant (gets oxidized))

↳ Cathode → Reduction reaction

$O + e^- \rightarrow R$

+ Electrodes must be connected via an electrolyte (any phase in which ions can be transported)

+ Transition between electric/ionic conduction occurs @ electrode/electrolyte interface

+ Electroneutrality states that currents @ electrodes must be equal and opposite

$$I_{anode} + I_{cathode} = 0$$

Convention: $I_{anode} \geq 0$

+ Cell potentials (Galvanic + Electrolytic Cell)

↳ Galvanic cell : Half-reactions sum to an overall spontaneous rxn ($\Delta G_r < 0$)

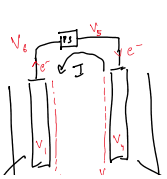
+ We can extract work from these types of cells (batteries)

+ H_2-fuel cell : $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$

↳ Electrolytic cell : electrical power from an external source is required to make rxns proceed ($\Delta G_r > 0$)

+ Put work into the cell

+ Charging a battery, water electrolysis.



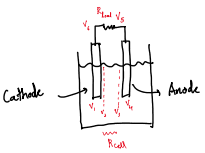
+ Requires power supply (PS)

+ Current is reversed

+ Now RHS electrode is cathode

+ PS increases $(V_6 - V_5) > V_{oc}$

Current flows from $V_6 \rightarrow V_5$ (high to low potential)



+ Anode and cathode are at different potentials since diff rxns occur @ them

+ I_{anode} → positive (convention)

∴ current flows opposite of electrons

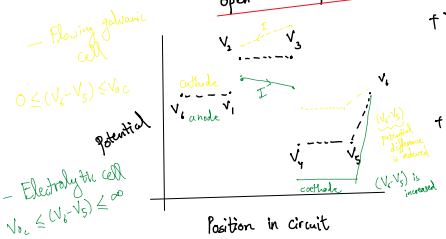
+ Ohm's law: $V = IR$

$(V_6 - V_5) = I R_{load}$

↳ If $R_{load} \gg R_{cell}$ → No current in external circuit

↳ $(V_6 - V_5)$ then describes/defines open circuit potential (VOC)

Measured by connecting e^- w/ a large known resistor & measuring small current



+ If conductive wires:

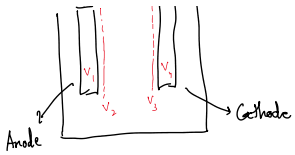
1) $V_6 \approx V_1$

2) $V_5 \approx V_4$

+ @ open circuit, $V_2 = V_3$

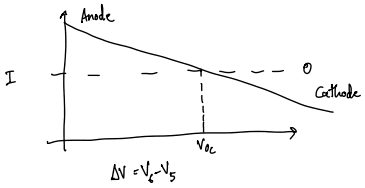
↳ No resistance to flow [$(V_6 - V_5) \approx 0$]

Max current but no work extracted → short circuit/closed circuit



+ Now RHS electrode is cathode
+ PS increases $(V_6 - V_5) \geq V_{oc}$
  Current flows from $V_6 \rightarrow V_5$ (high to low)
  [Opp. to potential]
  ↓
  open-circuit
  (at open-circuit there is no net current flow $I=0$)

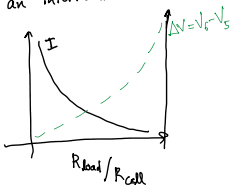
→ Cathode & Anode switching is confusing

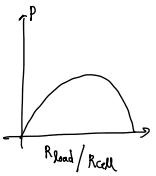


So, even though $I_{anode} > 0$, there can be both oxidation & reduction reactions that can occur, but just the net should be oxidation.

↳ No resistance to flow [V_6 "5"]
  Max current but no work extracted → short circuit / closed circuit

↳ Therefore, to achieve max power ($P = I(V_6 - V_5)$) we need an intermediate load





Lecture 4

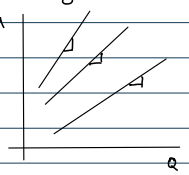
Thursday, 19 January 2023 1:32 PM

Open circuit potential, V_{oc}, potential diff b/w electrodes when $R_{load} \gg R_{cell}$, $I=0$

Faraday's law of electrolysis

1) Mass of product formed/consumed in e- chem rxn is directly proportional to charge passed

$\frac{m}{Q} = z$



linear irregardless of material

2) For a given amt of charge passed on electrode of different material, amt of product formed/consumed is proportional to electrochem equiv. weight of that product; $E = \frac{M}{n \cdot N_A}$

$m = x\,E = \frac{x\,M \cdot e}{n N_A \cdot e}$; e is fundamental charge

↓ Amount of e^- passed

ex is amount of charge passed Q

eN_A is Faraday's constant

$F = 96485.3 \frac{C}{mol\,e^-}$

$\therefore \boxed{m = \frac{QM}{nF}}$ → If we know amount of charge passed, we know how much mass is formed

$\frac{-\beta F V}{RT}$

$\frac{C}{mol \cdot e} \cdot \frac{K\,mol}{J} \cdot \frac{1}{K} \cdot V \Rightarrow \frac{C \cdot V}{J}$

Rate:

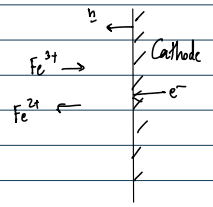
$I = \frac{\text{Charge}}{\text{time}}$ [=] $\frac{C}{s}$ [=] A

Faraday's law as a rate: $\frac{\frac{m}{M}}{s} = \frac{Q/s}{nF} \Rightarrow \boxed{r = \frac{I}{nF}}$ [=] $\frac{mol}{time}$

$\frac{mol\ e^-}{mol\ of\ product}$; $96485 \frac{C}{mol \cdot e}$

Flux @ electrode surface

↓ vector quant.



$Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$

↳ Flux balance: $Flux_{Fe^{3+}} = Flux_{e^-}$ @ SS

$(\bar{N}_3)$ $(\bar{N}_e)$

↳ Molar Flux: $\left[\frac{mol}{cm^2 \cdot s}\right] = N_j$

$\vec{n} \cdot \bar{N}_3 = -\vec{n} \cdot \bar{N}_e = -\vec{n} \cdot \bar{N}_3$; $i = \frac{I}{Area}$ ⇒ Current Density

i/F ⇒ Cathodic current < 0

For a generic rxn, w/ stoich coeffs s_j:

$\sum s_j N_j \rightleftharpoons n \cdot e$

(+ for products)

$\boxed{\vec{n} \cdot \bar{N}_j = \frac{-s_j}{nF} i}$ [=] $\frac{mol}{cm^2 \cdot s} = r' = \frac{-s_j\, i}{nF}$

→ Faraday's Law as flux

The minus sign for i would then mean it is cathodic current density.

→ Faraday's Law as flux density.

Electrode Potentials

$\Delta G = -nFV$; potential ~ ΔG } of an entire cell

↓ reversible cell potential

→ We care about potential of one electrode (usually working electrode)

↳ Counter electrode must exist, but it may complicate our potential measurements (since if we don't know the reactions happening at our counter, then we don't know if we are referencing to what)

↳ Therefore, we have to reference our WE potential to a 3rd electrode w/ a stable + defined potential → RE

Ex: $Ag_{(s)} + Cl^-_{(aq)} \rightarrow AgCl_{(s)} + e^-$

↓ does not participate in the cell reactions

Electrochemical Potential

~ similar concept to chemical potential ; $\mu = \left.\frac{\partial G}{\partial n_j}\right|_{T,P,n_{j\neq i}}$

Charged species involved interact w/ electric field → Electrochemical potential

Fick's Law (Mass Transport)

Flux = convection + diffusion + electrical migration (for charged species)

↳ In dilute soln.: diffusion is very imp & defined by Fick's Law

$$\bar{N}_j = -D_j \cdot \nabla c_j \quad ; \quad D \approx 10^{-5}\ cm^2/s \text{ (in aq. systems)}$$

D = diffusivity

Ohm's Law (Current transport)

In well-mixed electrolyte, ohm can describe current flowing thru electrolyte

$$\bar{i} = -\kappa \nabla \phi$$

κ: conductivity; ϕ: electric potential

Electrolyte Conductivity

$$\kappa = F^2 \sum z_j^2 u_j c_j$$

z_j: charge on species; u_j: ionic mobility

$$u_j = \frac{D_j}{RT}$$ → Nernst Einstein equation

Electroneutrality

$$\sum z_j c_j = 0$$

↳ Excellent (and necessary) approximation everywhere

↳ Only on length scales (~nm) can charges sufficiently separate (@ electrode/electrolyte interface)

Lecture 5

Tuesday, 24 January 2023 1:29 PM

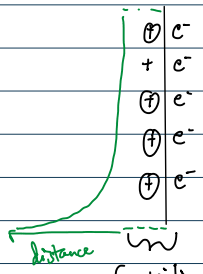
Double Layer

The region very near the electrode surface, where some charge separation occurs, but the charge built up will be exactly balanced by electrode

Due to brownian motion, this is not an exact single layer in solution, but instead net charge decreases exponentially away from electrode

↓

defined by debye length



Capacitive double layer

permittivity

$$\lambda = \sqrt{\frac{\varepsilon RT}{F^2 \sum_j z_j^2 c_{j,\infty}}}$$

; ε = (dielectric constant)(permittivity of free space)

$K = 78$ (for water) $\qquad \varepsilon_0 = 8.854 \times 10^{-14}\ F/cm$

Electrochemical Rates

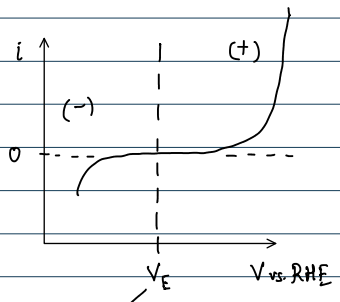
+ If we increase current density, we could reduce total # of devices needed

+ We need to understand the driving force required to achieve such an increase
  - ↳ deriving electrochemical rate laws

↳ Changing potential has an impact on the current → Varying potential is called polarizing an electrode

i vs. E

$\frac{mA}{cm^2}$ (V)



+ Voltage below & above V_{oc} leads to flow of charge one way vs other

+ Relationship b/w i & V appears exponential

+ $i \propto$ rxn rate (by Faraday's Law)

↓

Arrhenius equation ($k \propto e^{-E_a/RT}$) → originally an empirical relationship

Represents equilibrium of a single reversible electrode rxn

$[R \underset{k_c}{\overset{k_a}{\rightleftharpoons}} O + e^-]$

i.e. the rate of forward & reverse are equal resulting in 0 current

Same activation energy approach for echem rxns:

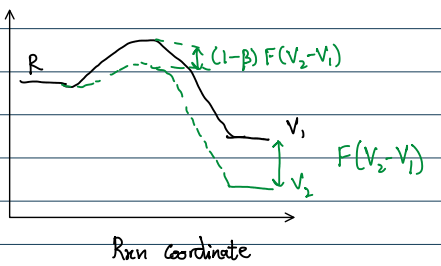
$W_{rev} = -F\Delta V$ ← work req in moving 1 mol e^- } ∝ Energy is associated

forward & reverse are equal resulting in 0 current @ the electrode

$r = k_a C_R - k_c C_o$

ex) $R \geq 0 + e^-$

$W_{rev} = -F\Delta V$ ← work req in moving 1 mol e^- across energy barrier ΔV } ∼ Energy is associated w/ e^-



β represents fraction of ΔV required to reach a new transition state $(0 \leq \beta \leq 1)$

So,

$$E_{A_{forward}}(V_2) = E_{A_{forward}}(V_1) - (1-\beta)F(V_2-V_1)$$

$$E_{A_{forward}}(V) = E_{A_{forward}}(0) - (1-\beta)FV \quad \text{(forward rxn is anodic)}$$

$$E_{A,a}(V) = E_{A,a}(0) - (1-\beta)FV$$ → change in potential of 1 electrode (for can just replace w/ V)

$$E_{A,c}(V) = E_{A,c}(0) + \beta FV \quad \text{(Cathodic)}$$

$$k_{forward} = A\exp\left[\frac{-E_{A,a}(0) + (1-\beta)FV}{RT}\right] \quad \text{(Arrhenius dependence)}$$

$$= k_a \exp\left[\frac{(1-\beta)FV}{RT}\right] \quad ; \quad k_a = \exp\left(\frac{-E_{A,a}}{RT}\right)$$

$$k_{rev} = k_c \exp\left[\frac{-\beta FV}{RT}\right]$$

$$\therefore \boxed{r = \frac{i}{F} = C_R k_a \exp\left[\frac{(1-\beta)FV}{RT}\right] - C_o k_c \exp\left[\frac{-\beta FV}{RT}\right]}$$

$$r = \frac{i}{F} = C_R k_a \exp\left[\frac{(1-\beta)FV}{RT}\right] - C_o k_c \exp\left[\frac{-\beta FV}{RT}\right]$$

We characterize exchange current density as rate of forward (or reverse) rxn when overall rxn is @ equil
↳ We define the potential @ which this occurs as U

$$i \Rightarrow 0 \Rightarrow \cancel{F} C_R k_a \exp\left[\frac{(1-\beta)FU}{RT}\right] = \cancel{F} C_o k_c \exp\left[\frac{-\beta FU}{RT}\right] = i_o$$

↓

Solving for U:

$$\exp\left[\left(\frac{(1-\beta)F}{RT} + \frac{\beta F}{RT}\right)U\right] = \frac{C_o k_c}{C_R k_a}$$

$$U = \frac{\ln\left(\frac{C_o k_c}{C_R k_a}\right)}{\frac{(1-\beta)F}{RT} + \frac{\beta F}{RT}} = \ln\left(\frac{C_o k_c}{C_R k_a}\right)\frac{RT}{F}$$

$$\therefore \boxed{U = \frac{RT}{F}\ln\left(\frac{k_c}{k_a}\right) + \frac{RT}{F}\ln\left(\frac{C_o}{C_R}\right)}$$

Introduce: $V = U + \eta$
↑ Overpotential → plugging into i expression

$$\therefore i = \underbrace{F C_R k_a \exp\left[\frac{(1-\beta)UF}{RT}\right]}_{i_o} \exp\left[\frac{(1-\beta)\eta F}{RT}\right] - \underbrace{F C_o k_c \exp\left[\frac{-\beta FU}{RT}\right]}_{i_o} \exp\left[\frac{-\beta F\eta}{RT}\right]$$

$$\boxed{i = i_o\left(\exp\left[\frac{(1-\beta)F\eta}{RT}\right] - \exp\left[\frac{-\beta F\eta}{RT}\right]\right)} \longrightarrow \text{Butler-Volmer Eq}$$

For rxns where multiple e^- transfers occur, we use α instead of β:

$$i = i_o\left(\exp\left[\frac{\alpha_a F\eta}{RT}\right] - \exp\left[\frac{-\alpha_c F\eta}{RT}\right]\right) \qquad 0 \le \alpha \le n$$

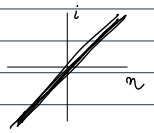
↑ transfer coeff.

Simplifications:

↳ @ small overpotential $\eta \to 0$; linearize B-V w/ 1st order Taylor-expansion

$$\boxed{i = i_o\frac{(\alpha_a + \alpha_c)F\eta}{RT}}$$

linear kinetics:



↳ @ large overpotential, ex $\eta \gg 0$
Total current contribution from cathodic rxn becomes small relative to anodic rxn,

$$i = i_o\left(\exp\left[\frac{\alpha_a F\eta}{RT}\right] - \exp\left[\frac{-\alpha_c F\eta}{RT}\right]\right)$$

positive, large — negative, small

$$\therefore \boxed{i = i_o \exp\left[\frac{\alpha_a F\eta}{RT}\right]}$$ Anodic Tafel equation

large small

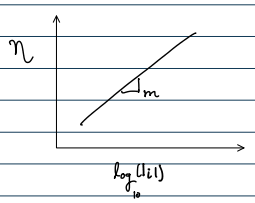
$$\therefore \boxed{i = i_o \exp\left[\frac{\alpha_a F \eta}{RT}\right]}$$ Anodic Tafel equation

↳ Taking $\log_{i_o}$ to linearize:

$$\log_{10}(i) - \log_{10}(i_o) = \frac{\alpha_a F \eta}{RT} \qquad \log_{10}(e) = \frac{\alpha_a F \eta}{2.303 RT}$$

$$\eta = m \cdot \log_{10}(i) + b$$

$m = \frac{2.303 RT}{F \alpha_a}$ $\quad b = -m \log_{10}(|i_o|)$



m = Tafel slope $\left[\frac{mV}{decade}\right]$

* analogous to Arrhenius plot

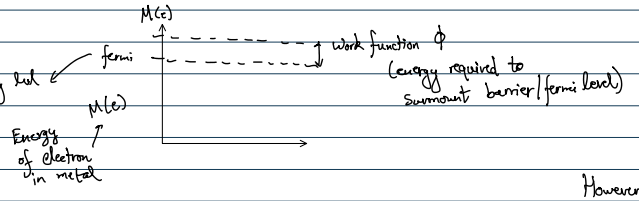
Slope → higher would mean that a small increase in η lead to ↑ i_o (higher rate?)

Actually, a lower slope means that a small overpotential causes a larger increase in current, therefore smaller slope is higher rate!!! (since overpotential is on the y axis) ✓

→ **β (Symmetry factor)**

Fraction of $F(V_2 - V_1)$ required to overcome energy barrier → What does it physically represent?

+ β characterizes the quantum mechanical transfer of e^- from electrode to chem species
↳ Classical mechanics of e^- predicts orders of mag. lower i than observed for potentials we apply;

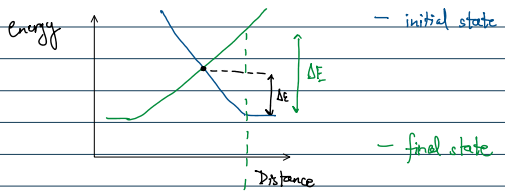


e^- needs to escape metal and then get transferred to another fermi level → However, recorded energies were for lower than predicted

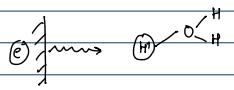
↳ QM allows e^- to tunnel rather than overcome energy barrier → this occurs w/out radiation energy (No light is seen)
↳ this requires initial + final state have to have the same energy

ex: adsorption of proton @ electrode

$$M(e^-) + H^+ - OH_2 \rightleftharpoons M-H + H_2O$$



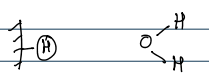
Initial



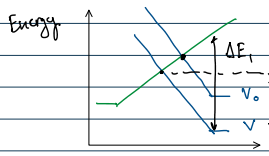
ΔE → large energy barriers req. to transfer e^- from ground state (no tunneling possible)

Δε → represents activation barrier for tunneling

Final:



When we apply a potential we vertically shift initial state & thus change $\Delta\varepsilon$:



$$\Delta(\Delta\varepsilon) = \Delta\varepsilon_1 - \Delta\varepsilon_0$$

$$\frac{\Delta\varepsilon_0}{\Delta E_0} = \frac{\Delta\varepsilon_1}{\Delta E_1} \quad \text{(straight line)}$$

$$= \frac{\Delta\varepsilon_0 + \Delta(\Delta\varepsilon)}{\Delta E_0 + \Delta V F}$$

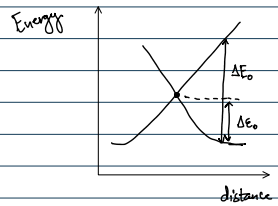
$F(V_1 - V_0)$ gives energy associated w/ voltage change

?

$$\frac{\Delta\varepsilon_0}{\Delta E_0} = \frac{\Delta(\Delta\varepsilon)}{\Delta V F}$$

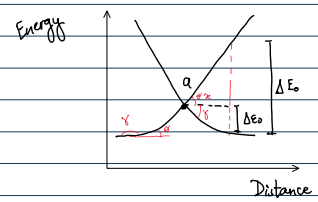
↳ represents the fraction of work applied that required to change E_a

$$= \beta$$



$$\frac{\Delta\varepsilon_0}{\Delta E_0} = \beta$$

Get value of β from trig.



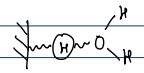
$$\beta = \frac{\Delta\varepsilon_0}{\Delta E_0} = \frac{x \tan\gamma}{x(\tan\gamma + \tan\theta)}$$ } Cathodic

$$(1-\beta) = \frac{\tan\theta}{\tan\gamma + \tan\theta}$$ } Anodic

Straight line assumption → validity?

↳ has to

ⓐ a → excited state



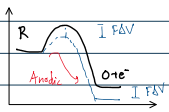
$R \rightleftharpoons O + e^-$

β → Fraction of energy for ΔV change going into activation energy change

If slopes ≈ equal: $\tan\gamma \approx \tan\theta$

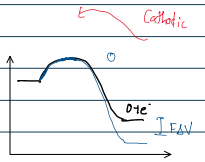
$\beta = \frac{1}{2}$

$\beta \approx 0$



$\beta=0$ → Anodic process is greatly impacted by ΔV

→ Cathodic process is not!



Anodic process is not impacted by ΔV!

Lecture 7

Tuesday, 31 January 2023 1:29 PM

Electrochemical Mechanisms

Similar process as heterogeneous catalysis → have to modify approach using echem kinetics

1) Series of elementary steps (1 e^- at a time)
2) Write rate law for each step
3) Assume RLS (rate limiting step), set all other rules to quasi-equilibrium
   (i) Eliminate intermediates / sites
4) Compose overall RL
5) Determine reversible potential (U), introduce η, & determine i_o
6) Compare i_o, α_a, α_c, and rxn order dependence to experiment

Ex: Water Electrolysis → hydrogen evolution half rxn

Mechanism 1

Volmer: $S + H^+ + e^- \underset{k_{1,a}}{\overset{k_{1,c}}{\rightleftharpoons}} H_{ads}$

Tafel: $2H_{ads} \underset{k_{2,r}}{\overset{k_{2,f}}{\rightleftharpoons}} H_2(g) + 2S$

Mechanism 2

Volmer: $S + H^+ + e^- \rightleftharpoons H_{ads}$

Heyrovsky: $H_{ads} + H^+ + e^- \rightleftharpoons H_2(g) + S$

+ Tafel → non-faradaic
+ Both steps involve adsorbed species (i.e. surface sites) → [S]

Something we can measure (i.e. BET isotherm)

Volmer-Tafel Mechanism:

$$r_1 = k_{1,c}[H^+][S]\exp\left[\frac{-\beta FV}{RT}\right] - k_{1,a}[H_{ads}]\exp\left[\frac{(1-\beta)FV}{RT}\right]$$

C_s = # unoccupied sites $= C_t(1-\theta)$; $C_{H_{ads}} = C_t\cdot\theta$

fractional coverage $\left[\frac{C_{H_{ads}}}{C_t}\right]$

$$\rightarrow r_1 = k_{1,c}C_{H^+}C_t(1-\theta)\exp\left[\frac{-\beta FV}{RT}\right] - k_{1,a}C_t\theta\exp\left[\frac{(1-\beta)FV}{RT}\right]$$

$$r_2 = k_{2,f}[H_{ads}]^2 - k_{2,r}[H_2][S]^2$$

$$= k_{2,f}C_t^2\theta^2 - k_{2,r}C_{H_2}[C_t(1-\theta)]^2$$

$$\rightarrow r_2 = k_{2,f}(C_t\theta)^2 - k_{2,r}P_{H_2}[C_t(1-\theta)]^2$$

Solving for θ:

r_2 is rate limiting, then r_1 is in quasi-equil.: $r_1 = 0 \Rightarrow k_{1,c}C_{H^+}C_t(1-\theta)\exp\left[\frac{-\beta FV}{RT}\right] = k_{1,a}C_t\theta\exp\left[\frac{(1-\beta)FV}{RT}\right]$

$$\theta = \frac{k_{1,c}C_{H^+}}{k_{1,a}\exp\left[\frac{FV}{RT}\right] + k_{1c}C_{H^+}}$$

$$\theta = \frac{\frac{k_{1,c}C_{H^+}}{k_a}\exp\left[\frac{-FV}{RT}\right]}{1 + \frac{k_{1c}C_{H^+}}{k_a}\exp\left[\frac{-FV}{RT}\right]}$$

→ similar to Langmuir isotherm $\theta = \frac{K}{1+K}$

Net rate:

$$r_{net} = r_2 = \frac{i}{2F} = k_{2,f}(C_t\theta)^2 - k_{2,r}P_{H_2}[C_t(1-\theta)]^2$$

Net rate:

$$r_{net} = r_2 = \frac{i}{2F} = k_{2,f}(C_t\theta)^2 - k_{2,r} P_{H_2} [C_t(1-\theta)]^2$$

Even though has no e^-, it is still used to define r_{net} and we use this Faraday's eq. always.

$\theta = f(V)$ [depends on applied potential] !

↳ Non-faradaic processes can also be indirectly affected by potential ★

This expression is not consistent w/ B-V kinetics

↳ There might be deviations from B-V (& rare exceptions)

Could be other

- Volmer (rds) - Tafel
- Volmer (rds) - Heyrovsky
- Volmer - Heyrovsky (rds)

} These all have e^- transfer on rds ↓ will take the look of B-V !

Ex: $Cu \rightarrow Cu^{2+} + 2e^-$

1. $Cu \underset{k_c}{\overset{k_{1a}}{\rightleftharpoons}} Cu^+ + e^-$

2. $Cu^{2+} \underset{k_{2c}}{\overset{k_{2a}}{\rightleftharpoons}} Cu^{2+} + e^-$ [RDS] ; $\beta = 0.5 \rightarrow \alpha_a = 1.5 \rightarrow i_0 \sim C_{Cu^{2+}}^{3/4}$ ← Show this

$\alpha_c = 0.5$

Lecture 8

Thursday, 2 February 2023 1:30 PM

What is α, the transfer coeff?

For a multi-step/multi-e^- electrode rxn, $A + ne^- \rightarrow Z$, composed of n e^- transfer steps where $O + re^- \rightleftharpoons R$ is RLS (stoichiometric coeff.)

w/ stoichiometry $Z = \gamma P$, and assume s e^- transferred before RLS and s' e^- are transferred after RLS

$r = 0 \rightarrow$ chemical step; $r = 1 \rightarrow$ e-chem step; N_r: # of e^- in rate limiting step

$|i_c| = F \cdot c_o \cdot \exp\left[\frac{-r\beta\eta F}{RT}\right]$, and after applying quasi-equ. for all other steps:

$$i = i_0 \left[\exp\left(\left(\frac{n-s}{\gamma} - r\beta\right)\frac{F\eta}{RT}\right) - \exp\left(-\left(\frac{s}{\gamma} + r\beta\right)\frac{F\eta}{RT}\right)\right] \quad ; \quad \alpha_a = \frac{n-s}{\nu} - r\beta$$

$n_r = s + s' + r$

Independent rxn:

$$\alpha_c = \frac{s}{\gamma} + r\beta$$

If RLS is non-faradaic, $r = 0 \rightarrow \beta$ does not appear in α_a, α_c but will depend on potential

$$\alpha_a + \alpha_c = \frac{n}{\nu}$$

Kinetics of Multiple Rxn

↳ More than 1 rxn can occur @ electrode simultaneously; ex: CO_2 electrochemical reduction to CO

$CO_2 + H_2O + 2e^- \rightarrow CO + 2OH^-$ ← This reaction occurs in the same potential regime as water electrolysis $[2H_2O + 2e^- \rightarrow H_2 + 2OH^-]$

↑ The total current observed would be sum of partial current:

$i_T = i_{CO} + i_{H_2}$ → we can model each partial current w/ B-V expression, assuming they are non-interacting

↳ We might also assign a "current efficiency" or Faradaic efficiency to understand selectivity

Is every process Faradaic?

How can non-faradaic process occur?

VdW? But then that is not a bond?

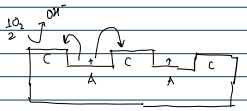
e^- coming from the same element)

Corrosion → special type of multiple rxn

↳ What must net current of corroding metal be?

$I_{total} = 0$, $i_{ox} = i_{red}$; other reactions: $H_2O + 2e^- \rightarrow H_2 + 2OH^-$

$H_2O + \frac{1}{2}O_2 + 2e^- \rightarrow 2OH^-$

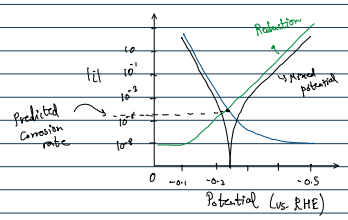


We care about the potential at which this process happens:

→ We polarize our electrode to find this point

$Fe \rightarrow Fe^{2+} + 2e^-$

$2H_2O + 2e^- \rightarrow 2OH^- + H_2$



→ Reducing potential → helping the reduction reaction

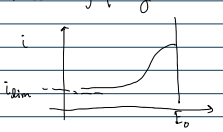
→ For non-zero currents → you need counter electrode??

Alloy electroplating

↳ We desire multiple rxns

Non-idealities that deviate from B-V

1) Mass Transport (ch. 5): We can have large changes in i w/ small change in V so we frequently move b/w rxn-limited + MT limited regions



2) Passivation: Certain oxidized metals are not soluble, & therefore they do not react further

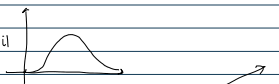
ex: $Cu \rightarrow Cu^{2+} + 2e^-$

Stability of electrodes this

2) Passivation : Certain oxidized metals are not soluble, & therefore they do not react further

ex: $Cu \rightarrow Cu^{2+} + 2e^-$

$Cu \rightarrow CuOH_{(s)} + e^-$



Stability of electrodes is determined from this

We can understand regimes w/ Potential - pH or Pourbaix diagram

3) Organic additives : Often used in electroplating to achieve desirable feature filling

Not taking part in e-chem reaction but are in fact affecting the rate (need to incorporate these non-reacting species into C_i term)

4) Electrode poisoning : Chem species bind so strongly that they eat up active sites + don't leave $CO + Pt \rightarrow Pt-CO$

Lecture 9 (Thermo)

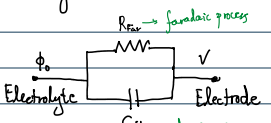
Tuesday, 7 February 2023 1:39 PM

Contributions of the Double Layer to Current

+ B-V describes current due to charge transfer rxns (faradaic)

+ Charge separation occurs in the double layer (charge accumulation) → polarizing an electrode
↓
manifesting as current (No flow of e^- so cannot be captured by B-V kinetics)

+ Qualitatively model using a circuit:



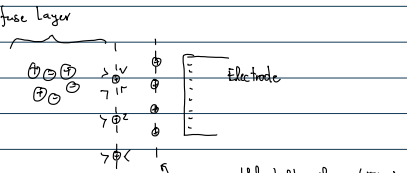
Capacitance [=] $F = \frac{C}{V}$;

$$i_{total} = i_{Far} + i_{dl}$$
$$= \frac{V-\phi_e}{R_{Far}} + C_{dl}\frac{\partial(V-\phi_e)}{\partial t}$$

; R_{Far} is given by $\frac{\partial \eta}{\partial i}$ (Ohm's law) ← B-V

@SS: $i_{dl} = 0$ → capacitive current = 0
↳ But DL still exist
} can impact transport of ions just by its presence

* How do we calculate C_{dl}?
↳ contains distribution of charge
↳ common model for a more accurate DL is the Helmholtz model



* Total charge must be balanced:

Gouy-Chapman Theory:

$$q_{electrode} = -(q_{IHP} + q_{OHP} + q_d)$$

We also know: $\frac{1}{C_{dl}} = \frac{1}{C_{IHP}} + \frac{1}{C_{OHP}} + \frac{1}{C_d}$,

where min $C_d = \frac{\varepsilon}{\lambda}$ ← permittivity; ↑ debye length

Thermodynamics

Voltage difference is an (electrochemical potential) / F ; What is electrochemical potential?
(That we measure across a load attached to EC)

Chemical Potential: $\mu_A = \left.\frac{\partial G}{\partial N_A}\right|_{T,P,N[i]}$; Relative values have meaning (i.e. to standard states)

work/energy associated w/ moving chemical species from one place to another.

$\mu_i = f(T, P, x_i)$

→ $W_{rev} = \mu_A(2) - \mu_A(1) = -\Delta G$

Electrical Potential:

Electrical Potential :

Energy associated w/ moving idealized point charges from one place to another (regardless of chemical state)

$$f = \frac{q_1 q_2}{4\pi\varepsilon_0 r^2} \quad ; \quad W_{rev} = -\Delta G = F(V_2 - V_1) = \mu_e(2) - \mu_e(1)$$

A need for something more:

Issue: 1) Energy associated w/ moving e^- from one sphere to another: $\sim 10^8$ V [2cm Cu sphere, moving 1×10^9 e^-] (Case studies) but $\Delta\mu_e \approx 0$ (Adding small e^- to very large mass/mol of e^-)

2) Volta battery (uncharged electrodes to make electric current) → μ_e can't describe this since it ignores chemical state

but

μ can be used to understand the change.

∴ E-chem systems : need charged species in chemical state whose energies are affected by: + changes in P, T, composition + Electrical state → Electrochemical Potential!

$$\mu_j = \mu_j^c + z_j F \phi$$

μ_j^c: chemical potential; z_j: charge no.; ϕ: quasi-electrostatic potential

This is a formal description of echem potential

However,

$$\mu_j = RT \ln(\gamma_j) \quad ; \quad \gamma_j = c_j f_j a_j$$

(γ_j: Electrochemical activity; a_j: activity; f_j: activity coefficient; c_j: constant of proportionality)

← This definition utilizes difference in reference/standard states → More useful

↳ This must depend on electrical state for ionic species

(But specific comb. of f_j are not dependent on electrical state) ←

$MX_2 \rightleftharpoons M^{2+} + 2X^-$

$$C_{MX_2} = C = \frac{C_{M^{2+}}}{1} = \frac{C_{X^-}}{2} \quad ; \quad \mu_{MX_2} = \mu_{M^{2+}} + 2\mu_{X^-}$$

$$\gamma_{MX_2} = \gamma_{M^{2+}} \gamma_{X^-}^2 = (C_{M^{2+}} f_{M^{2+}} a^\circ_{M^{2+}})(C_{X^-} f_{X^-} a_{X^-})^2$$

$$\gamma_{MX_2} = C^3 (a^\circ_{M^{2+}})(2a^\circ_{X^-})^2 (f_{M^{2+}} f_{X^-}^2)$$

All these species are not dependent on electrical state → So not dependent on electrical state! (Although individually they are)

Mean Activity Coeff

In general, $\bar{f}_\pm^{\bar{\nu}} = f_+^{\nu_+} f_-^{\nu_-}$ for a general electrolyte

* Neutral comb. of ions will always have mean activity coeff independent of electric state

Electrolyte: $M_{\nu_+}X_{\nu_-} \rightleftharpoons \nu_+ M^{z+} + \nu_- X^{z-}$, $z_+\nu_+ + z_-\nu_- = 0$ (Electroneutrality)

Electroneutrality

of electric static

Why?

+ Physical props & rates → defined in terms of activities / not conc.

+ framework of echem sys

Lecture 10

Thursday, 9 February 2023 1:32 PM

Electrochem Rxn Equilibrium

- Phase Equilibrium: chemical species in different phases are in equilibrium if their chem. potentials are equal

$$\mu_j^{\alpha} = \mu_j^{\beta}$$

Eg: O_2 dissolved in water: $\mu_{O_2}^{aq} = \mu_{O_2}^{g}$ → Henry's Law $P_{O_2} = K_H [O_2]$ → Chemical potential equivalence b/w gas phase & liquid solubility of a species

- Chemical Rxn Equilibrium: A chem rxn is in equil when chem. potential of reactants & products are equal

$$A + B \rightleftharpoons C$$

$$\mu_A + \mu_B = \mu_C$$

Conc. @ standard conditions

↳ For ideal solutions (activity coeff = 1): $\mu_j = \mu_j^{\circ} + RT \ln \frac{c_j}{c_j^{\circ}}$

(μ_j°: chem. potential @ std. conditions)

Plugging this in for our reaction:

$$\underbrace{\frac{C_C^{\circ}}{C_A^{\circ} C_B^{\circ}} \exp\left[\frac{\mu_A^{\circ} + \mu_B^{\circ} - \mu_C^{\circ}}{RT}\right]}_{K} = \frac{C_C}{C_A C_B}$$

$$K = \frac{C_C}{C_A C_B}$$

For an echem rxn, we can write: $\sum s_j \mu_j \rightleftharpoons n e^-$

@ equil,

$$\sum s_j \mu_j = n \mu_{e^-}$$

→ This is what we measure via voltmeter (electrochem potential of e^-)

Ex: Consider the full cell rxn:

α | β | γ_1 || γ_2 | δ | α'

$Pt \mid H_2(g) \mid H_2SO_4(aq) \parallel CuSO_4(aq) \mid Cu(s) \mid Pt(s)$

Lecture 11

Tuesday, 14 February 2023 1:29 PM

Phase + Reaction Equilibrium for Electrochemical systems

α β γ_1 γ_2 δ α'

$Pt(s) \mid H_2(g) \mid H_2SO_4(aq) \,\|\, CuSO_4(aq) \mid Cu(s) \mid Pt(s)$

→ When the potential difference b/w RHS + LHS electrode ($V^{\alpha'} - V^{\alpha}$) creates equilibrium, we define this as:

$$U_{cell} \equiv V^{\alpha'} - V^{\alpha}$$

→ We also know that measured potential differences across electrodes describe the electrochemical potential of e^-

$$\mu_e^{\alpha} - \mu_e^{\alpha'} = \frac{-F(V^{\alpha} - V^{\alpha'})}{n} \quad [=] \frac{J}{mol\, e^-} \quad ; \text{ Thus } \boxed{\mu_e^{\alpha} - \mu_e^{\alpha'} = FU} \text{ @ equilibrium}$$

Left − Right

$$W_{rev} = \Delta G_r = -F\Delta V$$

→ Applying phase equilibrium to our examples means that species in adjacent phases must be in equilibrium:

So on RHS: δ α'

$Cu(s) \mid Pt(s)$

$$\mu_e^{\delta} = \mu_e^{\alpha'} \text{ @ equil.}$$

Rxn: δ γ_2 δ

$$Cu^{\circ} \rightleftharpoons Cu^{2+} + 2e^-$$

$$2\mu_{e^-}^{\delta} = \mu_{Cu}^{\delta} - \mu_{Cu^{2+}}^{\gamma_2}$$

$$\therefore 2\mu_e^{\alpha'} = 2\mu_e^{\delta} = \mu_{Cu^{\circ}} - \mu_{Cu^{2+}}^{\gamma_2}$$

→ Repeating this process for LHS (now only e^- conducting phase α)

Rxn: $H_2(g) \rightleftharpoons 2H^+_{(aq)} + 2e^-$ (α)

$$2\mu_e^{\alpha} = \mu_{H_2}^{\beta} - 2\mu_{H^+}^{\gamma_1}$$

→ Combining w/ cell eq. potential:

$$U_{cell} = \frac{1}{F}(\mu_e^{\alpha} - \mu^{\alpha'}) = \frac{1}{2F}(2\mu_e^{\alpha} - 2\mu_e^{\alpha'})$$

$$\boxed{U_{cell} = \frac{1}{F}\left(\mu_{H_2}^{\beta} - 2\mu_{H^+}^{\gamma_1} - \mu_{Cu}^{\delta} + \mu_{Cu^{2+}}^{\gamma_2}\right)}$$

Don't typically measure μ → apply def. of activities to get a usable form:

$$\mu_{H_2}^{\beta} = \mu_{H_2}^{*} + RT \ln P_{H_2}^{\beta}$$

(μ^*: standard chem potential for a gas)

μ^{δ} . . . μ°

$\mu_{H_2}^{r} = \mu_{H_2}^{\ominus} + RT \ln P_{H_2}$ for a gas

$\mu_{Cu}^{s} = \mu_{Cu}^{\circ}$

$\mu_{H^+}^{\gamma_1} = RT \ln(\lambda_{H^+}^{\gamma}) = RT \ln(c_{H^+} f_{H^+} a^{\circ}_{H^+})$

$\mu_{Cu^{2+}}^{\gamma_2} = RT \ln(c_{Cu^{2+}}^{\gamma_2} f_{Cu^{2+}}^{\gamma_2} a^{\circ}_{Cu^{2+}})$ ⟶ Independent of conc.

$$V_{cell} = \frac{1}{2F}\left[\mu_{H_2}^{\ominus} + RT \ln(P_{H_2}^{\beta}) - 2(RT \ln(a^{\circ}_{H^+})) + RT \ln(C_{H^+} f_{H^+}) - \mu_{Cu}^{\circ} + RT \ln(a^{\circ}_{Cu}) + RT \ln(C_{Cu^{2+}}^{\gamma_2} f_{Cu^{2+}})\right]$$

V°_{cell} → standard equilibrium potential ⇒ $V^{\circ}_{cell} = \frac{1}{2F}\left(\mu^{\circ}_{H_2} - 2RT \ln(a^{\circ}_{H^+}) - \mu^{\circ}_{Cu} + RT \ln(a^{\circ}_{Cu^{2+}})\right)$

$= \underbrace{-\frac{1}{2F}\left(\mu^{\circ}_{Cu} - RT \ln(a^{\circ}_{Cu^{2+}})\right)}_{V^{\circ}_{RHS}\ (red)} - \underbrace{\left(-\frac{1}{2F}\left(\mu^{\circ}_{H_2} - 2RT \ln a^{\circ}_{H^+}\right)\right)}_{V^{\circ}_{LHS}\ (ox)}$

Factor out minus signs so we can write $V^{\circ}_{cell} = V^{\circ}_{red} - V^{\circ}_{ox}$ s.t. $V > 0$

$$\boxed{V^{\circ}_{cell} = V^{\circ}_{RHS} - V^{\circ}_{LHS}}$$

Standard potentials, when all species are @ standard state : 1 M for solutes, 1 bar for gases.

$$\therefore V_{cell} = V^{\circ}_{cell} + \frac{RT}{2F}\left(\ln(P_{H_2}^{\beta}) - 2\ln(C_{H^+}^{\gamma_1} f_{H^+}^{\gamma_1}) + \ln(C_{Cu^{2+}}^{\gamma_2} f_{Cu^{2+}}^{\gamma_2})\right)$$

* assume all $f_j = 1$:

$$\boxed{V_{cell} = V^{\circ}_{cell} + \underbrace{\frac{RT}{2F} \ln\left(\frac{P_{H_2}^{\beta} C_{Cu^{2+}}^{\gamma_2}}{(C_{H^+})^2}\right)}_{\text{Nernstian correction}}}$$

$\eta = V - U$

This was a half cell potential (for an electrode) → Not the same as V_{cell}

To discuss half cell potentials we reference to a standard Hydrogen Electrode (SHE) $\left[V^{\circ}_{H_2/H^+}\right]$

an ideal solution @ equil exposed to 1 bar H_2 & 1 M H^+ (aq) [pH = 0]

Cr: $AgCl + e^- \rightleftharpoons Ag + Cl^-$ in ref to SHE

$\overset{\alpha}{Pt} \mid \overset{\beta}{H_2(g)} \mid \overset{\gamma}{HCl(aq)} \mid \overset{s}{AgCl(s)} \mid \overset{\beta'}{Ag(s)} \mid \overset{\alpha'}{Pt(s)}$

Phase eq : $\mu_{e^-}^{\alpha'} = \mu_{e}^{\beta'} = \mu_{e}^{s}$

$AgCl + e^- \rightleftharpoons Ag + Cl^-$ ⟶ This would be an equivalent reaction to look at (thermodynamic analysis will lead to an answer both ways)

Phase eq: $\mu_{e^-} = \mu_e - \mu_e$

$AgCl + e^- \rightleftharpoons Ag + Cl^-$ → This won't be equivalent reaction to keep as (thermodynamic analysis will lead to answer both ways)

RHS Rxn: $Ag \rightleftharpoons Ag^+ + e^-$ → μ_{e^-}

$\mu_{Ag}^{\beta'} = \mu_{Ag^+}^{\delta} + \mu_{e^-}^{\alpha'}$

LHS Rxn: $H_2 \rightleftharpoons 2H^+ + 2e^-$

$\mu_{H_2}^{\beta} = 2\mu_{H^+}^{\gamma} + 2\mu_e^{\alpha}$

$\mu_e^{\alpha} - \mu_e^{\alpha'} = F V_{cell}$

$\Rightarrow (\mu_{H_2}^{\beta} - 2\mu_{H^+}^{\gamma}) - (\mu_{Ag}^{\beta'} - \mu_{Ag^+}^{\delta}) = F V_{cell}$

$$V_{cell} = \frac{1}{2F}(2\mu_e^{\alpha} - \mu_e^{\alpha'})$$

Half-reaction values

Note 1: If we tabulate all rxns vs. SHE, then we can determine cell potentials + Nernst correction for any comb. of half rxns

$V^\circ_{cell} = V^\circ_{Ag/Ag} - V^\circ_{Cu/Cu^{2+}}$

$$\boxed{V^\circ_{cell} = (V^\circ_{Ag/Ag} - V^\circ_{H_2/H^+}) - (V^\circ_{Cu/Cu^{2+}} - V_{Cu/Cu^{2+}})}$$

tabulated — tabulated → Standard potentials (or standard reduction potential)

↓ ↓

There are half-reactions

Note 2: If we experimentally measure electrode rxns @ standard conditions then we can just use the Nernst correction for deviations from tabulated standards.

ex: $Cu \rightleftharpoons Cu^{2+} + 2e^-$ in 0.0001M HCl in cell w/ SHE

$$V_{cell} = V^\circ_{cell} + Nernst_{cell} = V^\circ_{cell} + \frac{RT}{2F}\ln Cu^{2+}$$ ★

$V^\circ_{cell} = (V^\circ_{Cu/Cu^{2+}} - V_{SHE})$ ↳ set to zero

If Pt is electrode (then electrolyte conc. matters) → That is why we correct for that

The tabulated values are particularly useful when conducting experiments, bc it's often inconvinient to experimentally set up SHE

↳ Indeed we can set up a convinient electrode like Ag/AgCl in KCl & ref our WE to that:

$(V^\circ_{Ag/AgCl} - V^\circ_{H_2/H}) = 0.222\ V$

or, $V^\circ_{Ag/AgCl} = 0.222V$ vs. SHE ||

Eg: Ref. will have liquid junction potential (across the membrane)

Ag/AgCl Ref; $Cu \rightleftharpoons Cu^{2+} + 2e^-$

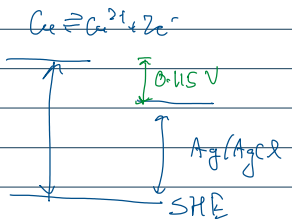
↑ want this to be in eq.

$Cu \rightleftharpoons Cu^{2+} + 2e^-$ → Finicky

Ag/AgCl Ref ; $Cu \rightleftharpoons Cu^{2+} + 2e^-$ ← want this to be in eq.

0.222 V

$U^\circ_{Cu/Cu^{2+}} = 0.337$



$V_{Cell} = 0.337 - 0.222\ V = 0.115\ V$

A potentiostat provides ΔV b/w WE and Reference electrode → measures the I and then provides a ΔV at CE to match that i.

Tafel kinetics:

$$i_a = i_0 \exp\left(\frac{\alpha_a F \eta}{RT}\right)$$

$$\frac{15 \times 10^{-3}}{1 \times 10^{-4}} = \exp\left(\frac{(1.5)96485\,\eta}{RT}\right)$$

$$\frac{\ln(15 \times 10)\,RT}{(1.5)(96485)} = \eta$$

η vs. REF

$\eta = V - U$

$V_{SHE} = \eta + 0.222\ V$

Lecture 12 (Nernst)

Thursday, 16 February 2023 1:33 PM

$$\sum_j s_{ij} M_j = n e^-$$

Nernst Equation

$$U_i - U^\circ_{H_2/H^+} = \left(U_i^\circ - U^\circ_{H_2/H^+}\right) - \frac{RT}{F}\left(\sum_j \frac{s_{ij}}{n} \ln c_{ij}^*\right)$$

(s_{ij}: stoichiometric coefficient; n: e^- mol of product; SHE; SHE)

$c_{ij}^* = c_{ij}$ for solute

$c_{ji}^* = P_{ij}$ for gas

Reversible Hydrogen Electrode

It is more convinient to reference a hydrogen electrode $\neq$ pH = 0

↳ eg: V_{cell} for H_2 fuel cell/water electrolyzer does not depend on H^+ conc. (pH)

But V_{red} & V_{ox} are both dependent on pH (i.e. the half reactions)

↳ If we ref a single half reaction to SHE, it would appear that a large potential is required @ pH > 0

Is E_{OC} @ equilibrium? i.e. V_{eq} value

To avoid confusion, we employ a RHE which is an SHE w/ a nernstian correction for pH shift

$$H_2 \rightleftharpoons 2H^+ + 2e^-$$

$$n_e = 2e^-$$

$$\text{Nernst: } V_{H_2/H^+} = \left(\cancel{U^\circ_{H_2/H^+}}^{\,0} - U^\circ_{H_2/H^+}\right) - \frac{RT}{2F} \ln\left(\frac{P_{H_2}}{C_{H^+}^2}\right) \quad ; \quad pH = -\log_{10} C_{H^+}$$

$$= \frac{RT}{F} \ln C_{H^+}$$

$$= -\frac{RT}{F} \ln\left(10^{-pH}\right)$$

$$\boxed{V_{H_2/H^+} = -0.0591\ V \text{ vs RHE}}$$

Spontaneity: 1) $(V_1$ vs. SHE $- V_2$ vs. SHE$) > 0$ } This will give a spontaneous (galvanic cell)

If < 0, take $V_2 - V_1$ }

Rxn @ more negative potential proceeds in oxidation direction

→ The more negative the potential, the more likely oxidation is to occur.
→ The no. of e^- involved does not matter to determine V_{cell} (since they are all normalized by the SHE)

Nernst Corrosion + Pourbaix diagrams

↳ Tell us about stability/passivation of metal electrodes as a function of potential + pH

+ Constructed w/ Nernst eq.
+ Tell us how susceptible metals are to corrosion via H_2 or O_2

eg: $P_{H_2} = 1$ bar, $P_{O_2} = 1$ bar ; V_{H_2/H^+} & V_{O_2/H_2O} as function of pH

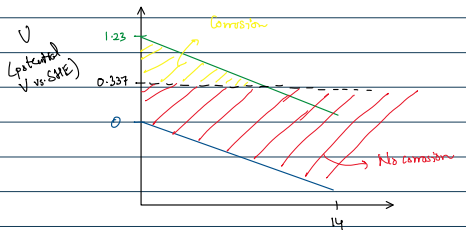
$H^+ + 2e^- \rightarrow \frac{1}{2} H_2$

$$(V_{H_2/H^+} - V^\circ_{H_2/H^+}) = (V^\circ_{H_2/H^+} - V^\circ_{H_2/H^+}) - \frac{RT}{F} \ln\left(\frac{P_{H_2}^{1/2}}{C_{H^+}}\right)$$

($V^\circ_{H_2/H^+} = 0$)

$V_{H_2/H^+} = -0.0591 \times pH$ V vs. SHE

Similarly, $V_{O_2/H_2O} = 1.23 - 0.0591\ pH$ V vs. SHE



Will a metal corrode?

$Cu \rightarrow Cu^{2+} + 2e^-$; $V_{Cu/Cu^{2+}} = \left(V^\circ_{Cu/Cu^{2+}}\ vs.\ SHE\right) - \frac{RT}{2F} \ln C_{Cu^{2+}}^{-1}$

$= 0.337 + \frac{RT}{F} \ln C_{Cu^{2+}}^{1/2}$ ← Not dependent on pH

Stability: • $(V_{Cu/Cu^{2+}} - V_{H_2/H^+}) > 0$ Spont ✓

↑ This one is more negative (i.e. this is oxidized)

∴ Cu does not corrode

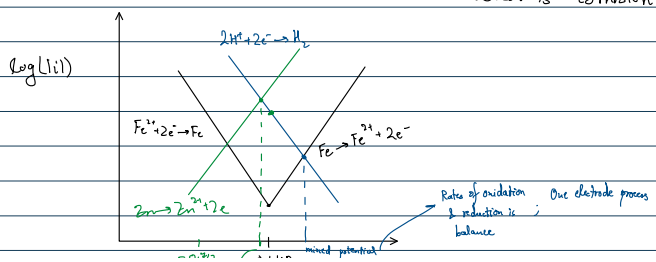
• $(V_{O_2/H_2O} - V_{Cu/Cu^{2+}}) > 0$ Spont ✓

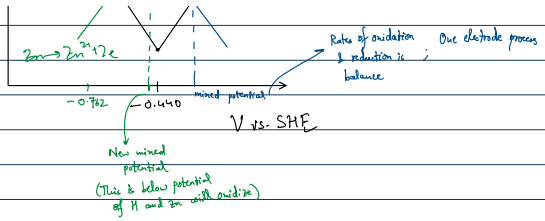
↓ More neg (oxidation)

∴ Cu does corrode

Current density vs. potential

What is corrosion potential & corrosion rate?





Electrochemical Transport

(1) Species can move via convection, diffusion and electrical migration

$$\bar{N}_j = \underbrace{-z_j u_j F c_j \nabla\phi}_{\text{migration}} \underbrace{- D_j \nabla c_j}_{\text{Diffusion}} + \underbrace{c_j \bar{v}}_{\text{convection}}$$

u_j: Ionic mobility; $\nabla\phi$: gradient of potential; $\bar{v}$: fluid velocity (Navier Stokes)

↳ migration: · how do things move in response to electric field ($-\nabla\phi$)
- ϕ represents electrostatic potential
- u_j describes avg. velocity of species acted upon by 1 N/mol force
- $z_j F$ → charge per mole of species

$$\therefore \underbrace{(u_j F z_j \nabla\phi)}_{cm/s} \cdot \underbrace{c_j}_{mol/cm^3} \quad ; \text{ for neutral speis: } z_j = 0$$

↳ Diffusion: · how do things move in response to conc. driving force?

↳ Convection: · how do things move in response to bulk motion of medium

Valid for dilute solutions → D_j constant and equal D_j of j @ infinite dilution
↳ No frictional forces on j besides solvent (molten salt)

(2) Flow of a charged species constitutes an electric current

↳ Current in electrolyte is due to motion (flux) of charged particles

$$\bar{i} = F \sum_j z_j \bar{N}_j \quad ; \quad \bar{i} \text{ is a vector quantity}$$

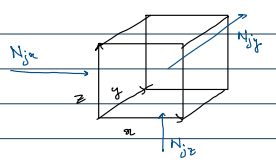
(3) Mass is conserved

$$\text{Acc} = \text{In} - \text{out} + \text{gen} - \text{cons.}$$

$$\frac{dc_j}{dt} = -\nabla \cdot \bar{N}_j + R_j$$

R_j: ↳ generation in homogenous soln (not electrode process) i.e. $R_j = 0$ } No reaction in the bulk

electrode reactions will be BC



In 1D: $\lim_{\Delta x \to 0} \frac{N_{jx}|_x - N_{jx}|_{x+\Delta x}}{\Delta x} = -\frac{\partial N_j}{\partial x}$

(4) Electroneutrality is imposed everywhere

$$\sum_i z_j c_j = 0$$

These principles enable us to describe current flow in electrolyte

$i = -F^2 \nabla\phi \sum z_j^2 u_j c_j - F \sum z_j D_j \nabla c_j + F v \underbrace{\sum z_j c_j}_{0}$; Convection does not constitute a current

For a well-mixed $\nabla c_j = 0 \Rightarrow i = -F \nabla\phi \sum z_j^2 u_j c_j$; $\kappa = F^2 \sum z_j^2 u_j c$

$\therefore \boxed{i = -\kappa \nabla\phi}$ ← Ohm's law

Fluid flow →

High pot. high conc. — ⊕ → Conv. diff. mig. — ⊖ ← mig. Conv. diff. — Low potential Low conc.

Even in the case of migration only, ions move differently depending on chemical identity, conc, charge → carry diff. currents

↳ We can characterize the fractions of current carried by a species as the transference #

$t_i \cdot i = -F^2 z_i^2 u_i c_i \nabla\phi = \frac{z_i^2 u_i c_i}{\sum_j z_j^2 u_j c_j} i$; i = individual species

$\frac{z_i^2 u_i c_i}{\sum_j z_j^2 u_j c_j} = t_i$

→ When modelling reaction - conv - diff - mig. ∴ We use material balances that adhere to concepts of mass + charge

$\frac{\partial c_j}{\partial t} = -\nabla \cdot \bar{N}_j + R_j$ (often 0 but...)

- → precipitation
- → $H^+ + OH^- \rightleftharpoons H_2O$
- ↳ any buffer reaction
  - ↳ bicarbonate

} Any homogeneous reaction

SEI → faradaic process actually (heterogeneous)

$\nabla \cdot N_j = \nabla \cdot (-z_j u_j F c_j \nabla\phi - D_j \nabla c_j + c_j \bar{v})$

$= \nabla \cdot (z_j u_j F c_j \nabla\phi - D_j \nabla c_j) + \nabla c_j \cdot \bar{v}$ ← Nernst-Einstein $u_j = \frac{D_j}{RT}$

$\boxed{\nabla \cdot N_j = -D_j \left(\frac{zF}{RT} \nabla \cdot (c_j \nabla\phi) + \nabla^2 c_j \right) + \nabla c_j \cdot \bar{v}}$

↳ We can write one of these eqn. for each species in our system (i.e. n eqs for n species)

↳ But we have (n+1) variables, considering ϕ

↳ Final equation → comes from electroneutrality

↳ This forces the electric field to take on whatever value satisfies this equation

$\frac{\partial}{\partial t} \underbrace{F \sum z_j c_j}_{0} = -\nabla \cdot F \sum z_j \bar{N}_j$ ← definition of current (flux of charged species) ; $\frac{\partial c_j}{\partial t}$

$\boxed{-\nabla \cdot i = 0}$

↳ Charge is conserved (divergence is zero) → since charge is carried by physical particles (& these are conserved)

Since $i = -\kappa \nabla\phi$

$-\nabla \cdot i = -\nabla \cdot (\kappa \nabla\phi) = 0$

$\boxed{\nabla^2 \phi = 0}$ → Laplace eq. (Valid for no conc. gradients)

These n+1 (flux + electroneutrality) give system of partial differential equations, need BC's:

↳ BC @ (insulating surface): normal comp. of flux is zero for all species

$\bar{n} \cdot \bar{N}_j = 0$ (n of these BC's + electroneutrality)

$\bar{n} \cdot \bar{i} = 0$

↳ BC @ (electrode surfaces): * Faraday's law as flux

↳ Flux of species @ electrode is proportional to current density

$\therefore \bar{n} \cdot \bar{N}_j = \frac{-S_j}{nF} i_n$ ← defined by B-V (or other relevant kinetic expression)

η is at electrode surface (must account for potential in electrolyte formed by MJ)

$\eta = (V - \phi) - (U - \phi^0)$

- V: applied potential
- ϕ: electrolyte potential
- U: electrolyte potential @ equil. ($i = 0$, $\phi^0 = 0$)
- η: user-defined

$\eta = V - \phi - U$ ← Thermo

variable to solve for PDE ($\nabla^2\phi = 0$ → or other forms)

↳ i_0 will also depend on c_j^{surf} → These are also solved for @ surface

Limiting cases for BC's:

↳ Limiting current density: @ large η, rxn may be fast that $c_{ion} = 0$ @ surface.
in this case current density @ electrode is defined completely by N_j

↳ Chemically limited rxn rate: @ slow rxn rates; $c_j^{surf} = c_j^{bulk}$ and we can neglect conc. gradients

↳ our only variable is ϕ, defined by $\nabla^2\phi = 0$

* and if ohmic resistance to current flow is much larger than charge transfer resistance (eg: poorly conducting electrode), then $V = \phi$ @ electrode surface

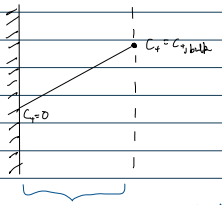
↳ Slow charge transfer rate: charge transfer resistance is much larger than ohmic resistance in solution, then we have uniform current density

$i_n = \frac{I}{\text{area}}$ @ all positions

(ohmic resistance: poorly conducting electrolyte)

Transport of Binary Electrolyte

↳ M^+X^- of a single salt species is present (migration is most important)
eg: Cu plating @ electrode where $C_{Cu^{2+}} = 0$ @ surface (fast rxn)



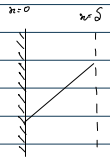
diffusion BL ($\delta \sim 10\,\mu m$) → Fick's law

$N_+ = -D \frac{(c_{+,bulk} - c_s)}{\delta}$

Lecture 15

Tuesday, 28 February 2023 1:31 PM

Nernst-Fickian Transport



For $M_{\nu_+}X_{\nu_-}$ where,

$z_+\nu_+ + z_-\nu_- = 0$

$c = \frac{c_+}{\nu_+} = \frac{c_-}{\nu_-}$

* Start w/ writing flux of each species:

1) $\nabla \cdot N_+ = \nabla \cdot (-z_+ u_+ F c \nu_+ - D_+ u_+ \nabla c) + \nu_+ \nabla c \cdot \bar{v}$ (convection)

(conc. of cation species)

2) $\nabla \cdot N_- = \nabla \cdot (-z_- u_- F c \nu_- - D_- u_- \nabla c) + \nu_- \nabla c \cdot \bar{v}$

* Material balance:

3) $\nu_+ \frac{\partial c}{\partial t} = -\nabla \cdot N_+ \Rightarrow \frac{\partial c}{\partial t} = -\frac{1}{\nu_+}(\nabla \cdot N_+)$

4) $\nu_- \frac{\partial c}{\partial t} = -\nabla \cdot N_- \Rightarrow \frac{\partial c}{\partial t} = -\frac{1}{\nu_-}(\nabla \cdot N_-)$

* Subtract (3-4): $0 = (z_+u_+ - z_-u_-)(F \nabla \cdot (c \nabla \phi)) + (D_+ - D_-)\nabla^2 c$

rearrange ↓

$$\nabla \cdot (c \nabla \phi) = -\frac{(D_+ - D_-)}{F(z_+u_+ - z_-u_-)} \nabla^2 c$$

* Plug this into: (either 3 or 4) $\frac{\partial c}{\partial t} = D \nabla^2 c - \bar{v} \cdot \nabla c$, where $D = \frac{(z_+u_+D_- - z_-u_-D_+)}{z_+D_+ - z_-D_-}$

Apply Nernst Einstein $D_j = RTu_j$

$$\boxed{D = \frac{(z_+ - z_-) D_+ D_-}{z_+D_+ - z_-D_-}}$$

@ SS, $\frac{dc}{dt} = 0$ in 1D, $\bar{v} = 0$ (Only diff. & migration)

$$D \frac{d^2c}{dx^2} = 0$$

$c = Ax + B$; BC's: 1) @ $x = 0$, $c = 0$

$$\boxed{c = \frac{C_{bulk}\, x}{S}}$$

2) @ $x = S$, $c = C_{bulk}$

$\therefore$ N_+ in BL ($\bar{v} = 0$) $= -z_+\nu_+u_+Fc\frac{d\phi}{dx} - D_+\nu_+\frac{dc}{dx}$ (we know this; solve for this)

Since this is Cu plating, the anion does not play any role: $N_- = 0$

$\therefore$ $0 = -z_-\nu_-u_-Fc\frac{d\phi}{dx} - D_-\nu_+\frac{dc}{dx}$ ↓ plug in for $\frac{d\phi}{dx}$ in N_+

$$\boxed{N_+ = -\nu_+\left(\frac{z_-u_-D_+ - z_+u_+D_+}{z_-u_-}\right)\frac{dc}{dx}}$$

For $CuSO_4$, $z_+ = 2$, $z_- = -2$

$$\boxed{N_{Cu^{2+}} = -2D_{Cu^{2+}}\frac{C_{bulk,Cu^{2+}}}{S}}$$

↳ Exactly 2x the fickian diffusion

Why? + Conc. of anions has to also vary linearly in BL — i.e. anions create E-field
even though flux is 0, which imposes an electric — s.t. Cu has 2x flux
field, which Cu^{2+} ions migrate within

From transference #,

$$t_+ = \frac{z_+^2 u_+ c_+}{\sum c_j u_j z_j^2} \quad \text{(Nernst Einstein)} \qquad \text{for binary electrolyte} \Rightarrow \frac{z_+ D_+}{z_+ D_+ - z_- D_-}$$

$$t_+ = \frac{z_+ u_+}{z_+ u_+ - z_- u_-}, \quad t_- = \frac{-z_- u_-}{z_+ u_+ - z_- u_-}$$

Applying this,

$$D = \frac{z_+ u_+ D_- - z_- u_- D_+}{z_+ u_+ - z_- u_-}$$

$$\frac{N_+}{\nu_+} = -\frac{dc}{dx}\left(\frac{z_- u_- D_+ - z_+ u_+ D_-}{z_- u_-}\right)$$

$$= \frac{D(z_+ u_+ - z_- u_-)}{z_- u_-}\frac{dc}{dx}$$

$$= \frac{-D}{t_-}\frac{dc}{dx}$$

$t_- + t_+ = 1$ (fraction of total current)

$$\boxed{\frac{N_+}{\nu_+} = \frac{-D}{(1-t_+)}\frac{dc}{dx}}$$

If t_+ is small (migration becomes unimportant) !!! ★

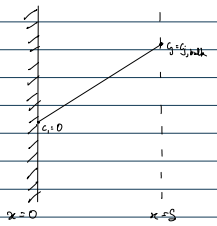
+ Often we want to add extra ions to solution (called supporting electrolyte)

↳ increase the conductivity (recall V)

↳ effect on migration → decreases electric field, means migration is less important (decrease flux to surface!)

Supporting Electrolyte (Ternary Electrolyte)

↳ Add H_2SO_4, complete dissociation to $2H^+ + SO_4^{2-}$; Cu-plating w/ $CuSO_4 + H_2SO_4$ @ $C_{Cu^{2+}} = 0$ @ surface



1 = Cu^{2+}
2 = SO_4^{2-}
3 = H^+

Dilute solution (No ion-ion, just ion-solvent)

Assume:
1) SS
2) $\bar{v} = 0$
3) Only Cu^{2+} reacts @ electrode ($N_2 = N_3 = 0$)

+ Flux of 2 and 3: (Nernst Einstein: $u \to D$)

$$\not{D}\frac{dc_j}{dx} = -\frac{c_j z_j F}{RT}\not{D}\frac{d\phi}{dx}$$

$\frac{d\ln c_j}{dx}$

$$\frac{1}{c_j}\frac{dc_j}{dx} = -\frac{z_j F}{RT}\frac{d\phi}{dx}$$

Integrate (

$$\ln c_j = -\frac{z_j F}{RT}\phi + A \quad ; \quad BC: @\, x = \delta,\ \phi = 0,\ c_j = c_{j,bulk} \quad ; \quad A = \ln c_{j,bulk}$$

$$c_j = \exp\left[-\frac{z_j F \phi}{RT}\right]\exp\left[\ln c_{j,bulk}\right]$$

$$c_j = c_{j,bulk}\exp\left[-\frac{z_j F \phi}{RT}\right]$$

↳ Apply electroneutrality to get C_1:

$$z_1 C_1 = -\sum_{j=2}^{3} z_j C_j = -\sum_{j=2}^{3} z_j C_{j,bulk} \exp\left(\frac{-z_j F \phi}{RT}\right)$$

↳ Material balance for Cu:

$$\frac{\partial C_1}{\partial t} = \frac{\partial N_1}{\partial x} + R_1$$

$$\frac{\partial N_1}{\partial x} = 0 \rightarrow N_1 = \text{constant}$$

$$N_1 = -z_1 \frac{D_1 F}{RT} C_1 \frac{d\phi}{dx} - D_1 \frac{dC_1}{dx}$$

* from electroneutrality we saw $C_1 = f(\phi)$

$$\Rightarrow N_1 = D_1 \frac{d\phi}{dx}\left(-z_1 \frac{F C_1}{RT} - \frac{dC_1}{d\phi}\right) \qquad \left\{ \frac{dC_1}{dx} = \frac{dC_1}{d\phi} \cdot \frac{d\phi}{dx} \right.$$

$$\int_{x=0}^{x=\delta} \frac{N_1}{D_1} dx = \int_{\phi=0}^{\phi} \left(z_1 \frac{F C_1}{RT} + \frac{dC_1}{d\phi}\right) d\phi$$

$$\frac{N_1(\delta - x)}{D_1} = \left[\frac{z_1 F}{RT}\left(\frac{-z_2}{z_1} C_{2,bulk}\left(\frac{-RT}{F z_2}\right)\exp\left(\frac{-z_2 F\phi}{RT}\right) + \frac{-z_3}{z_1} C_{3,bulk}\left(\frac{-RT}{F z_3}\right)\exp\left(\frac{-z_3 F\phi}{RT}\right)\right) + C_1\right]_0^{\phi}$$

$\frac{-z_2}{z_1} \cdot C_{2,bulk} \cdot \exp\left(\frac{z_2 F\phi}{RT}\right) + \frac{-z_3}{z_1} C_{3,bulk} \exp\left(\frac{-z_3 F\phi}{RT}\right)$

* Use, $-\frac{z_2 C_{2,bulk} - z_3 C_{3,bulk} + C_{1,bulk}}{z_1}$

$$\boxed{\frac{N_1(\delta - x)}{D_1} = \left(1 - \frac{z_2}{z_1}\right) C_{2,bulk} \exp\left(\frac{-z_2 F\phi}{RT}\right) + \left(1 - \frac{z_3}{z_1}\right) C_{3,bulk} \exp\left(\frac{-z_3 F\phi}{RT}\right) - C_{1,bulk} - C_{2,bulk} - C_{3,bulk}}$$

Need to only evaluate N_1 at one position (since its a constant) and satisfy $C_1 = 0$ @ $x = 0$,

$$C_1 = 0 = \frac{-1}{z_1}\left(z_2 C_{2,bulk} \exp\left(\frac{-z_2 F\phi}{RT}\right) + z_3 C_{3,bulk} \exp\left(\frac{-z_3 F\phi}{RT}\right)\right)$$

$$\phi = \frac{-RT}{F(z_2 - z_3)} \ln\left(\frac{-z_3 C_{3,bulk}}{z_2 C_{2,bulk}}\right) \quad @\ x = 0$$

↳ Plug in to integrated N_1 expression @ $x = 0$ to get N in terms of constants only.

Define a useful parameter $\Rightarrow \gamma = \frac{-z_3 C_{3,bulk}}{z_2 C_{2,bulk}}$; $\phi_0 = \frac{-RT}{F(z_2 - z_3)} \ln\gamma$

@ $x = 0$: $\frac{N_1(\delta - x)}{D_1} = \left(1 - \frac{z_2}{z_1}\right) C_{2,bulk} \exp\left(\frac{-z_2 F}{RT}\left(\frac{-RT}{F} \ln \gamma^{\frac{1}{(z_2 - z_3)}}\right)\right) + \left(1 - \frac{z_3}{z_1}\right) C_{3,bulk} \exp\left[\frac{-z_3 F}{RT}\left(\frac{-RT}{F}\right) \ln \gamma^{\frac{1}{(z_2 - z_3)}}\right] - C_{1,bulk} - C_{2,bulk} - C_{3,bulk}$

$$= \left(1 - \frac{z_2}{z_1}\right) C_{2,bulk} \gamma^{\frac{z_2}{(z_2 - z_3)}} + \left(1 - \frac{z_2}{z_1}\right) C_{3,bulk} \gamma^{\frac{z_3}{(z_2 - z_3)}} - C_{1,b} - C_{2,b} - C_{3,b}$$

$$\frac{N_1 \delta}{D_1 C_{1,b}} = \frac{\frac{-z_1}{z_2}\left[\left(1 - \frac{z_2}{z_1}\right)\gamma^{\frac{z_2}{(z_2 - z_3)}} - \frac{z_2}{z_3}\left(1 - \frac{z_3}{z_1}\right)\gamma^{1 + \frac{z_3}{z_2 - z_3}} + \frac{z_2}{z_1}(1 - \gamma) - 1 + \frac{z_2}{z_3}\gamma\right]}{(1 - \gamma)}$$

For our electrolyte, (z_j):

$$\boxed{\frac{N_1 \delta}{D_1 C_{1,b}} = \frac{(-2 - \gamma + 3\gamma^{2/3})}{(1 - \gamma)}} \quad ; \quad \gamma = \frac{-z_3 C_{3,bulk}}{z_2 C_{2,bulk}}$$

$\gamma = 0 \rightarrow$ reduces to binary electrolyte — Have to apply L'Hopital!

↳ $\frac{N_1 \delta}{D_1 C_{1,b}} = -2$

$\gamma = 1 \rightarrow$ "[illegible] electrolyte" $\Rightarrow N_1 \delta = -1$ — Main reason would be to

↳ $\frac{N_i \delta}{D_i C_{i_b}} = -2$

$\gamma = 1 \rightarrow$ "excess supporting electrolyte" $\Rightarrow \frac{N_i \delta}{D_i C_{i_b}} = -1$

↳ Expected result for diffusion only! (migration is not important)

Main reason would be to increase conductivity (improve EE of cell)
↓
You still want to keep the same kinetics by using the same electrolyte (but just reducing ohmic resistan)

So generally, when is migration important?

↳ Solve our combined current flux, eq. for $\nabla\phi$

$$\bar{i} = -F \sum_j \left(z_j^2 u_j F c_j \nabla\phi + z_j D_j \nabla c_j \right)$$

$$\nabla\phi = \frac{-1}{F \sum z_j^2 u_j c_j} \left(\frac{\bar{i}}{F} + \sum z_j D_j \nabla c_j \right)$$

Pg 120

$j \rightarrow$ species

$k \rightarrow$ all other species

$t_j \rightarrow 0$

↳ Plug into general flux expression for species k

$$N_k = \frac{t_j}{z_j F} \left(\bar{i} + F \sum_i \right)$$

* When $t_k \rightarrow 0$, migration can be neglected (for the reacting species at electrode)

Diffusion BL:

$$\delta_{diff} = \delta_{mom.} \, Sc^{-1/3} \; ; \quad Sc = \frac{\nu}{D_j}$$

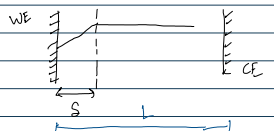
↑ navier stokes

Limiting cases for governing Eqs:

↳ Solving our transport eqs can be cumbersome and require numerical techniques to solve but we can more easily determine transport behaviour in limiting cases that are often good first approx.

1. Uniform Electrolyte composition (low current)

↳ When current is applied @ electrode, reactants are consumed, changing composition



If $\frac{i_{avg}}{i_{lim}} \ll 1$, consider uniform electrolyte

↑ current @ which $C_{surf} = 0$

* even if $\frac{i_{avg}}{i_{lim}} = 0.2$

$C_{surf} = 80\% \, C_{bulk}$

↳ If composition is uniform, how many eqs do we solve?
electroneutrality → to solve for ϕ potential in a solution

↳ Stated before $\nabla c_j = 0$, potential is defined by Laplace eq: $\nabla^2\phi = 0$

In 1D ex.: $\frac{d^2\phi}{dx^2} = 0$, subject to BC's @ electrodes

BC1 : WE @ $x=0$ $\quad i_{avg} = -F \sum z_j N_j$

$$= -F \sum z_j \left(z_j u_j F c_j \frac{d\phi}{dx} + z_i D_j \nabla c_j - c_j \bar{v} \right)$$

($\nabla c_j \to 0$, $\bar{v} \to 0$)

$$i_{avg} = -\kappa \frac{d\phi}{dx}$$

Assume cathodic Tafel kinetics:

$$-\kappa \frac{d\phi}{dx} = -i_0 \exp\left(\frac{-\alpha_c F (V - \phi - U)}{RT} \right)$$

often U gets dropped (assuming that ref electrode is of same comp. as WE)

$$-\kappa \frac{d\phi}{dx} = -i_0 \exp\left(\frac{\alpha_c F (V - \phi)}{RT} \right)$$

BC2 (CE): @ $x = L$, $\phi = 0$

Integrating (x2) $\Rightarrow$ $\phi = Ax + B$

BC $\hookrightarrow$ $\phi = \frac{i_{avg}}{\kappa}(L - x)$, where $i_{avg} = -i_0 \exp\left[\frac{-\alpha_c F (V - \frac{i_{avg} L}{\kappa})}{RT} \right]$

We can use ϕ to tell us something about resistances that are imp. for echem performance

↳ Using ohm's law, we can define: + Charge transfer resistance (R_{ct}) → associated w/ Faradaic process @ electrode

$$R_{ct} = \left. \frac{\partial (V - \phi_s)}{\partial i} \right|_{i_{avg}}$$

$$i_{avg} = -i_0 \exp\left(\frac{-\alpha_c F (V - \phi)}{RT} \right)$$

$$(V - \phi) = \frac{RT}{\alpha_c F} \ln\left(\frac{i}{i_0} \right)$$

$\frac{R_{ct}}{R_\Omega}$ → charge dist.

* Slope of polarization curve, how much does current change in response to small change in overpotential

+ Ohmic resistance (solution resistance) @ electrode:
↳ associated w/ charged species moving in electrolyte

$$R_\Omega = \left. \frac{\partial (\phi_0)}{\partial i} \right|_{i_{avg}} = \frac{\partial}{\partial i} \left(\frac{i_{avg}}{\kappa} (L - x) \right) = \frac{L}{\kappa}$$

2. Limiting Current Density (high current)

↳ $c_{i,lim} = 0$ @ electrode surface

* Assume $t_{i,lim} \to 0$, $\bar{v} = 0$

$$\text{SS} \quad \nabla^2 c_{i,lim} = 0 \qquad \text{subject to: } c_{i,lim} = 0 \text{ @ } x = 0, \quad c_{i,lim} = c_{i,bulk} \text{ @ } x = \delta$$

$c_{i,lim}$

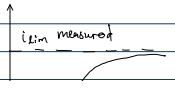
$$\bar{n} \cdot \bar{N}_j = \frac{-s_j}{nF} i_{lim}$$

$$i_{lim} = \frac{nF}{-s_j} N_j$$

$$i_{lim} = \frac{nFD}{s_j} \left. \frac{dc_j}{dx} \right|_{x=0}$$

$$i_{lim} = \frac{nFD_{lim}}{\delta} c_{i,bulk}$$

In practice we can compute this calc. to a measured i_{lim}



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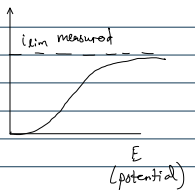
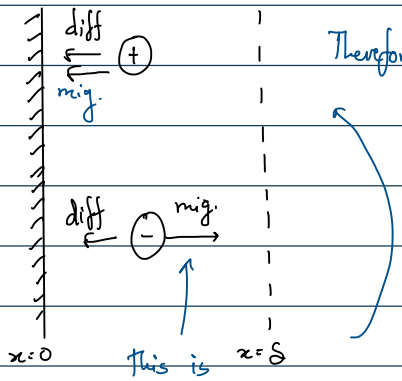
In practice, we can compute this calc. to a measured i_{lim}



Lecture 16

Thursday, 2 March 2023 1:31 PM

Exam 1: Open book, 1 sheet front-side notes, calculator



Therefore, the E-field is sit. cation will have an extra migration from.

This is here because we have imposed no flux of anion @ electrode

Since its being pushing & pulling in both sides → overall net flux is zero !

Notes cont. in previous lecture

Ternary electrolyte: $\frac{N_1\delta}{D_1 c_{1,bulk}} = \frac{-2-\gamma+3\delta^{2/3}}{1-\delta}$, $\gamma = \frac{-z_3 c_{3,bulk}}{z_2 c_{2,bulk}}$

↳ $\gamma=0 \rightarrow$ binary electrolyte $\Rightarrow \frac{N_1\delta}{D_1 c_{1,bulk}} = -2$

↳ $\gamma=1 \rightarrow$ XS supporting electrolyte $\Rightarrow \frac{N_1\delta}{D_1 c_{1,bulk}} = -1$

(when $t_j \to 0$, we neglect migration)

Limiting cases: 1) Uniform composition $\frac{i_{avg}}{i_{lim}} << 1$ ← corresponds maximum change in conc. in the BL

$\nabla^2\phi = 0 \Rightarrow \phi = \frac{i_{avg}}{\kappa}(L-x)$

→ Broke this down into charge transfer + ohmic resistances

$R_{ct} = \left.\frac{\partial(V-\phi)}{\partial i}\right|_{i_{avg}}$

$R_\Omega = \left.\frac{\partial \phi}{\partial i}\right|_{i_{avg}}$

R_{ct}/R_Ω ratio would tell you what current regime you would in

↳ Limiting current density ($c_{i,lim} = 0$ @ surface): $\nabla^2 c_{i,lim} = 0 \Rightarrow c_{i,lim} = \frac{c_{i,bulk}}{\delta} x$

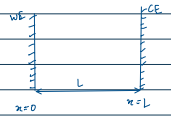
$i_{lim} = \frac{nF D_{i,lim}}{s_{i,lim}} \frac{c_{i,bulk}}{\delta}$

Using $R_{ct} \& R_\Omega$ (for uniform composition) → what cell length L, would cause current density to be "ohmically limited"

(Tafel kinetics) $\frac{R_{ct}}{R_\Omega} = \frac{RT}{\alpha_c F i_{avg}}\left(\frac{\kappa}{L}\right)$ If $\frac{R_{ct}}{R_\Omega} << 1 \rightarrow$ dominated by ohmic resistance

comes from differentiating B-V i expression

Assume $\kappa = 0.1\ S\ cm^{-1}$
$i_0 = 0.1\ mA\ cm^{-2}$
$\alpha_c = 0.5$
@ $i_{avg} = 100\ mA/cm^2$
} $L = 5\ cm$ ← characteristic cell length



If we instead assume linear kinetics, how does V and Q_0 vary w/ $\frac{R_{ct}}{R_\Omega}$?

linear. $i_{avg} = \frac{i_0 F(\alpha_a+\alpha_c)}{RT}(V-\phi_0)$ ↳ potential at the electrode

$\phi_0 = \frac{i_{avg} L}{\kappa}$

$\left.\frac{\partial(V-\phi)}{\partial i}\right|_{i_{avg}} = \frac{RT}{i_0 F(\alpha_a+\alpha_c)} = R_{ct}$

$\left.\frac{\partial \phi}{\partial i}\right|_{i_{avg}} = \frac{L}{\kappa} = R_\Omega$

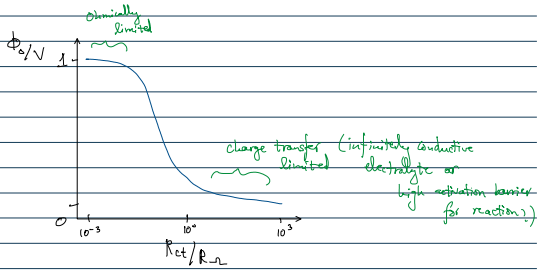
Independent of i_{avg}

$\frac{R_{ct}}{R_\Omega} = \frac{RT}{(i_0)F(\alpha_a+\alpha_c)} \cdot \left(\frac{\kappa}{L}\right)$

plugging into $\phi_0 = i_{avg}\frac{L}{\kappa}$

$\phi_0 = \frac{i_0(\alpha_a+\alpha_c)FL}{RT\kappa}(V-\phi_0)$

$\frac{\phi_0}{(V-\phi_0)} = \frac{i_0(\alpha_a+\alpha_c)FL}{RT\kappa} = \frac{R_\Omega}{R_{ct}}$



$$\frac{\phi_a}{V} = \frac{1}{w_a} \cdot \quad w_a = \frac{R-n}{R_{ct}}$$

Linear kinetics

Limiting Currents:

$\nabla^2 c_{i,lim} = 0$, BC's : @ $x=0$, $c_{i,lim} = 0$

@ $x=\delta$, $c_{i,lim} = c_{i,bulk}$

$$c_{i,lim} = \frac{c_{i,bulk}\, x}{\delta} \quad ; \quad i_{lim} = \frac{nF D_{i,lim} c_{i,bulk}}{\delta_{lim}\ \delta}$$

What will be i, if $c_{surf} \neq 0$, assuming $t_{i,lim} = 0$, i.e. diffusion only?

$\nabla^2 c = 0$, BC @ $x=0$, $c = c_{surf}$

@ $x=\delta$, $c = c_{bulk}$

$c = Ax + B$

$$\boxed{c = \frac{c_b - c_{surf}}{\delta} x + c_{surf}}$$

↳ Faraday's law: $i = \frac{nF D_j}{-s_j} \left.\frac{dc_j}{dx}\right|_{x=0} = \frac{nF D_j}{-s_j}\left(\frac{c_{bulk} - c_{surf}}{\delta}\right)$ ← If we measure i & then have δ (from BL analysis) ↳ then can find c_{surf} → Conc of H^+ will give local pH

Lecture 18

Tuesday, 21 March 2023 1:29 PM

Silent Spring, Rachel Carson → "lucid" communication

Current Distributions

Reaction Rate is a key design parameter for any chemical rxn eng. analysis and we know for non-chem rxns, it depends on c_j, T

$$r_j = f(c_j, T, P) \text{ for non-chem}$$

In e-chem systems, $r = \frac{i}{nF}$, and we know that i is additionally dependent on V & ϕ

$$i = f(c_j, T, P, V, \phi)$$

∴ this can create spatial differences of rxn rate that we might not predict from classic rxn eng. (have to solve PDE's for each finite element)

↳ Need some simplifications that arrive at approx. for real behaviour.

+ understand scenarios that lead to specific regimes

+ " governing eqs + BC's within each regime

+ Generalities about current dist., based on solutions to gov. eqs.

Two dim. parameters:

① $\frac{i_{avg}}{i_{lim}}$ → spatial avg (@ each position) ⇒ describes imp. of conc. gradients

↳ $\frac{i_{avg}}{i_{lim}} \to 1$ (MT-controlled limit)

↳ ∇c is important

↳ $\frac{i_{avg}}{i_{lim}} \to 0$ (uniform composition)

↳ ∇c not important

② $W_a = \frac{R_{ct}}{R_e}$ ⇒ describes imp. of charge transfer vs. Ohmic resistance

↳ different dependence on i, depending on kinetic expression (tafel, or linear)

General case:

Governing Eqs: $N_j = -z_j u_j F c_j \nabla\phi - D_j \nabla c_j + v c_j$

$\sum z_j c_j = 0$

"Tertiary Current distribution" → Solving this gives us the complete picture

1) This occurs when $\nabla c_j \neq 0$ ($i_{avg} \not\approx i_{lim}$)
2) Rct an Re are both imp
3) $t_j \to 0$ ⇒ convection diffusion (fast kinetics) → i_{lim}
4) $t_j \neq 0$ ⇒ full governing eqs (no simplifications)
5) $W_a > 0.1$
6) MT BC's typically

Can ∇c_j be neglected? ($\frac{i_{avg}}{i_{lim}} = ?$)

No → $\frac{i_{avg}}{i_{lim}} \sim 1$

Yes ($\frac{i_{avg}}{i_{lim}} \to 0$) → Governing Eq: (Laplace) $\nabla^2\phi = 0$

$t_j > 0$

No → Can electric field be neglected ($t_j = ?$)

Yes, $t_j \to 0$ → Gov. eqs: Convection diff. eq.

$$\frac{\partial c_j}{\partial t} + \bar{v} \cdot \nabla c_j = D_j \nabla^2 c_j$$

Further considerations: Peclet No.

Governing Eq: (Laplace) $\nabla^2\phi = 0$ → Should Rct be considered? ($W_a = ?$)

Yes ($W_a > 0.1$) → Secondary Current distribution

1) $\nabla c_j = 0$
2) Ohmic + Ct resistances are both imp. ($W_a > 0.1$)
↳ (i-η) type BC [tafel, linear etc] (kinetic BC)
↳ Always more uniform than primary

No ($W_a \to 0$) → Primary Current Distribution

1) Occurs when $\nabla c_j = 0$ (uniform conc)
2) Ohmic resistance dominates
↳ Result in constant potential BC's → linear BC's + governing eqs

↳ Result in constant potential BC's → linear BC's + governing eqs

& non-uniform current dist. (has the maximum non-uniformity)

Further considerations: regarding convective transport → Peclet No.

Regions:

Polarization curve → There is i_{lim} for anodic & for cathodic (these two are different)

i_{lim}

III → high kinetics (chemically dominated) R_{ct} is small [Primary]

II → Secondary?

I

→ large WE

→ electrolyte solution has poor conductivity $\kappa \downarrow$ $R_\Omega \uparrow$, } moving towards primary

Don't have enough info about R_Ω to make the general claim

Large electrodes + high conc'n

↳ significant non-uniformity

Wagner No.:

↳ We can make good assumptions about uniformity

$$W_a = \frac{R_{ct}}{R_\Omega}$$

electrodes dealt w/ B-V kinetics

- Charge transfer resistance (R_{ct}) → This is usually at the electrode (since no charge transfer is happening with @ the electrode)
  - ↳ Resistance to current flow due to charge transfer kinetics @ electrodes ($\eta = V - \phi_s - U$)
  - ↳ Last chapter we said $R_{ct} = \left.\frac{\partial(V-\phi)}{\partial i}\right|_{i_{avg}}$ or $\left.\frac{\partial \eta_c}{\partial i}\right|_{i_{avg}}$ ★ ; i_{avg} → average current on the entire electrode

↑ Kinetics → i_{avg} ↑ → R_{ct} ↓ → $W_a \to 0$ (primary)

$$i = i_o \frac{(\alpha_a + \alpha_c) F \eta}{RT} \Rightarrow \frac{\partial \eta}{\partial i}$$

  - ↳ Since dependent on kinetics → will change w/ the given process: · small overpotential (linear kinetics: $R_{ct} = \frac{RT}{(\alpha_a + \alpha_c) F i_o}$)
    - · Large overpotential (Tafel kinetics: $R_{ct} = \left.\frac{\partial \eta}{\partial i}\right|_{i_{avg}} = \frac{RT}{\alpha_a F i_{avg}}$) ↳ does depend on i_{avg}

Anodic Tafel: $\left\{ i = i_o \exp\left[\frac{\alpha_a F \eta}{RT}\right] \right.$

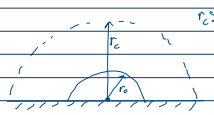
    - * Any other type of kinetics → solve for $\eta = f(i)$ & differentiate, or get slope of

- Ohmic Resistance (R_Ω):
  - ↳ Resistance of current flow due to property of electrolyte + cell

for 1D case, $R_\Omega = \frac{L}{K} = \frac{\text{characteristic length}}{\text{conductivity}}$ → Need to select this carefully

  - ↳ Does not depend on kinetics
  - ↳ Depends on K (type of electrolyte), L (geometry)

ex:



Hemispherical electrode (spherical coord since in that it becomes a 1D problem!)

If uniform composition:

$$\nabla^2 \phi = 0$$

$$\Rightarrow \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial \phi}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial \phi}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2 \phi}{\partial \theta^2} = 0$$

$$\Rightarrow \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial \phi}{\partial r}\right) = 0 \qquad \text{BC's: } \phi = V \text{ @ } r = r_o,\ \phi = 0 \text{ @ } r = \infty$$

$$\therefore \frac{d^2\phi}{dr^2} + \frac{2}{r}\frac{d\phi}{dr} = \frac{r}{2}\frac{\partial^2 \phi}{\partial r^2} + \frac{\partial \phi}{\partial r} = 0$$

$$\phi = \frac{c_1}{r} + c_2 \xrightarrow{\text{BC's}} \boxed{\phi = \frac{r_o V}{r}}$$

$$i = -K \left.\frac{d\phi}{dr}\right|_{r=r_o} \quad (\nabla c = 0)$$

$$i_\Omega = \frac{V}{r_o} K \rightarrow i_\Omega = \text{ohmically limited current (controlled by potential & not kinetics)}$$

$$R_\Omega = \frac{V}{i_\Omega} \text{ (ignoring c.t.)}$$

$$\therefore \boxed{R_\Omega = \frac{r_o}{K}} \rightarrow \therefore \text{This is the characteristic length.}$$

ex: for hemicylindrical electrode

$$R_\Omega = \frac{r_o}{K} \ln\left(\frac{r_{CE}}{r_o}\right) \qquad r_{CE} = \text{counter electrode}$$

↓ Need this for W_a

↓ Need W_a for current dist.

↓ change values for r_{CE} and r_o to determine R_Ω dependence (what dominates)

Putting together R_{ct} and R_Ω: $W_a = R_{ct,L} = \frac{RT}{F} K$) Linear

Putting together R_{ct} and R_Ω: $W_{a,L} = \frac{R_{ct,L}}{R_\Omega} = \frac{RT}{(\alpha_a+\alpha_c)F i_o l}$ } Linear kinetics

$W_{a,T} = \frac{R_{ct,T}}{R_\Omega} =$

Lecture 20

Tuesday, 28 March 2023 1:29 PM

W_a → tells us uniformity of a electrochemical system

↳ $R_{ct} = \frac{\partial \eta}{\partial i}$ (depends on kinetics) ← W_{a_L}

↳ $R_e = \frac{\partial \phi_2}{\partial i}$ (essentially $\frac{\text{characteristic length}}{\text{cond.}}$) ← W_{a_T}

Low W_a → primary current dist.

Characteristics & Solutions of Primary & Secondary current dist.

↳ $\nabla i_j = 0$ $W_a \to 0$, the max non-uniformity in current distribution

↳ In practice, we get governing eq. and boundary condition

This is where the ref matters

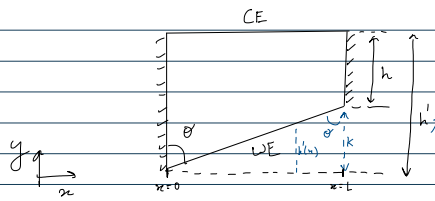
$\eta = V - \phi - \tilde{V}$

$\nabla^2 \phi = 0$; BC's: @ WE surface $\phi = V$

@ CE $\phi = 0$

This is an arbitrary constant value

Ex: Hull cell → has specific geometry to allow experimental determination of current non-uniformity



determined to see the throwing potential of electrolyte

based on how much mass accumulates @ CE at each distance

$K = \frac{L}{\tan\theta}$; $k'(x) = \frac{x}{\tan\theta}$

For primary current distribution: $\nabla^2 \phi = 0$

(Rect. coord.) $\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$

BC's: @ $y = h'$, $\phi = 0$

@ $y = K - k'$ $\phi = V$

@ $x = 0$ $\frac{\partial \phi}{\partial x} = 0$

@ $x = L$ $\frac{\partial \phi}{\partial x} = 0$

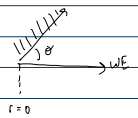
} insulating BC

$i = -\kappa \nabla \phi$

No i through an insulator ⇒ $\frac{\partial \phi}{\partial x} = 0$

Symmetry condition

Conformal mapping solution (to solve a partial diff. eq.): parameters are curve fit by realizing the property of primary current @ junction w/ insulator



$\frac{i(x)}{i_{avg}} \propto r^{\left(\frac{\pi}{2\theta} - 1\right)}$

If $\theta > \frac{\pi}{2} \to i \to \infty$

$\theta < \frac{\pi}{2} \to i \to 0$

$\theta = \frac{\pi}{2} \to i \to$ finite

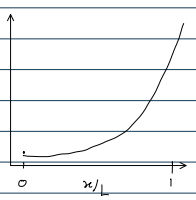
Current flowing from every where

Current line, can't flow into a narrow corner

We get:

$$\frac{i(x)}{i_{avg}} = \frac{\left(\frac{x}{L}\right)^a}{\left(1 - \frac{x}{L}\right)^b}\left(c - d\frac{K}{L}\right)$$

where a, b, c, d are constants



Secondary Current Dist.

$\nabla c_j = 0$; $Wa > 0.1$, always more uniform than primary ; kinetics are no longer negligible
(Go ϕ at electrodes should include this)

$\nabla^2 \phi = 0$;

Boundary Cond: $i = f(V, \phi)$ → BV kinetics (only describes current normal to the electrode)

↓ for uniform composition $i = -\kappa \nabla \phi$

$$i = -\kappa \left.\frac{\partial \phi}{\partial y}\right|_{y=0}$$

↳ For linear kinetics:

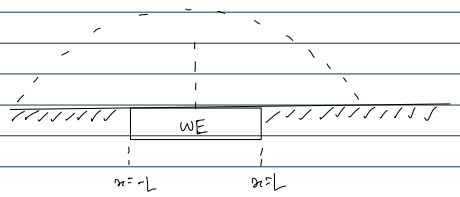
$$i = \frac{(\alpha_a + \alpha_c) i_o F}{RT} (V - \phi)$$

$$\boxed{\frac{\partial \phi}{\partial y} = \frac{-(\alpha_a + \alpha_c) i_o F}{\kappa RT} (V - \phi)}$$

BC @ WE for secondary current dist. (linear kinetics)

↳ Tafel kinetics:

ex: "Line Electrode"



Secondary current dist., linear kinetics : $\nabla^2 \phi = 0 \Rightarrow \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$

BC's: @ CE : $\phi = 0$
$x^2 + y^2 = a^2$

$$\frac{\partial \phi}{\partial y} = \frac{i_o}{\kappa} \frac{(\alpha_a + \alpha_c) F}{RT} (V - \phi)$$

Non-dimensionalizing : let $\bar{x} = \frac{x}{L}$, $\bar{y} = \frac{y}{L}$, $\bar{\phi} = \frac{\phi}{V}$

Then $\frac{\partial^2 \bar{\phi}}{\partial \bar{x}^2} + \frac{\partial^2 \bar{\phi}}{\partial \bar{y}^2} = 0$

@ $\bar{y} = 0$ $\frac{\partial \bar{\phi}}{\partial \bar{y}} = -\frac{i_o (\alpha_a + \alpha_c) F L}{\kappa RT} (1 - \bar{\phi})$; $\frac{1}{Wa_c}$

$$\boxed{\frac{\partial \bar{\phi}}{\partial \bar{y}} = \frac{-(1 - \bar{\phi})}{Wa_c}}$$ BC for linear kinetics

↳ For Tafel kinetics : $\frac{\partial \phi}{\partial y} = -\frac{i_o}{\kappa} \exp\left[\frac{\alpha_a F (V - \phi)}{RT}\right]$ @ $y = 0$

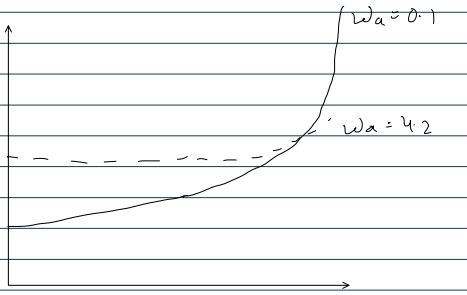
Non-dim using term in exp : $\bar{\phi} = \frac{\alpha_a F \phi}{RT}$

Non-dim using term in exp: $\bar{\phi} = \frac{\alpha_a F \phi}{RT}$

$$\frac{\partial \bar{\phi}}{\partial \bar{y}} = \frac{-i_o L F}{\kappa RT} \exp\left[\frac{\alpha_a F V}{RT} - \frac{\alpha_a F \phi}{RT}\right]$$

$$\frac{\partial \bar{\phi}}{\partial \bar{y}} = -A \exp\left[-\bar{\phi}\right], \text{ where } A = \frac{i_o \alpha_a F L}{\kappa RT} \exp\left[\frac{\alpha_a F V}{RT}\right]$$

The current dist. as $W_a \to 0$ (primary



As electrode gets larger ↓ greater non-uniformity

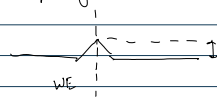
below $W_a = 0.1$ you are basically in the primary regime

We can use this understanding of W_a to get non-uniform. info

↳ eg: Dendrite formation in batteries seriously limit efficiency + development of new battery tech.

↳ uncontrolled electrode deposition

↳ Defect/asperity could create significant non-uniformity



$W_a = \frac{RTK}{\alpha_c F |i_{avg}| \ell}$; if $i_{avg} = -50\ mA/cm^2$, $K = 0.1\ \Omega^{-1} cm^{-1}$, $\alpha_c = 0.5$, $\ell = h_a$

@ $W_a = 0.1$; $h_a = 1\ cm$ ← This is high and unlikely to happen

@ $W_a = 1$ $h_a = 1\ mm$

Lecture 21

Thursday, 30 March 2023 1:27 PM

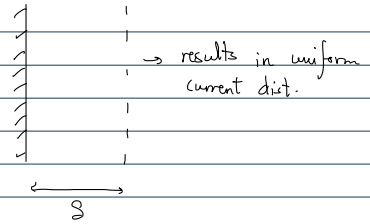
Mass Transfer effects on current distributions

↳ Mass transfer limited current distribution : + Tertiary current dist. $\nabla c \neq 0$, $t_i \to 0$ (neglect $\nabla\phi$)
(also called "convective transport" problems)

Previously, $i_{lim} \Rightarrow c_{lim} \Rightarrow 0$ @ electrode. ; constant BL thickness (δ)

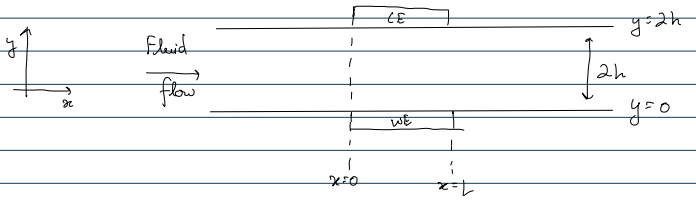
Flux is not same as conc.!

Conc = 0 @ electrode
[But there will be flux]



But geometry + flow conditions of the cell may not always result in constant δ need to
↳ we understand how M-T impacts this + current dist.

Ex: Microfluidic echem cell, w/ 2 line electrodes



in z dir, electrodes take up full width and w>>h

+ Assume laminar flow, we only have x-velocity

$$v_x(y) = \frac{3}{2} U \left(1 - \left(\frac{h-y}{h}\right)^2\right) \quad ; \quad U = \frac{Q}{2hw}$$

↳ Governing eq for $t_j = 0$: convection-diffusion (mig. neglected)

$$v_x \left(\frac{\partial c_j}{\partial x}\right) = D \left(\frac{\partial^2 c_j}{\partial x^2} + \frac{\partial^2 c_j}{\partial y^2}\right)$$

For fast kinetics: $c_{lim} = 0$ @ all $y = 0$ on electrode

Flow in the x-direction is much greater than diff. in x-direction ⇒ Microfluidic device (small Re) (convection dominates in flow direction, according to Pe)
↓
large Sc

$$Pe = \frac{\text{convective transport}}{\text{diffusive transport}} = Sc\, Re$$

Pe → large [Convection dominates]

$$Sc = \frac{\nu}{D_{lim}} = \sim 10^3 \text{ for small molecules in water}$$

$$\therefore \quad v_x \frac{\partial c_{lim}}{\partial x} = D_{lim} \frac{\partial^2 c_{lim}}{\partial y^2} \quad ; \quad \text{BC's: } \begin{cases} @\ x = 0, & c_{lim} = c_\infty \quad \text{(no rxn yet)} \\ @\ y \to \infty, & c_{lim} = c_\infty \end{cases}$$

Conc. should only vary w/ δ near electrode, so we can approx velocity there w/ Taylor expansion

$$v_x = \cancel{v_x}\Big|_{y=0} + \left.\frac{\partial v_x}{\partial y}\right|_{y=0} y$$

(No slip)

Plugging in laminar flow eq:

$$\frac{\partial v}{\partial y} = \frac{3U}{2}\left(-2\left(\frac{h-y}{h}\right)\left(-\frac{1}{h}\right)\right)$$

$$@\ y=0 \quad \frac{\partial v_x}{\partial y} = \frac{3U}{h} \Rightarrow v_x = \frac{3Uy}{h}$$

$$\boxed{\frac{3U}{h} y \frac{\partial c}{\partial x} = D \frac{\partial^2 c}{\partial y^2}}$$

This is a PDE that can be solved w/ "comb. of variables" (Similarity solution!)

Let $\xi = y\left(\frac{U}{3hDx}\right)^{1/3}$ → this will make the equation w/ just ind. var.

↳ make them an ODE

$$\frac{\partial c}{\partial x} = \frac{\partial c}{\partial \xi}\cdot\frac{\partial \xi}{\partial x} \quad ; \quad \frac{\partial \xi}{\partial x} = y\left(\frac{U}{3hD}\right)^{1/3}\cdot\left(-\frac{1}{3}\right)x^{-4/3} = -\frac{\xi}{3x} \Rightarrow \frac{\partial c}{\partial x} = \frac{\partial c}{\partial \xi}\cdot\left(-\frac{\xi}{3x}\right)$$

$$\frac{\partial c}{\partial y} = \frac{\partial c}{\partial \xi}\cdot\frac{\partial \xi}{\partial y} \qquad \frac{\partial \xi}{\partial y} = \left(\frac{U}{3hDx}\right)^{1/3} = \frac{\xi}{y}$$

$\frac{\partial^2 c}{\partial y^2}$ ← Both product & chain rule

$$= \frac{\partial}{\partial y}\left(\frac{\partial c}{\partial y}\right) = \frac{\partial}{\partial y}\left(\frac{\partial c}{\partial \xi}\frac{\partial \xi}{\partial y}\right)$$

$$= \frac{\partial c}{\partial \xi}\frac{\partial}{\partial y}\left(\frac{\partial \xi}{\partial y}\right) + \frac{\partial \xi}{\partial y}\frac{\partial}{\partial y}\left(\frac{\partial c}{\partial \xi}\right)$$

$$= \frac{\partial \xi}{\partial y}\left(\frac{\partial}{\partial \xi}\left(\frac{\partial c}{\partial \xi}\right)\frac{\partial \xi}{\partial y}\right)$$

$$\frac{\partial^2 c}{\partial y^2} = \frac{\partial^2 c}{\partial \xi^2}\cdot\left(\frac{\partial \xi}{\partial y}\right)^2 = \frac{\partial^2 c}{\partial \xi^2}\cdot\left(\frac{U}{3hDx}\right)^{2/3}$$

Plugging into original PDE:

$$\frac{3U}{h} y \frac{\partial c}{\partial \xi}\left(\frac{-\xi}{3x}\right) = D\frac{\partial^2 c}{\partial \xi^2}\left(\frac{U}{3hDx}\right)^{2/3} \div \left(\frac{U}{3hDx}\right)^{2/3}$$

$$3y\left(\frac{U}{3hDx}\right)^{1/3}\frac{\partial c}{\partial \xi}(-\xi) = \frac{\partial^2 c}{\partial \xi^2}$$

($y\left(\frac{U}{3hDx}\right)^{1/3} = \xi$)

$$-3\xi^2\frac{\partial c}{\partial \xi} = \frac{\partial^2 c}{\partial \xi^2} \Rightarrow \boxed{\frac{d^2c}{d\xi^2} + 3\xi^2\frac{dc}{d\xi} = 0}$$

BC's: @ $x = 0$, $c_{lim} = c_\infty \Rightarrow \xi \to \infty$ $\quad c_{lim} = c_\infty$

@ $y \to \infty$, $c_{lim} = c_\infty \Rightarrow \xi \to \infty$ $\quad c_{lim} = c_\infty$

@ $y = 0$, $c_{lim} = 0 \Rightarrow @\xi = 0$, $c_{lim} = 0$

↓

Solving using Forbenius method

$$\boxed{\frac{c_{lim}}{c_\infty} = \frac{1}{\Gamma\left(\frac{1}{3}\right)}\int_0^{\xi} e^{-x^3}\,dx}$$

→ Now we know conc. profile in BL

↓

Solving for flux: $N_y = -D\frac{\partial c}{\partial y} = D\frac{\partial c}{\partial y}\Big|_{y=0} = D\frac{\partial c}{\partial \xi}\Big|_{\xi=0}\frac{\partial \xi}{\partial y}\Big|_{y=0}$

(Laminar flow
(No convection in y))

$$\frac{\partial}{\partial x_u}\int_0^{x_u} f(x)\,dx = f(x_u)$$

$$N_y = \frac{D c_\infty}{\Gamma\left(\frac{1}{3}\right)}\underbrace{\frac{\partial}{\partial \xi}\left(\int_0^{\xi} e^{-x^3}\,dx\right)\Big|_{\xi=0}}_{\text{Fundamental theorem of Calculus}}\left(\frac{U}{3kDx}\right)^{1/3}$$

$$N_y = \frac{D c_\infty}{\Gamma\left(\frac{1}{3}\right)} e^{-\xi^3}\Big|_{\xi=0}$$

$$N_y = -\frac{c_\infty D}{\Gamma\left(\frac{1}{3}\right)}\left(\frac{U}{3kDx}\right)^{1/3}$$

↓ Find current

$$\frac{Si}{nF} = -N_y$$

Plot $\frac{i(x)}{i_{avg}}$; $i_{avg} = \frac{1}{L}\int_0^L i_{avg}\,dx$

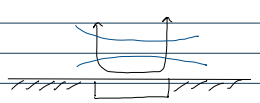
Lecture 22

Tuesday, 4 April 2023 1:29 PM

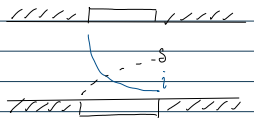
Impact of electrode Conductivity on Current dist.

So far, we have shown current non-uniformities arise from: • electric field / geometry (primary, secondary)

We can also encounter current variations due to imperfect conductors (i.e. resistive substrates)

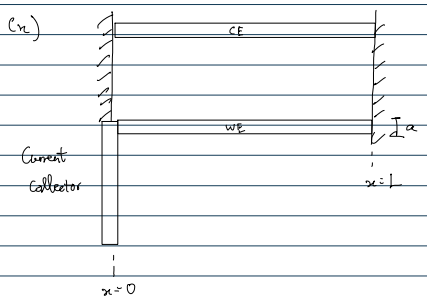


• MT effects during fast kinetics



Current dist. in Resistive Substrate:

↳ Encountered frequently during metallization of Si chips



* previously we set $V=\phi$ @ $y=0$ (1D problem w/ uniform current dist.)

potential of metal

Now because substrate is resistive: all current must flow through $x=0$, $\phi_m = f(x)$

$\eta = \phi_m - \phi_s$

↑ potential of metal; ↳ potential in solution

To solve, we assume $\nabla c_j = 0$, then

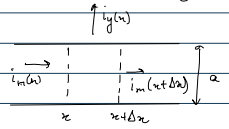
$$\nabla^2 \phi_s = 0$$

$$\frac{\partial^2 \phi_s}{\partial x^2} + \frac{\partial^2 \phi_s}{\partial y^2} = 0$$

, for BC's: Assume Tafel kinetics @ electrode

$$i(x) = -\kappa \left.\frac{\partial \phi_s}{\partial y}\right|_{y=0} = -i_o \exp\left[\frac{-\alpha_a F}{RT}(\phi_m - \phi_s)\right]$$

$= f(x)$ → We get this from charge balance



In − out + gen = Acc.

$$a\left[i_m(x) - i_m(x+\Delta x)\right] - i_y(x)\Delta x = 0$$

$$i_y(x) = \frac{(i_m|_x - i_m|_{x+\Delta x})\,a}{\Delta x}$$

$$i_y(x) = -a\frac{di_m}{dx}$$

, From Ohm's law: $i_y(x) = \frac{d}{dx}\left(\frac{1}{R_s}\frac{d\phi_m}{dx}\right)$

$i_y(x) = -a \frac{di_m}{dx}$; From Ohm's law: $i_y(x) = \frac{d}{dx}\left(\frac{1}{R_s}\frac{d\phi_m}{dx}\right)$

R_s: sheet resistance

Combining:

$$-a\frac{di_m}{dx} = \frac{d}{dx}\left(\frac{1}{R_s}\frac{d\phi_m}{dx}\right)$$

$$\boxed{i_m = \frac{-1}{aR_s}\frac{d\phi_m}{dx} = -\sigma\frac{d\phi_m}{dx}}$$

σ: analogous to κ [$\Omega^{-1}\,cm^{-1}$]

gradient is zero perpendicular to the insulator position

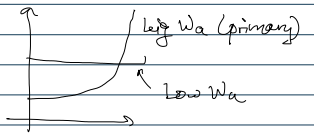
; BC's: @ $x = L$, $\frac{d\phi_m}{dx} = 0$

@ $x = 0$, $\phi_m = V$

Can be solved numerically or analytically (limiting cases):

+ Depends on $W_a = \frac{RT\kappa}{\alpha_c F i_{avg} L}$; $K_0 = \frac{\kappa L}{a\sigma}$ (ratio of electrolyte to substrate conc.)

If $\frac{W_a}{K_0}$ → large ; } more uniform



$K_0 \downarrow$, $\sigma \uparrow$ (electrode cond. goes up) → tend to be more uniform

Summary:

1) Spatial variations of current on electrode are caused by:

→ $W_a = \frac{R_{ct}}{R_\Omega} \to 0$ when, large

$i_{avg} << i_{lim}$

→ MT conditions when $i_{avg} \approx i_{lim}$

→ Resistive electrodes (even when $W_a > 0.1$)

Uniform current:

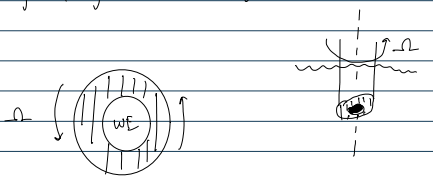
1) Conductive electrolyte
2) Small electrodes
3) Uniform MT if $i \sim i_{lim}$
4) Conductive electrodes

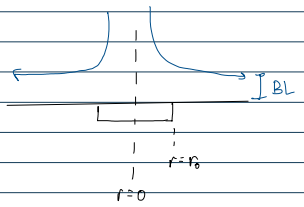
Fluid Flow to a rotating disk (RDE)

→ Even under i_{lim} we have non-uniformity of current due to MT (Boundary layer problems)

→ A rotating disk electrode provides a MT scenario where an exact solution to fluid flow is obtained

→ A rotating disk electrode provides a MT scenario where an exact solution to fluid flow is obtained





↳ Solution to flow flow reveals a momentum BL of thickness: $\sqrt{\frac{\nu}{\Omega}}$ ← Not dependent on radial position
(Thicker than diffusion BL)

↳ Based on this thickness, we can define dimensionless axial variable

$$\xi = \frac{z}{\sqrt{\frac{\nu}{\Omega}}}$$

Then we can define each velocity component in non-dim form as func. of ξ

Radial · $v_r = r\Omega f(\xi)$

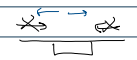
Angular · $v_\theta = r\Omega G(\xi)$

Axial · $v_z = \sqrt{\nu\Omega}\, H(\xi)$; No slip condition @ $\xi = 0 \Rightarrow F = H = 0$; $G = 1$ → whatever is at surface keeps rotating w/ it

↳ Axial velocity is not a function of radial position
(Uniform diffusion layer thickness) → uniform current @ i_{lim}

↳ Radial velocity is outward (positive r) → negligible flux from large radial positions to smaller radial positions → minimal edge effects

} These enable its use (v. diff. from earlier case)



Now solving for current distribution:

↳ Assume limiting current @ disk, and $t_i = 0$ } Convection-diffusion equation

$$v_z\left(\frac{\partial c}{\partial z}\right) = D\left(\frac{\partial^2 c}{\partial z^2} + \frac{\partial^2 c}{\partial r^2}\right)$$

0 Diffusion layer constant in r direction (even though we have flow in r-direction) → No conc. gradients in the r-direction

From hydrodynamic analysis:

$$v_z = -a z^2 \left(\frac{\Omega}{\nu^{1/2}}\right)^{3/2}$$

* This is the 1 term Taylor expansion of $v_z = \sqrt{\nu\Omega}\cdot H(\xi)$

$a = 0.51023$ (fit parameter)

↓

Non-dimensionalize: $\bar{z} = \left(\frac{\Omega}{\nu^{1/4} D^{1/3}}\right)^{1/2} z$,

$$\therefore \frac{\partial c}{\partial z} = \frac{\partial c}{\partial \bar{z}}\left(\frac{\Omega}{\nu^{1/6} D^{1/2}}\right)^{1/2} \quad \frac{\partial^2 c}{\partial z^2} = \frac{\partial^2 c}{\partial \bar{z}^2}\left(\frac{\Omega}{\nu^{1/3} D^{2/3}}\right)$$

Plugging in:

$$-a z^2 \frac{dc}{d\bar{z}} = \frac{d^2 c}{d\bar{z}^2}$$

BC's: @ $\bar{z} = 0$, $c = 0$

Plugging in:

$$-a\bar{z}^2\frac{dc}{d\bar{z}} = \frac{d^2c}{d\bar{z}^2}$$

BC's: @ $\bar{z}=0$, $c=0$

@ $\bar{z}=\infty$, $c=c_\infty$

Lecture 23

Thursday, 6 April 2023 1:32 PM

RDE: $-a\bar{z}^2\frac{dc}{d\bar{z}} = \frac{d^2c}{d\bar{z}^2}$, $\bar{z} = \left(\frac{\Omega^{1/2}}{\nu^{1/6}D^{1/3}}\right)z$

BC's: @ $\bar{z}=0$, $c=0$; @ $\bar{z}=\infty$, $c=c_\infty$; $g = \frac{dc}{d\bar{z}}$

$-a\bar{z}^3 g = \frac{dg}{d\bar{z}} \Rightarrow -a\bar{z}^2 d\bar{z} = \frac{1}{g}dg$

$-\frac{a\bar{z}^3}{3} = \ln g$

$\frac{dc}{d\bar{z}} = g = c_1 \exp\left(\frac{-a\bar{z}^3}{3}\right)$ → Use gamma functions to solve for C

↓ Add in the BC's

$c_\infty = c_1\left(\frac{\Gamma(\frac{1}{3})}{3^{1/3}a^{1/3}}\right) \Rightarrow c_1 = \frac{c_\infty 3^{2/3} a^{1/3}}{\Gamma(\frac{1}{3})}$

$\frac{dc}{dz} = \frac{c_\infty 3^{2/3} a^{1/3}}{\Gamma(\frac{1}{3})}\exp\left[\frac{-a}{3}z^3\right]$

@ $\bar{z}=0$ $\left.\frac{dc}{dz}\right|_{\bar{z}=0} = 0.62\, c_\infty$

$$\left.\frac{dc}{dz}\right|_{z=0} = \left(\frac{\Omega^{1/2}}{\nu^{1/6}D^{1/2}}\right)(0.62)\,c_\infty$$

From Ch 5: $N = -D\left.\frac{dc}{dz}\right|_{z=0}$

$N = -D c_\infty\left(\frac{0.62\,\Omega^{1/2}}{D^{1/3}\nu^{1/6}}\right)$

Normally, Flux @ $i_{lim} \Rightarrow N = -D\frac{c_\infty}{\delta}$

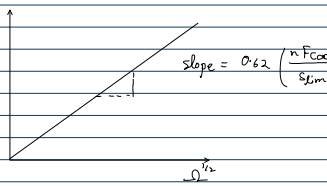
$\therefore$ Similarity $\Rightarrow \delta = \left(\frac{0.62\,\Omega^{1/2}}{D^{1/3}\nu^{1/6}}\right)^{-1}$

From Faraday's Law:

$$\frac{s_{lim}\, i_{lim}}{nF} = \frac{Dc_\infty}{\delta} = 0.62\, D^{2/3}\nu^{-1/6}\Omega^{1/2}c_\infty$$

Levich Equation!

If we can accurately vary Ω (rotation rate), we can understand species diffusion



; more spin, more convection, smaller diffusion layer (higher c going through → higher i_{lim})

Note: we can also define "Levich-type" eq. for non-lim current, as long as current dist. is uniform ($W_a \rightarrow$ large)

(You will just have to change the bounds of integration)

Then, $\frac{s\, i_{ang}}{nF} = \frac{D(c_2 - c_1)}{\delta} = 0.62\, D^{2/3}\nu^{-1/6}\Omega^{1/2}(c_\infty - c_0)$

Often, we have charge transfer resistance

↳ This means current has contributions from MT + kinetics

↳ To separate out kinetics + MT effects on current, use "Koutecky-Levich" analysis

↳ To understand kinetic effects, we generally define current as function of potential

$i = nF c_0 K(V)$ ← surface conc.; ↳ potential dependent rate constant

$K(V) = K_0 \exp(f(\eta))$

We need to get an expression for c_0 in terms of i_{lim}:

$\frac{s\, i_{ang}}{nF} = \frac{D(c_\infty - c_0)}{\delta} = \frac{Dc_\infty}{\delta} - \frac{Dc_0}{\delta}$

$= i_{lim} - Dc_0$;

$$\frac{S\, i_{ang}}{nF} = \frac{D(C_\infty - C_0)}{\delta} = \frac{DC_\infty}{\delta} - \frac{DC_0}{\delta}$$

$$= i_{lim} - \frac{DC_0}{\delta}$$

$$i = i_{lim}\left(\frac{C_\infty - C_0}{C_\infty}\right)$$

$$C_0 = C_\infty - \frac{i}{i_{lim}} C_\infty = C_\infty\left(1 - \frac{i}{i_{lim}}\right)$$

$$\boxed{i = nF\, K(V)\, C_\infty \left(1 - \frac{i}{i_{lim}}\right)}$$

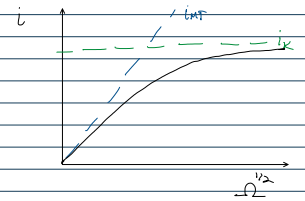
We can call this i_K → kinetic contr. to current

$i_{lim} = i_{MT}$ → mass transport contr. to current

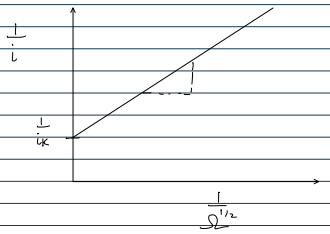
$$i = i_K\left(1 - \frac{i}{i_{MT}}\right) \Rightarrow i = \frac{i_K\, i_{MT}}{i_{MT} + i_K} \Rightarrow \boxed{\frac{1}{i} = \frac{1}{i_K} + \frac{1}{i_{MT}}}$$

"Summing up resistors"

↳ Koutecky-Levich Equation



If we tried to do Levich analysis, D might not be accurate → We can plot $\frac{1}{i}$ vs. $\Omega^{-1/2}$ (Linearize the system)



Primary Current Distribution for Rotating Disks

RDEs typically used near i_{lim}, but not always (w/ cases not near i_{lim})

↳ Does rotation rate of RDE affect primary current analysis?

No, because $\nabla C_j = 0$ for primary (conver. + diffusion don't matter, we just solve $\nabla^2\phi$)

+ Only difference is geometry, (solved by Neumann analytically):

$$\frac{i(r)}{i_{ang}} = \frac{0.5}{\sqrt{1 - (r/r_0)^2}} \qquad @\ z=0 \quad r > r_0 \Rightarrow \frac{\phi}{V} = 1 - \frac{2}{\pi}\tan^{-1}\sqrt{\left(\frac{r}{r_0}\right)^2 - 1}$$

+ Potential varies rapidly outside of disk close to disk

The potential drop in solution will affect CE & RE?
↳ How?

That is why to have

Position of CE is determined by this sort of analysis

If you are 3 disk radii away you have already gone through 90% of

Rotating Ring Disk Electrode (RRDE):

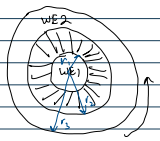
↳ We can add a second WE co-planar w/ disk to give added functionality to well-defined hydrodynamics of RDE

a measured potential drop

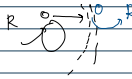
you have
a compensation!

↓

Luggin Cappillary



Most common application: "generator-collector" mode



Can get exactly how much / partial current going towards the desired reaction

Lecture 24

Tuesday, 11 April 2023 1:35 PM

Demo: RDE ; Au disk (WE)

Hg/Hg_2SO_4 RE (0.62 V vs. SHE)

Pt wire CE

0.1 M K_2SO_4 (+ H_2SO_4) → pH ≈ 4

Relevant Rxns:

(1) Hydrogen evolution reaction (HER)

$2H^+ + 2e^- \rightarrow H_2$; $U^\circ = 0$ vs. SHE @ pH = 0

$U = 0$ V vs. RHE ← built-in pH correction (shifts w/ pH of solution)

@ pH ≈ 4, electrolyte is proton-limited

HER still occurs when we run out of protons

via $2H_2O + 2e^- \rightarrow H_2 + 2OH^-$ $U^\circ = -0.828$ V vs. SHE @ pH = 14

$U = 0$ V vs. RHE

(2) Oxygen Reduction Reaction (since we are not purging w/ Ar so O_2 will also react @ electrode)

$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$ $U^\circ = 1.23$ V vs. SHE @ pH = 0

$U = 1.23$ V vs. RHE

$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$ $U^\circ = 0.401$ V vs. SHE @ pH = 14

$U = 1.23$ V vs. RHE

Nernst

$V_{applied} = V_{RHE} - V_{RE} - (0.0591\ V \times pH)$

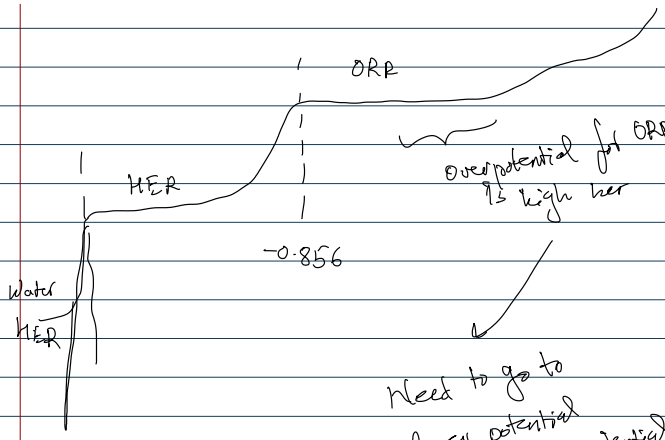
$V_{applied} = 0 - 0.62 - 0.0591\,(4) = -0.856$ V vs.

@ 0 V vs. RHE, $\eta_{HER} = V_{applied} - U = 0$ We are giving it no "extra" potential (i.e. no reaction should be driven)

$\eta_{ORR} = -1.23$ V ← very large driving force for this to happen so @ $V_{applied} = 0$ it will be dominating

We want current density on CE to be very small (thus high SA)

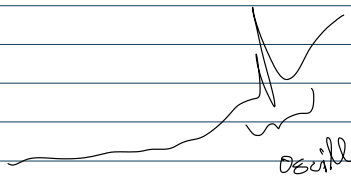
CV → linear change b/w two vertexes



Overpotential for ORR is high here

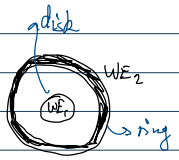
Need to go to lower potential to have overpotential for HER (But ORR is still happening as well)

Diff coeff → tells you spacing b/w them (resulting in diff. slopes)



Oscillation is due to potentiostat switching it's I-range (since it is Auto)

Rotating Ring Disk Electrode



→ "Generator - collector" mode

↳ Redox species generated @ disk, and that species does not exist in bulk electrolyte

↳ Ring potential is held s.t. generated specie conc. is zero @ ring surface

If not known, need to do sweep & measure i to see where / what could be possible

For, $C_{ring} = 0$, the MT equations solved to yield i_{ring} as function of i_{disk}

$$i_{ring} = \frac{nF}{s} D \int_{r_2}^{r_3} \left.\frac{\partial c}{\partial y}\right|_{y=0} r \cdot dr$$

Solution provides:

$$i_{ring} = -N_c \, i_{disk}$$

↳ collection (only a function of

$i_{ring} = -N_c\, i_{disk}$

↳ collection efficiency (fraction of species generated @ disk that reacts @ ring) (only a function of r_1, r_2, r_3)

↳ No function of Ω, D, C_∞

How to maximize N_c?

+ Small r_2 and + Large $(r_3 - r_2)$ } Small gap & large ring

[Since you want to collect all the species formed on the disk which are moving outwards]

→ "Detection Mode"

↳ Instead of generated species reacting @ ring; avg conc of species @ ring induces change in ring equil. potential

↳ This can be useful to detect surface pH changes if ring is potentiometric pH sensor

↳ Hydrodynamics are the same, but now we are not reacting on ring

$$(C_{ring} - C_\infty) = N_D (C_{disk} - C_\infty)$$

↳ detection efficiency (fraction of species generated @ disk, detected as surf conc. @ ring

↳ only depends on r_1, r_2, r_3

Maximizing N_D?

Small ring — Don't want to "average" out the measurement

Lecture 25

Thursday, 13 April 2023 1:31 PM

Research Presentation: Exxon Labrage CO_2 capture tech (Largest in the world)

→ Transform CO_2 to make it economically viable

→ pH @ surface is imp.

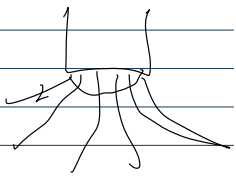
→ Buffers have MT issues & actually might partake in the mechanism

→ Measure normalized OCP @ ring → measure the current density at ring (calculate then what is happening at the disk)

↳ There is current flow by the ring } Contribution due to the primary current dist → Have to correct for this Ohmic drop.

Autoprotolysis equilibrium

→ Ohmic drop gets larger w/ decreasing r_2 (distance between disk & ring) } Since its being affected by the V dist. at the disk



Lecture 26 (CA+CP+CV)

Tuesday, 18 April 2023 1:31 PM

HW #4 → Thursday

Final report → Friday

Exam 2 → 4/28 } HW material #4

HW 0 pt 2

Fundamental Techniques:

+ Chronoamperometry (CA)
+ Chronopotentiometry (CP)
+ Polarization curves (LSV, CV)
+ Electrochemical Impedence Spectroscopy (EIS)

→ CA's: Chrono → time, Amp → current } changing current (while holding potential constant)

Time dependent conc. profile can be solved w/ MT concepts

ex) Assume:

- Double layer current negligible (Since it takes time/charge for this double layer to form)
- No forced convection } Diffusion controlled
- $t_j \to 0$

Mass balance:

$$\frac{\partial c_j}{\partial t} = -\nabla \cdot N_j + R_j$$

($R_j \to 0$; N_j = migration + diff. + convection) } This may not be always valid (steady state voltammetry) → Application of current usually will usually result in gradients & "flow" → harder to reach steady state in these larger electrodes

PDE $\boxed{\frac{\partial c_j}{\partial t} = D \frac{\partial^2 c}{\partial x^2}}$

BC's: @ $t=0$; $c = c_\infty$

@ $x \to \infty$; $c = c_\infty$

@ $[t>0, x=0]$; $c = 0$ ← limiting current

Combination of variables or Laplace Transform to solve PDE:

$$\frac{c(x,t)}{c_\infty} = \frac{2}{\sqrt{\pi}} \int_0^{\frac{x}{\sqrt{4Dt}}} \exp[-y^2]\, dy$$

$$= \operatorname{erf}\left[\frac{x}{2(Dt)^{1/2}}\right]$$

from conc. profile we get flux to then get current (i)

Flux @ $x=0$:

$$\left.\frac{\partial c}{\partial x}\right|_{x=0} = \frac{2c_\infty}{\sqrt{\pi}} \exp\left[\frac{x^2}{4Dt}\right] \frac{1}{\sqrt{4Dt}} \Bigg|_{x=0}$$

$$= \frac{c_\infty}{\sqrt{\pi Dt}}$$

$$-N_{ox} = D\left.\frac{\partial c}{\partial x}\right|_{x=0} = \frac{D c_\infty}{\sqrt{\pi D t}}$$

$$\boxed{i(t) = \frac{nF}{s}\frac{D c_\infty}{\sqrt{\pi D t}}}$$ ← Cottrell Equation

Note: As: $t \uparrow$, $i \downarrow$ (depletion of reacting species near the surface) → : $t \to 0, i \to \infty$

↳ infinite driving force (potentiostat usually has current limits that will artificially cap current)

Since we are only considering diffusion (so the species close to the electrode will react & depleting this layer more & more)

* According to $c(x,t)$
  ↳ diffusion BL (defined as x @ $c = c_\infty$) grows w/ t.

CP's: Vary potential (constant current) w/ time } Same system as before

+ Material balance: $\frac{\partial c}{\partial t} = D\frac{\partial^2 c}{\partial x^2}$

@ $t=0$, $c = c_\infty$

@ $x \to \infty$, $c = c_\infty$

@ $x=0$, $t>0$, $i = \frac{nF}{s} D\frac{\partial c}{\partial x}$ [Flux BC]

↑ Constant!

* Will this result in $c_{lim}=0$ @ all $t>0$?
  ↳ no, initially $c_{lim}>0$, @ some time c_{lim} will get depleted @ electrode

* What happens after $c_{lim} \to 0$ when applying constant current? (Conc. of our limiting species goes to zero)
  ↳ Since we are applying a constant current, and all the species have been depleted
  ↓
  Secondary reaction will turn on which may cause abrupt change in potential } Something else has to make up the current (pass charges)

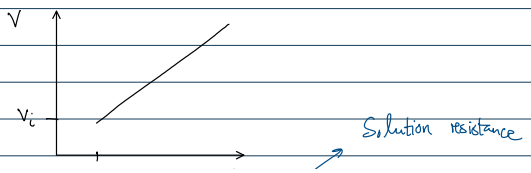
Sand Eq (for semi-infinite geometry)

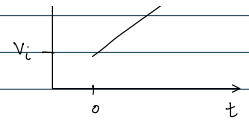
The time @ which $c_{lim} \to 0$ is given by solution to PDE w/ const. flux BC:

$$\boxed{\tau = \frac{\pi D}{4}\left(\frac{c_\infty n F}{s_{lim} q}\right)^2}$$

Polarization curves (LSV / CV)

→ A polarization (I vs. V) curve can be obtained by linear sweep voltametry (LSV)
(Apply linearly changing potential function + measure current)





Solution resistance

This measurement is affected by ohmic drop in solution
↳ potential gradient in solution resulting from current flow b/w WE & CE

$$\eta = (V - V_{ref}) - (U - U_{ref}) - \Delta\phi_{\Omega}$$

$(V - V_{ref})$: RE potential; $(U - U_{ref})$: ↙; $\Delta\phi_{\Omega}$: Ohmic drop

↳ We can estimate the value of $\Delta\phi_{\Omega}$ @ RE far away from disk electrode for primary current distribution

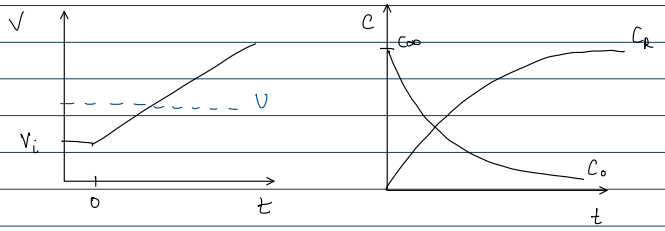
$$\Delta\phi_{\Omega} = \frac{\pi r_0}{4\kappa} i_{avg}$$

(r_0: WE radius)

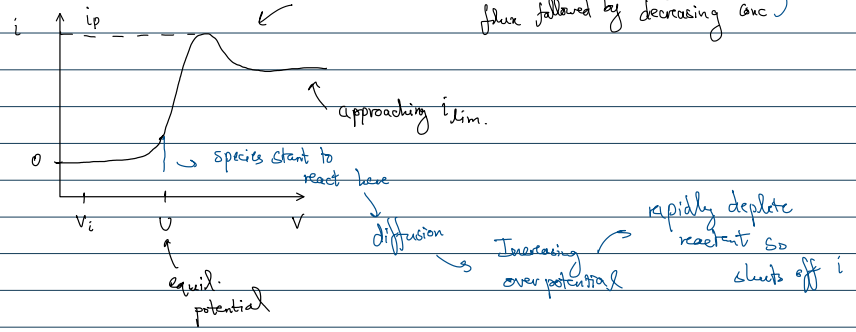
; Ohmic drop depends on electrode size & electrolyte conductivity [proportional to current]

↳ LSVs for reversible rxns, w/ only one half of redox couple in solution

ex: $O + ne^- \rightleftharpoons R$ w/ equil. potential U if solution contains O, but no R, and we apply an LSV



How does i vs. V look like?



we get a peak (non-SS) that results from increasing flux followed by decreasing conc } Non-SS process

approaching $i_{lim.}$

species start to react here

diffusion

Increasing overpotential

rapidly deplete reactant so shuts off i

For $V(t) = V_i + vt$ (No species O initially in the system)

v: Scan rate $(\frac{V}{s})$

$$i_p = -0.4463\,(nF)^{3/2}\left(\frac{D_O}{RT}\right)^{1/2} C_{OO}^{*}\, v^{1/2}$$

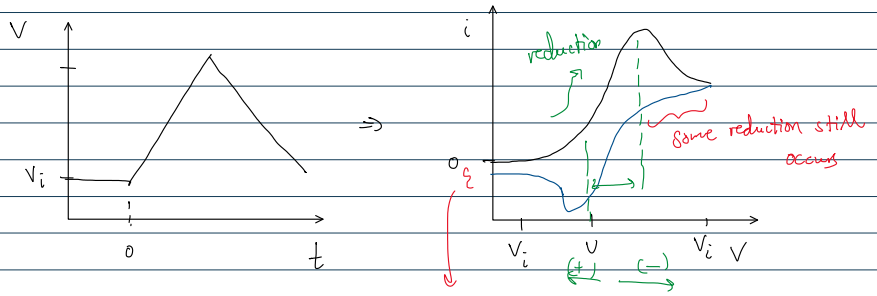
→ † For semi-infinite (like disk electrode)

† Diffusion only

Height changes but not the peak (as you sweep faster)

Often we want to reverse the potential sweep to create a cyclic voltammogram (CV)

Often we want to reverse the potential sweep to create a cyclic voltammogram (CV)

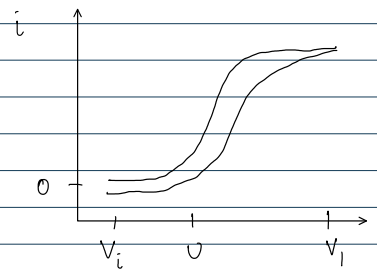


we don't reach V_i because we do have species that can be reduced

Comes from the math } ~60 mV (for reversible process) $\left[\frac{2.303RT}{F}\right]$

These techniques have complex math analysis due to time response but we can simplify some of these techiques by using "ultramicro electrodes" (UME)

↳ electrodes ~10μm → (establish SS in ecem synthesis @ much shorter time scales than standard electrodes)



No peaks!

(SS voltammetry)

Electrochemical Impedance Spectroscopy (EIS)

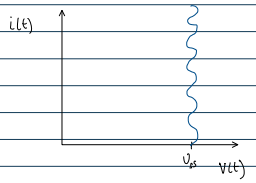
↳ Apply a small amplitude sinusoidal potential @ various frequencies (+ analyze current response)

↳ EIS can be used to interpret everything that affects flow of current in echem system:

+ Electrical resistance in current path
+ Interfacial capacitance
+ MT of reactants
+ Kinetics of relevant hetero/homog. rxns.

↳ But system/measurement is complex + it can be easy to over-interpret results.

↳ EIS measurement:
+ Apply small amplitude (~10mV) oscillating potential around some S.S. potential (V_{ss})



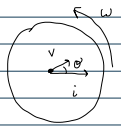
Potential represented w/ cos function

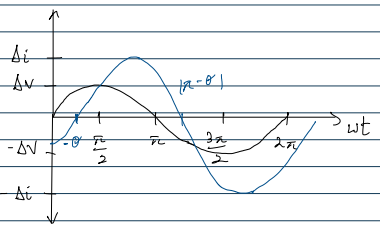
$V(t) = V_{ss} + \Delta V \cos(\omega t)$ ← angular frequency
↓ amplitude

* How does i(t) vary w/ V(t)

↳ w/ small amplitude, we can expect linear response but might have some phase shift

$i(t) = i_{ss} + \Delta i \cos(\omega t + \theta)$
↳ phase angle





We can imagine current response to potential is affected by real electrochem processes that can be modelled by resistances (eg. C-t/kinetics) and capacitances (eg. double-layer charging)

+ Resistor: Ohm's law;

$$i(t) = \frac{V(t)}{R} = \frac{\Delta V}{R}\cos(\omega t) \rightarrow \text{phase angle } (\theta = 0)$$

+ Capacitor: $q = CV$

$$i(t) = C\frac{dV(t)}{dt} = -\omega C \Delta V \sin(\omega t)$$

$$= \omega C \Delta V \cos\left(\omega t - \frac{\pi}{2}\right)$$
↳ phase angle $= \frac{\pi}{2}$

Its convinient to express relationship b/w i-V as complex numbers

↳ express y-component as imaginary by multiplying by $\sqrt{-1}$ (j) and x-component as real, then:

must have units of resistance.

↳ express y-component as imaginary by multiplying by $\sqrt{-1}$ (j) and x-component as real, then:

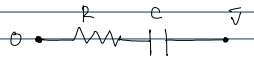
+ capacitor : $\bar{i}$ → , $\bar{v}$ ↓

$$\bar{V} = -j\bar{i}\,\frac{1}{\omega C}$$

($\frac{1}{\omega C}$ must have units of resistance → capacitive resistance; $\bar{i}$ = derivative)

+ Resistor : $\bar{v} \longrightarrow \bar{i}$ $\qquad \bar{V} = R\bar{i}$

Ex) If we have a resistor + capacitor in series w/ voltage $\bar{V}$ applied across them



↳ Total voltage must equal sum of voltage across each component

$$\bar{V} = \bar{V}_R + \bar{V}_C$$

$$= R\bar{i} - j\bar{i}\,\frac{1}{\omega C} = \bar{i}\left(R - \frac{j}{\omega C}\right) = \bar{i}\,\bar{Z}$$

this quantity we can define as complex impedence

Resistive contributions from both capacitors + resistors

$$\boxed{\bar{Z} = Z_{Re} + j\,Z_{Im} = \frac{\bar{V}}{\bar{i}}}$$

How does this help us understand e-chem system?

ex) Consider irreversible rxn w/ Tafel kinetics

+ current from double layer

$$i(t) = i_0 \exp\left[\frac{\alpha_a F}{RT}(V - \phi_0)\right] + C_{dl}\frac{\partial(V-\phi)}{\partial t}$$

Apply Euler's formula ($\exp(j\omega t) = \cos(\omega t) + j\sin(\omega t)$)

Plug

$$\begin{cases} V(t) = V_{ss} + \Delta V\cos(\omega t) \Rightarrow V_{ss} + Re\left(\bar{V}\exp(j\omega t)\right) \\ i(t) = i_{ss} + \Delta i\cos(\omega t) \Rightarrow i_{ss} + Re\left(\bar{i}\exp(j\omega t)\right) \end{cases}$$

+ Assume $\phi_0 = \underbrace{iR_\Omega}_{\text{ohmic drop}}$

+ Assume we can linearize exponentials

$$\bar{i} = \left(\frac{\alpha_a F i_{ss}}{RT} + j\omega C_{dl}\right)\left(\bar{V} - \bar{i}R\right)$$

Rearrange to get $\bar{Z} = \frac{\bar{V}}{\bar{i}}$; $\qquad \frac{\bar{i}}{\bar{V} - \bar{i}R_\Omega} = \left(\frac{\alpha_a F i_{ss}}{RT} + j\omega C_{dl}\right)$

$$\frac{\bar{V}}{\bar{i}} = R_\Omega + \left(\frac{\alpha_a F i_{ss}}{RT} + j\omega C_{dl}\right)^{-1} = \bar{Z}$$

$$Z^{\dagger} = \frac{R_\Omega + \frac{\alpha_a F i_{ss}}{RT}}{\left(\frac{\alpha_a F i_{ss}}{RT}\right)^2 + (\omega C_{dl})^2} + \frac{-j\omega C_{dl}}{\left(\frac{\alpha_a F i_{ss}}{RT}\right)^2 + (\omega C_{dl})^2}$$

$$Z^{\dagger} = \frac{R_{\Omega} + \frac{\alpha_a F i_{ss}}{RT}}{\left(\frac{\alpha_a F i_{ss}}{RT}\right)^2 + (\omega C_{dl})^2} + \frac{-j\omega C_{dl}}{\left(\frac{\alpha_a F i_{ss}}{RT}\right)^2 + (\omega C_{dl})^2}$$

Z_{Re} + $j \cdot Z_{im}$

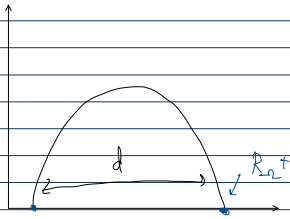
A common way to analyze EIS is to make a Nyquist Plot (Plot Z_{im} vs. Z_{Real})

Impedence: resistance to flow of current (units of ohms)

(ohmic drop)
Solution resistance

@ $\omega \to \infty$; $Z_{Re} = R_s$

Diameter $\propto \frac{RT}{\alpha_a F i_{ss}}$



(Solution resistance)

$d \propto \frac{RT}{\alpha_a F i_{ss}}$

As $d \uparrow$, $R_{ct} \uparrow$

$\omega \to \infty$ (R_Ω)
$\omega \to 0$ (

} Do analyses via the

Lecture 28

Tuesday, 25 April 2023 1:31 PM

Exam 2: Same format, open book, 1 pg front side,
Ch 5-7, conceptual 1-5, 8

→ 75 min, 100 pts

Electrolyte potential vs. Electrode potential

↓ This is what the R_Ω, R_{ct} analysis fit

↳ Resistive substrate (i.e. conductivity gradients)

(Electrolyte really close to electrode potential)

$t_+ \to 0$ assumption [fraction of current carried by species]

Binary electrolyte ← $t_+ : 1$

$t_+ : 0$ } Excess supporting electrolyte

↓ Lots of other ions in species (conduction / diff.)

Shows up in flux in migration term

↳ dependent on conc. gradient)

↓ But need for tertiary current dist.

Electrochemical Reaction Engineering:

CSTR: $V_0(C_{A_0} - C_A) - r_A = 0$ (@ SS, isothermal)

$r_A = \frac{-i_{ks}}{F} = 0$ ← surface area

$$\frac{-i_{ls}}{nF} = 0$$ → surface area

$$i = f\left(C_s, \frac{D}{\delta}\right)$$

$$\tau = \frac{V}{v_o} \quad ; \quad \sigma = \frac{s}{V}$$

σ: specific electrode area.

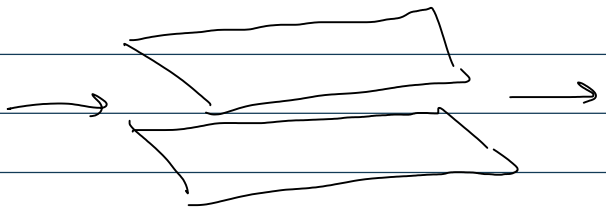
$$\frac{\sigma \tau i_A}{nF} = (C_{Ao} - C_A) \quad ; \quad \chi = \frac{C_{Ao} - C_A}{C_{Ao}}$$

K_c [=] $\frac{m}{s}$

$$\chi = \frac{\sigma \tau i_A}{C_A nF} \quad @ \text{ limiting current} \quad i_A = nF\left(\frac{D}{\delta}\right) C_A$$

$$\boxed{\chi = \frac{\sigma \tau K_L}{1 + \sigma i K_c}}$$

PFR:



$$\frac{i_A S}{nFL} = -v_o \frac{dC_A}{dx}$$

hw

Wednesday, 18 January 2023 6:23 PM

Hw 1

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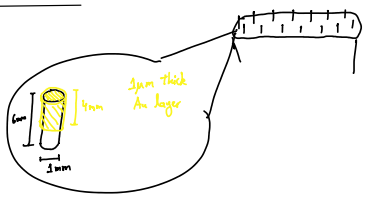
Problem 1.3



Q1] Cost of gold in connector?

Volume of Au in 1 pin: $\pi (r_{pin} + Au_{thickness})^2 (l + 1\mu m) - \pi (r_{pin})^2 \cdot l$

$= \pi [(0.5\,mm + 1\mu m)^2] (3\,mm + 1\mu m) - \pi [0.5\,mm^2] \cdot 4\,mm$

$= \pi [(0.5 \times 10^{-3} + 1 \times 10^{-6})^2 (3.001 \times 10^{-3})] - \pi (0.5 \times 10^{-3})^2 \cdot 4 \times 10^{-3}$

$= 1.258 \times 10^{-11}\ m^3$

Total Volume: $1.258 \times 10^{-11}\ m^3$

$1\ cm = 10^{-2}\ m$

Mass of Au = Density of Au · Volume of Au

@LBMA UK [lbma.org.uk]: $\rho_{Au} = 19.32\ g/cm^3 = 19.32 \times 10^6\ g/m^3$

$\therefore$ Mass of Au = $(19.32)(1.258 \times 10^{-5})\ g$

@MONEX Metals [monex.com/gold-prices/]: $\$61.38\ g^{-1}$

Total cost of gold for 1 pin = $(61.38)(19.32)(1.258 \times 10^{-5}) = \$0.015$ $\xrightarrow{\times 25}$ $\Rightarrow$ Cost of Connector $\Rightarrow$ \$0.38

Q2] Cost/l of Au electrolyte (conc. = 40 mM)?

$\Rightarrow$ Conc. = $\frac{mol}{l}$; 1 l of electrolyte has 40×10^{-3} mol of Au

$MW_{Au} = 196.967\ \frac{g}{mol}$

$\Rightarrow$ Mass of Au (per l electrolyte): $40 \times 10^{-3} \cdot (196.967) = 7.878\ g/l$

$\Rightarrow$ Cost of electrolyte: $\$61.38\ \frac{}{g} \cdot \frac{7.878\ g}{l}$

$= \$483.6$

Q3] Supporting electrolyte?

Adding another electrolyte will result in reduction in efficiency of the E-cell since another side reaction at counter electrode is likely to take place. This may allow for greater control of gold electroplating process?

Problem 1.4

Batteries vs Fuel Cells:

Similarities:

- Both batteries and fuel cells are devices that generate electricity via electrochemical reactions (i.e. transfer of electrons) at electrode interface. Both technologies depend on a species being oxidised at the anode resulting in flow of electrons around an external circuit that can then be used as voltage difference across a load.

Differences:

- A battery needs to be replenished by charging or adding in "electricity" to be stored for later use. A fuel cell needs chemical compounds (such as hydrogen in case of a hydrogen fuel cell) in order to provide electricity.
- A fuel cell operates as an open (continuous) system where it can produce electricity via the flowing fuel (hydrogen) and flow out of by product (such as water); a battery on the other hand, can provide DC current continuously until it runs (i.e. all lithium ions have been intercalated back into the cathode and need to be transported back to the graphitic anode to convert the battery into a charged state). Therefore a battery can be considered as a batch process, where it would need to ideally be stopped to charge again.

Links:
energy.gov
usdoe
Visual capitalist

Problem 1.7

Total Al production: 10^{11} lb/year
$= 4.54 \times 10^{10}$ kg/year

Faraday's law: relates charge passed to amount of product (Al) formed

$$M_i = \frac{-s_i (MW_i) Q}{nF}$$; Q = total charge, s_i = stoich coeff, M_i = total mass formed, F = Faraday's constant

Al half reaction: $Al^{3+} + 3e^- \rightarrow Al$ (Cathode)

In 1 year:

$$4.54 \times 10^{13}\ g = \frac{(1)(26.98\ g/mol)\ Q}{(1)(96485\ C/mol)}$$

$$Q = 1.623 \times 10^{17}\ C$$

Total current: $I = \frac{Q}{t} = \frac{1.623 \times 10^{17}\ C}{1 \cdot 12 \cdot 30 \cdot 24 \cdot 60 \cdot 60} = 5.218 \times 10^{9}\ A$

$\therefore P = I \cdot V = 5.218 \times 10^{9}\ A \cdot 5\ V = 2.61 \times 10^{10}\ W$
$\approx 26\ GW$

Power consumption in an year $\approx$ 26 GW.

1 Watt-hour = 1 Watt · 1 hour ; 1 year = 8760 hours

↳ Global electric-power production (in 2000): IEA report estimates 12,696 TWh $\approx$ 1.449×10^{12} W in 1 year [@ iea.org]

Therefore, Al production consumed around 1.79% of total world power production

↳ Capacity of a larger power plant: [Ourworldindata.org]: 38000 MWh/day (typical coal plant in USA) Circa 2021

$= 1387000$ MWh/year

$= 158.33 \times 10^6$ W in 1 year

This value is 0.61% of what is needed in Al production

Problem 1.8

Half-cell reactions in electrochemical cell.

(a) $Cu(s) \mid CuSO_4(aq) \parallel CuSO_4(aq) \mid Cu(s)$

Left: $Cu^{2+} + 2e^- \rightleftharpoons Cu$

Right: $Cu \rightleftharpoons Cu^{2+} + 2e^-$

} No potential drop since same material is electrode

Overall: $Cu \rightleftharpoons Cu$

Reduction: $Cu^{2+} + 2e^- \rightarrow Cu$

Oxidation: $Cu \rightarrow Cu^{2+} + 2e^-$

Reduction: $H_2O + 2e^- \rightarrow H_2 + 2OH^-$

Oxidation: $Ag \rightarrow Ag^{2+} + 2e^-$

$[Ag^{2+} + Cl^- \rightarrow AgCl(s)]$

Reduction: $H_2O + 2e^- \rightarrow O_2 +$

Oxidation: $Zn \rightarrow Zn^{2+} + 2e^-$

$[Zn^{2+}$

(b) $Ag(s) \mid AgCl(s) \mid HCl(aq) \mid H_2(g) \mid Pt(s)$

Left: $2Ag + 2Cl^- \rightleftharpoons 2AgCl_{(s)} + 2e^-$

Right: $H_2O + 2e^- + 2H^+ \rightleftharpoons H_{2(g)} + H_2O$

Overall: $2Ag + 2HCl \rightleftharpoons H_2 + 2AgCl$

(c) $Zn(s) \mid ZnO(s) \mid KOH(aq) \mid O_2(g) \mid Pt(s)$

Left: $2Zn_{(s)} + 4OH^- \rightleftharpoons 2ZnO + 4e^- + 2H_2O$

Right: $O_{2(g)} + 4e^- + 2H_2O \rightleftharpoons 4OH^-$

} Technically this should be a reduction?

Overall: $2Zn_{(s)} + O_{2(g)} \rightleftharpoons 2ZnO_{(s)}$

(d) $Pt(s) \mid O_2(g) \mid NaOH(aq) \mid NH_3(aq), NaNO_2(aq) \mid Pt(s)$

Left: $4OH^- \rightleftharpoons O_{2(g)} + 2H_2O + 4e^-$ $(\times \frac{3}{2})$

Right: $NO_2^- + 6e^- + 5H_2O \rightleftharpoons NH_3 + 7OH^-$

NH_4^+: N is +4, −3 ; NO_2^-: N is +3, −4

$[N^{3-} + 6e^- \rightleftharpoons N^{3+}]$

Overall: $NaNO_2 + 2H_2O \rightleftharpoons NH_3 + NaOH + \frac{3}{2}O_{2(g)}$

(e) $Pt(s) \mid FeCl_2(aq), FeCl_3(aq) \parallel SnCl_2(aq), SnCl_4(aq) \mid Pt(s)$

Left: $2FeCl_2 + 2Cl^- \rightleftharpoons 2FeCl_3 + 2e^-$ $[2Fe^{2+} \rightleftharpoons 2Fe^{3+} + 2e^-]$

Right: $SnCl_4 + 2e^- \rightleftharpoons SnCl_2 + 2Cl^-$ $[Sn^{4+} + 2e^- \rightleftharpoons Sn^{2+}]$

Overall: $2FeCl_2 + SnCl_4 \rightleftharpoons 2FeCl_3 + SnCl_2$

Problem 5

(a) Electron-conducting : Electrode 1
Electrode 2
Metal wire

Ion-conducting : Liquid Electrolyte

Insulating : Glass beaker

(b) Since e^- are flowing from V1 to V2 across a load R1, it is a galvanic cell

Oxidation @ V1 (loss of e^-) → Anode

Reduction @ V2 (gain e^-) → Cathode

Current flow in the opposite direction of e^- flow (V2 → V1)

(c) When $V1 \approx V2$, there is no resistance to flow, therefore, we have max current (short circuit / closed circuit)

" $V1 \gg V2$, there is min current (negligible), and we have max cell voltage (open circuit potential) → V_{OC}

Max cell power is achieved at an intermediate value, where $\boxed{0 \leq V1 \leq V_{OC}}$

Problem 6

Bockris 1.1 – 1.5, 1.12

a. What did you know about electrochemistry before reading the chapter?
I knew that electrochemistry is essential to understanding how electrochemical cells (such as batteries) function. There are two types of electrochemical cells: galvanic and electrolytic.

b. What were 3 things you learned by reading the chapter? Even if ideas were new, were the physical and chemical concepts familiar to you?

- Firstly, it was illuminating to learn about the history of electrochemistry and its profound impact on technology years down the line, starting from Volta, Debye, Huckel, Faraday and Tafel.

- Field of electrochemistry can be subdivided into:
  - Ionics: ions in solution, solid solutions. This sub-field deals with:
    - Ion-solvent interaction,
    - Ion-ion interaction,
    - Ion transport interaction
    - Ionic liquids interaction
  - Electrodics: region between electronic and ionic conductor;
    - Interfacial region: electrically controlled surface reactions

- Electrochemical reaction, in its bare form, is just a transfer of electrons across an interface that results in a chemical tr ansformation. However, this chemical transformation occurs in a very different way, fundamentally, than a conventional chemical reaction. Reactants no longer coll ide with each other but with separated "charge transfer catalysts" (i.e. electrodes). From an overall perspective, the net reaction may have been the same but fundamentally electrochemical reactions differ from conventional chemical reactions. Furthermore, this spatial separation (among other things) gives electr ochemical cells an edge over other reactors since separation may not be an added cost in the production of chemicals.

- Reaction rate is the rate at which the power source pushes out and receives back electrons after their journey around the cir cuit. Therefore, one can easily control the electrochemical reaction rate: increasing the potential across the power source!

Yes the concepts were familiar but it was helpful to read them in an interesting thought experiment form to understand the fu ndamentals.

c. What were 2 things you are still confused about?
It is described that electrons are "physically" pushed from the power source into the electrode and then "overflow" into the solution to strike and neutralize ions in the layer of solution in contact with the metal. This may be a little misleading since it is sort of describing the electrons as "flowing" and then building up to then overflow to the solution.

My understanding is that the potential difference (due to the power source) results in increasing (or decreasing) the highest occupied molecular orbital in the electrode and therefore electrons want to transfer to the highest unoccupied molecular orbital of ions in solution that is at a lower (or higher) energy. This results in a transfer of electrons. Higher the potential difference (from the power source), the greater the difference between this molecular orbitals and faster the electron wants to jump across to lower energy.

d. Which of the actual or potential (section 1.12) applications of electrochemistry are most intriguing to you

It was interesting to learn the electrical dependence of humans increasing as time goes on where entire towns and cities would be self-sufficient in producing all that is required from electrochemical devices including even "internal fuel-cell powered heart" and electrically functioning body parts that will lead to a "person-machine combination" i.e. cyborgs. This seems pretty cool!

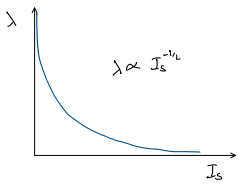
Problem 2.1

Debye length : $\lambda = \sqrt{\frac{\varepsilon RT}{F^2 \sum_j z_j^2 c_{j\infty}}}$; $I_s = \frac{1}{2}\sum_j z_j^2 c_{j\infty}$ (ionic strength) $\therefore \quad \lambda = \sqrt{\frac{\varepsilon RT}{F^2 \cdot 2I_s}}$

Net charge falling exponentially over distances on order of Debye length → i.e. distances over which electroneutrality breaks down

$\lambda = \sqrt{\frac{\varepsilon RT}{2F^2}} \cdot I_s^{-1/2}$

$\lambda \propto I_s^{-1/2}$



@ pH 7, pure water, $I_s = 1\times10^{-3}$ M ; assuming room temperature of 20°C

$K = \frac{\varepsilon}{\varepsilon_0}$; $\varepsilon_{H_2O} = 78\cdot(8.854\times10^{-14}\ \frac{F}{cm}) = 78\times(8.85\times10^{-12}\ \frac{F}{m})$

$\therefore \quad \lambda = \left(\frac{78\times8.854\times10^{-12}\times 8.314\times 293.15}{2\times(96485)^2}\right)^{1/2}\cdot\left(1\times10^{-3}\times10^{3}\right)^{-1/2}$

$\lambda = 9.5\times10^{-9}$ m $\approx$ 9.5 nm

+ Really small value ✓
+ Significance → Debye length just provides a general idea of the diffuse layer ↓

A really small value would mean that charge neutrality can be only broken over nm scale; As the conc. of electrolyte (or I_s ↑) increases. (This might have significance when dealing w/ double layer analysis)

@ same conditions but $I_s = 1$M : $\lambda = 3.01\times10^{-11}$ m $\approx$ 0.03 nm

As I_s↑, λ↓ ✓

Problem 2.2

↳ Conductivity of an electrolyte : $K = F^2 \sum z_j^2 y_j c_j$; $y_j = \frac{D_j}{RT}$ (Diffusion coefficient → Ionic mobility)

[K] = $\Omega^{-1}cm^{-1}$; c_j → ionic conc.

Water dissociates into H^+ and OH^- (H_3O^+) → @ pH 7 10^{-7} mol of H_3O^+ & OH^- ions

$K = F^2\left[(+1)^2\cdot 10^{-7}\cdot\frac{9.310\times10^{-9}\times10^3}{(8.314)(293.15)} + \frac{(-1)^2\cdot10^{-7}\cdot5.270\times10^{-9}\cdot10^3}{(8.314)\cdot(293.15)}\right]$

$K = 5.57\times10^{-9}\ \Omega^{-1}m^{-1} \approx$ 0.056 mS/cm

↳ Sea water : Na^+, Cl^- ; $I_s = \frac{1}{2}\sum z_j^2 \cdot c_{j\infty} = 0.5$M ; @ room temp

$K = F^2\cdot\left((+1)^2\cdot\frac{D_{Na^+}}{RT}\cdot c_{Na^+} + (-1)^2\frac{D_{Cl^-}}{RT}c_{Cl^-}\right)$; Assuming only Na^+, Cl^- in seawater

If we assume $D_{Na^+} \approx D_{Cl^-} \approx 10^{-9}\frac{m^2}{s}$; $D_{Na^+} = 1.33\times10^{-9}\ m^2/s$; $D_{Cl^-} = 2.03\times10^{-9}\ m^2/s$ (Somewhat valid assumption)

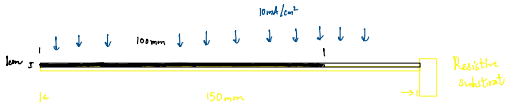
$K = F^2\cdot\frac{D_{Na^+}}{RT}\underbrace{\left((+1)^2\cdot c_{Na^+} + (-1)^2 c_{Cl^-}\right)}_{2\cdot I_s} = \frac{(96485)^2\cdot(1.0\times10^{-9})}{(8.314)(293.15)}(0.5)\cdot 2$

$= 38.2\times10^{-3}\ \Omega^{-1}m^{-1}$ or 38 mS/cm (1 S = 1 ohm^{-1})

0.5 M NaCl → 0.0479 S/cm ≈ 47.9 mS/cm
@ Engineering Toolbox

Our estimate is a factor of 0.79 of the actual value.

Problem 2.9



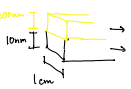
Substrate → Ru (resistive)

(a) Substrate: $R = \frac{\rho L}{A}$ (ρ: resistivity; Resistance is proportional to the no. of e⁻ that can pass through a cross section of wire)

$$R = \frac{(71\times10^{-9})(150\times10^{-3})}{(10\times10^{-9})(1\times10^{-2})}$$

$$\boxed{R = 106.5\ \Omega}$$

(b)



$$R = \frac{\rho L}{A}$$

$$R = \frac{(17\times10^{-9})(150\times10^{-3})}{(200\times10^{-9})(1\times10^{-2})}$$

$$\boxed{R = 1.3\ \Omega}$$

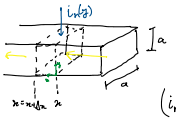
Ideally we would have to consider the resistivity of both Cu & Ru, but because Cu is much more conductive (and thicker) we only need to consider this layer

$\rho_{Cu} = 17\ n\Omega m$

(c) Constant deposition rate = $10\ mA\ cm^{-2}$; Ohms Law: $V = IR$

(current density (i))

Shell balance:



m = metal
s = substrate ; $V_m = \rho i_m L$

$$(i_m(x) - i_m(x+\Delta x))a^2 = -i_n \Delta x^2$$

$$a^2 \frac{d^2 i_m}{dx^2} = i_n \quad ; \quad i_m = \frac{V_m}{\rho_s L_s}$$

$$\frac{a^2}{\rho_s L_s}\frac{d^2 V_m}{dx^2} = i_n \quad ; \quad a^2 = L_s \cdot \lambda_s$$

$$\boxed{\frac{\lambda_s}{\rho_s}\left(\frac{d^2 V_m}{dx^2}\right) = i_n}$$

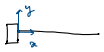
(d) Solve differential equation:

$$\frac{d^2 V_m}{dx^2} = \frac{\rho_s \cdot i_n}{\lambda_s} \neq f(x)$$

$$\frac{dV_m}{dx} = \frac{\rho_s \cdot i_n}{\lambda_s} x + c$$

$$V_m = \frac{\rho_s i_n}{2\lambda_s} x^2 + cx + d$$

$$\therefore V_m = \frac{\rho_s i_n L^2}{\lambda_s 2}$$



BC:

1) @ $x=0$, $V_m = 0$ (potential at current collector is zero)

2) @ $x=L$, $\frac{dV_m}{dx} = 0$ (electronic current at end is zero)

For 10nm Ru:

$$V_m = \frac{(71\times10^{-9})(100)(150\times10^{-3})^2}{2\cdot 10\times10^{-9}} = 8V$$

For 200nm Cu:

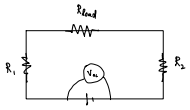
$$V_m = \frac{(17\times10^{-9})(100)(150\times10^{-3})^2}{2\cdot 200\times10^{-9}} = 0.095\ V$$

Although current flow majorly happens via Cu

↓

So Cu potential is insignificant

Problem 2.13



+ Simple, linear circuit
+ $R_{load} \to \infty$, $V_{oc} = 1V$ ← Open circuit (We assume that this is the voltage output if the cell and is constant)

(a) $R_1 = R_2 = 1\,\Omega$; $P_{load} = V_{load} \cdot I = I^2 R_{load}$.

$V_{load} = I \cdot R_{load}$

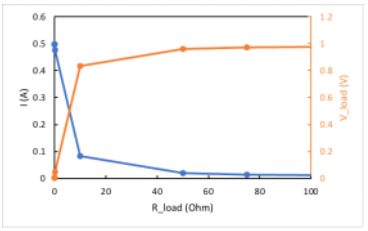
Linear/Series circuit: $R_{total} = \underbrace{R_1 \| R_2}_{2\,\Omega} + R_{load}$; $I = \dfrac{V_{oc}}{(R_{Total})}$

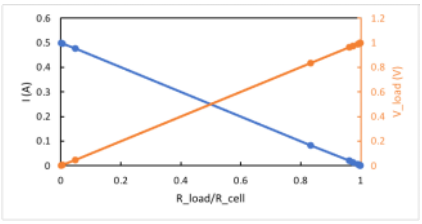
$V_{total} = V_1 + V_2 + V_{load}$

Bounds of I: @ $R_{load} = 1000\,\Omega$, $V_{load} = 1V$ (Very high R) } Open circuit

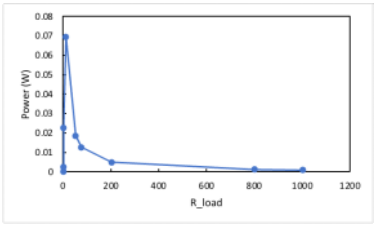
@ $R_{load} = 0.001\,\Omega$, $V_{load} = 0.01\,V$ (Very low R) } Short circuit

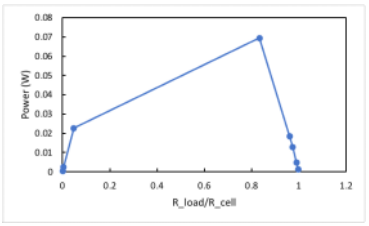
Plotting using discretized values :





(b)





* Graphs show the necessary trend: Max power occurs at a specific Rload well below that at V_{oc} value for R

(c) Current at which P_{max}? $P_{load} = V_{load} \cdot I$; $V_{load} = I\,R_{load}$

$P_{load} = I^2 \cdot R_{load}$; $P_{load} = \dfrac{V_{load}^2}{R_{load}}$

$P_{load} = \dfrac{V_{oc}}{(2+R_{load})^2} \cdot R_{load}$;

$$\left.\frac{\partial P_{load}}{\partial R_{load}}\right|_{V_{oc}} = V_{oc}\left[\frac{(2+R_{load})^2 - R_{load}\cdot 2\cdot(2+R_{load})}{(2+R_{load})^4}\right] = \frac{V_{oc}\left(4 + R_{load}^2 + 4R_{load} - 4R_{load} - 2R_{load}^2\right)}{(2+R_{load})^4} = \frac{V_{oc}(4-R_{load}^2)}{(2+R_{load})^4} = \frac{(2+R_{load})(2-R_{load})V_{oc}}{(2+R_{load})^{3+2}} = \frac{(2-R_{load})V_{oc}}{(2+R_{load})^3}$$

$P_{max} \Rightarrow \dfrac{\partial P_{load}}{\partial R_{load}} = 0 \Rightarrow \dfrac{(2-R_{load})V_{oc}}{(2+R_{load})^3} = 0$

$R_{load} = 2\,\Omega$ @ P_{max}

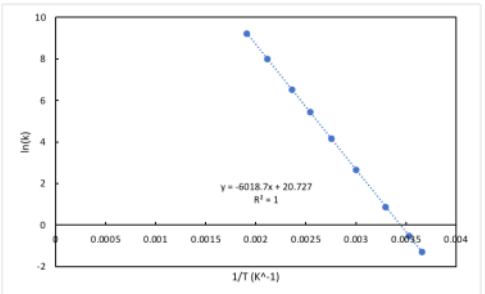
$\therefore I_{max} = \dfrac{V_{oc}}{(2+R_{load}^{min})} = \boxed{\dfrac{V_{oc} = I_{max}}{4}}$

Problem 3.1

Given kEJs^{-1} & TEJ°C

Arrhenius relationship: $k = A_0 \exp\left(-\dfrac{E_A}{RT}\right)$ (pre-exp. factor)

Linearized ⟶ $\ln k = \underbrace{\ln(A_0)}_{\text{y-intercept}} - \underbrace{\dfrac{E_A}{R}}_{\text{gradient}}\left(\dfrac{1}{T}\right)$



(Assuming the last point is a typo and should be 10,000)

Plotting $\ln k$ vs. $\dfrac{1}{T}$ gives:

$m = -6018.7$

$c = 20.727$

$\Rightarrow -E_a = -6018.7\,(R)$

$\boxed{E_a = 50039.5\,J}$

To achieve 10x reaction rate increase (i.e. increasing rate constant) [$r \propto k$]: (Room T = 20°C)

$$\frac{k_{new}}{k_o} = 10 \Rightarrow \frac{A_o \exp\left(\frac{E_a}{RT_{new}}\right)}{A_o \exp\left(\frac{E_a}{RT_{room}}\right)} = 10$$

$$\exp\left(-\frac{E_a}{RT_{new}} + \frac{E_a}{RT_{room}}\right) = 10$$

$$\frac{E_a}{R}\left(\frac{1}{T_{room}} - \frac{1}{T_{new}}\right) = \ln 10$$

$$\frac{1}{T_{new}} = \frac{1}{T_{room}} - \frac{R \ln 10}{E_a}$$

$$= \frac{1}{293.15} - \frac{8.314 \ln(10)}{50039.5}$$

$$T_{new} = 330.18 \text{ K}$$

$$\approx 57^\circ C$$

i.e. T increase by 37°C to achieve 10x rate increase from 20°C

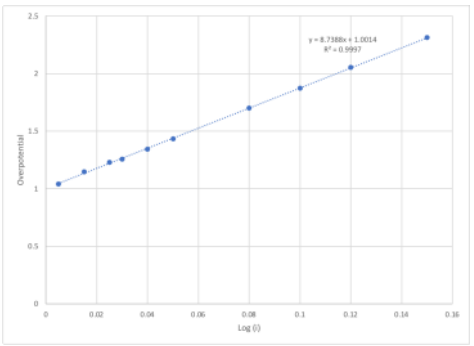
Problem 3.2

Tafel plot: Large overpotential ($\eta \gg 0$) $\Rightarrow$ $i = i_o \exp\left(\frac{\alpha_a F \eta}{RT}\right)$ ↓ Linearize

$$\log_{10} i = \underbrace{\log_{10} i_o}_{c} + \underbrace{\left(\frac{\alpha_a F \log_{10}(e)}{RT}\right)}_{m} \eta$$

(y) (x)

Plotting $\log_{10} i$ vs. η : (Inverse Tafel plot)



$m = 8.7388$ @ room temp 293.15 K

$$\frac{\alpha_a \cdot F}{RT} \cdot \log_{10}(e) = 8.7388 \quad ; \quad \log_{10} i_o = 1.0014$$

$$\alpha_a = \frac{(8.7388)\cdot(8.314)(293.15)}{(96485)(0.4343)}$$

$$\boxed{i_o = 10.03 \ \frac{A}{m^2}}$$

$$\boxed{\alpha_a = 0.508}$$

$$\frac{i_{new}}{i_{old}} = 10 \Rightarrow \frac{\cancel{i_o} \exp\left(\frac{\alpha_a F \eta_{new}}{RT}\right)}{\cancel{i_o} \exp\left(\frac{\alpha_a F}{RT} \eta_{old}\right)} = 10$$

Similar analysis as previous part: $\frac{\alpha_a F}{RT}\left(\eta_{new} - \eta_{old}\right) = \ln 10$ @ room temp: 293.15 K

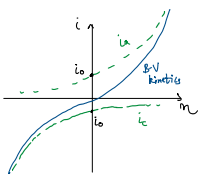
$$\boxed{\Delta\eta = 1.003 \text{ V}}$$

The overpotential increase of 1V seems a lot easier/economic to do than a ΔT increase of 37°C

Problem 7

B-V kinetics ; $\alpha_a = 0.5$; $\eta = ?$ @ 90% of total current is attributed to the anodic reaction → This means that anodic process is dominating & $0.9 = \frac{i_a}{i_o}$?

$$i = i_o\left(\underbrace{\exp\left(\frac{\alpha_a F \eta}{RT}\right)}_{\text{Anodic}} - \underbrace{\exp\left(-\frac{\alpha_c F \eta}{RT}\right)}_{\text{Cathodic}}\right)$$



i_o means the equilibrium case where anodic & cathodic reactions are happening at same rate

Since we are in the anodic regime;

$$i = i_o \exp\left(\frac{\alpha_a F \eta}{RT}\right)$$

$$\frac{i}{i_o} = \exp\left(\frac{\alpha_a F \eta}{RT}\right)$$

$$0.9 = \exp\left(\frac{(0.5)(96485)}{RT}\eta\right)$$

$$|\eta| = 5.32 \times 10^{-3}\,V$$

@ 99% of total current:

$$\frac{i}{i_0} = 0.99 = \exp\left(\frac{\alpha_a F \eta}{RT}\right)$$

$$|\eta| = 5.08 \times 10^{-2}\,V$$

It seems that we are in linearized B-V region i.e. use linear kinetics to model the system

Problem 8

① $Cu \rightleftharpoons Cu^+ + e^-$ (RDS) $\qquad i = f(V, C_k)$

② $Cu^+ + 2Cl^- \rightleftharpoons CuCl_2^-$

① Electrochem:

$$r_1 = k_{1,a}\exp\left(\frac{(1-\beta)FV}{RT}\right)[Cu] - k_{1,c}[Cu^+]\exp\left(\frac{-\beta FV}{RT}\right)$$

② Chemical:

$$r_2 = k_f[Cu^+][Cl^-]^2 - k_r[CuCl_2^-]$$

r_1 is rds, r_2 is QE: $\quad r_2 \approx 0 \Rightarrow \quad k_f[Cu^+][Cl^-]^2 = k_r[CuCl_2^-]$

$$[Cu^+] = \frac{k_r}{k_f}\frac{[CuCl_2^-]}{[Cl^-]^2}$$

$$r_{net} = r_1 = k_{1,a}[Cu]\exp\left(\frac{(1-\beta)FV}{RT}\right) - k_{1,c}[Cu^+]\exp\left[\frac{-\beta FV}{RT}\right]$$

$$\frac{i}{F} = k_{1,a}[Cu]\exp\left(\frac{(1-\beta)FV}{RT}\right) - \frac{k_{1,c}k_r}{k_f}\frac{[CuCl_2^-]}{[Cl^-]^2}\exp\left(\frac{-\beta FV}{RT}\right) \rightarrow$$ Matches B-V kinetics (w/ 1 e^- transfer)

Solving for exchange current: $i = 0$ when $V = U$

$$\therefore \quad k_{1,a}[Cu]\exp\left(\frac{(1-\beta)FU}{RT}\right) = \overbrace{\frac{k_{1,c}k_r}{k_f}\frac{[CuCl_2^-]}{[Cl^-]^2}}^{Z}\exp\left(\frac{-\beta FU}{RT}\right)$$

$$\frac{k_{1,a}[Cu]}{Z} = \exp\left(\frac{-\beta FU}{RT} - \frac{(1-\beta)FU}{RT}\right) \qquad -\beta - 1 + \beta$$

$$\frac{k_{1,a}[Cu]}{Z} = \exp\left(\frac{-FU}{RT}\right)$$

$$\frac{FU}{RT} = -\ln\left(\frac{k_{1,a}[Cu]}{Z}\right) \qquad \left[\exp\left(\frac{FU}{RT}\right) = \frac{Z}{k_{1,a}[Cu]}\right]$$

$$U = \frac{RT}{F}\ln\left(\frac{Z}{k_{1,a}[Cu]}\right)$$

$\eta = V - U$; $V = \eta + U \Rightarrow$ Plugging in the original expression:

$$\frac{i}{F} = k_{1,a}[Cu]\exp\left(\frac{(1-\beta)FV}{RT}\right) - \frac{k_{1,c}k_r}{k_f}\frac{[CuCl_2^-]}{[Cl^-]^2}\exp\left(\frac{-\beta FV}{RT}\right)$$

$$= k_{1,a}[Cu]\exp\left(\frac{(1-\beta)F}{RT}(\eta+U)\right) - Z\exp\left(\frac{-\beta F}{RT}(\eta+U)\right)$$

$$= k_{1,a}[Cu]\exp\left(\frac{(1-\beta)F\eta}{RT}\right)\exp\left(\frac{(1-\beta)FU}{RT}\right) - Z\exp\left(\frac{-\beta F}{RT}\eta\right)\exp\left(\frac{-\beta F}{RT}U\right)$$

$$= k_{1,a}[Cu]\exp\left(\frac{(1-\beta)F\eta}{RT}\right)\exp\left(\frac{FU}{RT}\right)\exp\left(\frac{-\beta FU}{RT}\right) - Z\exp\left(\frac{-\beta FU}{RT}\right)\exp\left(\frac{-\beta F\eta}{RT}\right)$$

$$= \frac{k_{1,a}[Cu]}{k_{1,a}[Cu]}Z\cdot\exp\left(\frac{-\beta FU}{RT}\right)\left[\exp\left(\frac{(1-\beta)F\eta}{RT}\right) - \exp\left(\frac{-\beta F\eta}{RT}\right)\right]$$

$$= z \cdot \exp\left(-\frac{\beta \cdot F}{RT} \cdot \frac{RT}{F} \ln\left(\frac{z}{k_{1,a}[Cu]}\right)\right) \left[\quad '' \quad \right]$$

$$= z \cdot \exp(-\beta) \frac{z}{k_{1,a}[Cu]} \left[\quad '' \quad \right]$$

$$i = \underbrace{\frac{z^2 \cdot \exp(-\beta)}{k_{1,a}[Cu]}}_{i_0 = \left[\frac{k_{1,c}\, k_r}{k_f} \frac{[Cu Cl^-]}{[Cl^-]^2}\right]^2 \cdot \frac{1}{k_{1,a}[Cu]} \exp(-\beta)} \left[\exp\left(\frac{(1-\beta) F\eta}{RT}\right) - \exp\left(-\frac{\beta F \eta}{RT}\right) \right]$$

$$i_0 \propto \frac{[Cu Cl^-]^2}{[Cl^-]^4 [Cu]}$$

Problem 4.2

Qs: Do elec. potentials equal each other over salt bridge? (Flow of ions is permitted?)

(a) $\overset{\alpha}{Cu(s)} \mid \overset{\beta}{0.001\,M\ CuSO_4(aq)} \parallel \overset{\beta'}{0.1\,M\ CuSO_4(aq)} \mid \overset{\alpha'}{Cu(s)}$

$U_{cell} = V^{\alpha'} - V^{\alpha}$

- Reversible potential @ equilibrium : $\mu_e^{\alpha} - \mu_e^{\alpha'} = FU$
- Phases in adjacent phases are at equilibrium

LHS: $Cu \rightleftharpoons Cu^{2+} + 2e^- \Rightarrow \mu_{Cu}^{\alpha} = \mu_{Cu^{2+}}^{\beta} + 2\mu_e^{\alpha} \Rightarrow \mu_e^{\alpha} = \frac{1}{2}\left(\mu_{Cu}^{\alpha} - \mu_{Cu^{2+}}^{\beta}\right)$

RHS: $Cu \rightleftharpoons Cu^{2+} + 2e^- \Rightarrow \mu_{Cu}^{\alpha'} = \mu_{Cu^{2+}}^{\beta'} + 2\mu_e^{\alpha'} \Rightarrow \mu_e^{\alpha'} = \frac{1}{2}\left(\mu_{Cu}^{\alpha'} - \mu_{Cu^{2+}}^{\beta'}\right)$

$\therefore \boxed{U_{cell} = \frac{1}{2F}\left(\mu_{Cu}^{\alpha} - \mu_{Cu^{2+}}^{\beta} - \mu_{Cu}^{\alpha'} + \mu_{Cu^{2+}}^{\beta'}\right)}$

(b) $\overset{\alpha}{Ag(s)} \mid \overset{\beta}{AgCl(s)} \mid \overset{\beta'}{HCl(aq)} \mid \overset{\alpha'}{H_2(g)}, Pt(s)$

$U_{cell} = V^{\alpha'} - V^{\alpha}$

- Reversible potential @ equilibrium : $\mu_e^{\alpha} - \mu_e^{\alpha'} = FU$
- Phases in adjacent phases are at equilibrium : $\mu_e^{\alpha} = \mu_e^{\beta}$

LHS: $AgCl + e^- \rightleftharpoons Ag + Cl^- \Rightarrow \mu_{AgCl}^{\beta} + \mu_e^{\alpha} = \mu_{Ag}^{\alpha} + \mu_{Cl^-}^{\beta'} \Rightarrow \mu_e^{\alpha} = \mu_{Ag}^{\alpha} - \mu_{Cl^-}^{\beta'} - \mu_{AgCl}^{\beta}$

RHS: $H_2 \rightleftharpoons 2H^+ + 2e^- \Rightarrow \mu_{H_2}^{\alpha'} = 2\mu_{H^+}^{\beta'} + 2\mu_e^{\alpha'} \Rightarrow \mu_e^{\alpha'} = \frac{1}{2}\left(\mu_{H_2}^{\alpha} - 2\mu_{H^+}^{\beta'}\right)$

$\therefore \boxed{U_{cell} = \frac{1}{F}\left(\mu_{Ag}^{\alpha} - \mu_{Cl^-}^{\beta'} - \mu_{AgCl}^{\beta} - \frac{1}{2}\left(\mu_{H_2}^{\alpha} - 2\mu_{H^+}^{\beta'}\right)\right)}$

(c) $\overset{\alpha}{Zn(s)} \mid \overset{\beta}{ZnO(s)} \mid \overset{\beta'}{KOH(aq)} \mid \overset{\alpha'}{O_2(g)}, Pt(s)$

$U_{cell} = V^{\alpha'} - V^{\alpha}$

- Reversible potential @ equilibrium : $\mu_e^{\alpha} - \mu_e^{\alpha'} = FU$
- Phases in adjacent phases are at equilibrium : $\mu_e^{\alpha} = \mu_e^{\beta}$

LHS: $Zn + 2OH^- \rightleftharpoons ZnO + H_2O + 2e^- \Rightarrow \mu_{Zn}^{\alpha} + 2\mu_{OH^-}^{\beta'} = \mu_{ZnO}^{\beta} + \mu_{H_2O}^{\beta'} + 2\mu_e^{\alpha} \Rightarrow \mu_e^{\alpha} = \frac{1}{2}\left(\mu_{Zn}^{\alpha} + 2\mu_{OH^-}^{\beta'} - \mu_{ZnO}^{\beta} - \mu_{H_2O}^{\beta'}\right)$

RHS: $4OH^- \rightleftharpoons O_2 + 4e^- + 2H_2O \Rightarrow 4\mu_{OH^-}^{\beta'} = \mu_{O_2}^{\alpha'} + 4\mu_e^{\alpha'} + 2\mu_{H_2O}^{\beta'} \Rightarrow \mu_e^{\alpha'} = \frac{1}{4}\left(4\mu_{OH^-}^{\beta'} - \mu_{O_2}^{\alpha'} - 2\mu_{H_2O}^{\beta'}\right)$

$\therefore U_{cell} = \frac{1}{2F}\left(\mu_{Zn}^{\alpha} + 2\mu_{OH^-}^{\beta'} - \mu_{ZnO}^{\beta} - \mu_{H_2O}^{\beta'} - \frac{1}{2}\left(4\mu_{OH^-}^{\beta'} - \mu_{O_2}^{\alpha'} - 2\mu_{H_2O}^{\beta'}\right)\right) = \boxed{\frac{1}{2F}\left(\mu_{Zn}^{\alpha} - \mu_{ZnO}^{\beta} + \frac{1}{2}\mu_{O_2}^{\alpha'}\right)}$

(d) $\overset{\alpha}{Pt(s), O_2(g)} \mid \overset{\beta}{NaOH(aq)} \parallel NH_3(aq), \overset{\beta'}{NaNO_2(aq)} \mid \overset{\alpha'}{Pt(s)}$

$\mu_{H_2O} = ?$

- Reversible potential @ equilibrium : $\mu_e^{\alpha} - \mu_e^{\alpha'} = FU$
- Phases in adjacent phases are at equilibrium : $\mu_{H_2O}^{\beta} = \mu_{H_2O}^{\beta'}$;

LHS: $4OH^- \rightleftharpoons O_2 + 2H_2O + 4e^- \Rightarrow 4\mu_{OH^-}^{\beta} = \mu_{O_2}^{\alpha} + 2\mu_{H_2O}^{\beta} + 4\mu_e^{\alpha} \Rightarrow \mu_e^{\alpha} = \frac{1}{4}\left(4\mu_{OH^-}^{\beta} - \mu_{O_2}^{\alpha} - 2\mu_{H_2O}^{\beta}\right)$

RHS: $NO_2^- + 6e^- + 5H_2O \rightleftharpoons NH_3 + 7OH^- \Rightarrow \mu_{NO_2^-}^{\beta'} + 6\mu_e^{\alpha'} + 5\mu_{H_2O}^{\beta'} = \mu_{NH_3}^{\beta'} + 7\mu_{OH^-}^{\beta} \Rightarrow \mu_e^{\alpha'} = \frac{1}{6}\left(\mu_{NH_3}^{\beta'} + 7\mu_{OH^-}^{\beta} - 5\mu_{H_2O}^{\beta'} - \mu_{NO_2^-}^{\beta'}\right)$

$\therefore \boxed{U_{cell} = \frac{1}{2F}\left(\frac{1}{2}\left(4\mu_{OH^-}^{\beta} - \mu_{O_2}^{\alpha} - 2\mu_{H_2O}^{\beta}\right) - \frac{1}{3}\left(\mu_{NH_3}^{\beta'} + 7\mu_{OH^-}^{\beta} - 5\mu_{H_2O}^{\beta'} - \mu_{NO_2^-}^{\beta'}\right)\right)}$

Problem 4.3 $\overset{\alpha}{\quad} \mid \overset{\beta}{\quad} \parallel \overset{\beta'}{\quad} \mid \overset{\alpha'}{\quad}$

Problem 4.3

α | β | β' | α'

$Pt_{(s)}$ | $FeCl_{2\,(aq)}$, $FeCl_{3\,(aq)}$ || $SnCl_{2\,(aq)}$, $SnCl_{4\,(aq)}$ | $Pt_{(s)}$

(a) + Reversible potential @ equilibrium : $\mu_e^{\alpha} - \mu_e^{\alpha'} = FU$

- Phases in adjacent phases are at equilibrium

LHS: $Fe^{3+} \rightleftharpoons Fe^{2+} + e^-$ $\Rightarrow$ $\mu_{Fe^{3+}}^{\beta} = \mu_{Fe^{2+}}^{\beta} + \mu_e^{\alpha}$ $\Rightarrow$ $\mu_e^{\alpha} = \mu_{Fe^{3+}}^{\beta} - \mu_{Fe^{2+}}^{\beta}$

RHS: $Sn^{2+} \rightleftharpoons Sn^{4+} + 2e^-$ $\Rightarrow$ $\mu_{Sn^{2+}}^{\beta'} = \mu_{Sn^{4+}}^{\beta'} + 2\mu_e^{\alpha'}$ $\Rightarrow$ $\mu_e^{\alpha'} = \frac{1}{2}\left(\mu_{Sn^{2+}}^{\beta'} - \mu_{Sn^{4+}}^{\beta'}\right)$

$$\therefore U_{cell} = \frac{1}{F}\left(\mu_{Fe^{3+}}^{\beta} - \mu_{Fe^{2+}}^{\beta} - \frac{1}{2}\left(\mu_{Sn^{2+}}^{\beta'} - \mu_{Sn^{4+}}^{\beta'}\right)\right)$$

(b) $\mu_{Fe^{2+}}^{\beta} = RT\ln\left(\lambda_{Fe^{2+}}^{\beta}\right) = RT\ln\left(C_{Fe^{2+}}^{\beta} f_{Fe^{2+}} a^{\circ}_{Fe^{2+}}\right)$

$\mu_{Fe^{3+}}^{\beta} = RT\ln\left(\lambda_{Fe^{3+}}^{\beta}\right) = RT\ln\left(C_{Fe^{3+}}^{\beta} f_{Fe^{3+}} a^{\circ}_{Fe^{3+}}\right)$

$\mu_{Sn^{2+}}^{\beta'} = RT\ln\left(\lambda_{Sn^{2+}}^{\beta'}\right) = RT\ln\left(C_{Sn^{2+}}^{\beta'} f_{Sn^{2+}} a^{\circ}_{Sn^{2+}}\right)$

$\mu_{Sn^{4+}}^{\beta'} = RT\ln\left(\lambda_{Sn^{4+}}^{\beta'}\right) = RT\ln\left(C_{Sn^{4+}}^{\beta'} f_{Sn^{4+}} a^{\circ}_{Sn^{4+}}\right)$

(c) $$U_{cell} = \frac{1}{F}\left[\left(RT\ln\left(\frac{a_{Fe^{2+}}^{\beta}}{a_{Fe^{3+}}^{\beta}}\cdot\left(\frac{a_{Sn^{4+}}^{\beta'}}{a_{Sn^{2+}}^{\beta'}}\right)^{1/2}\right)\right) + RT\ln\left(\frac{C_{Fe^{2+}}^{\beta} f_{Fe^{2+}}^{\beta}}{C_{Fe^{3+}}^{\beta} f_{Fe^{3+}}^{\beta}}\cdot\frac{\left(C_{Sn^{4+}}^{\beta'} f_{Sn^{4+}}^{\beta'}\right)^{1/2}}{\left(C_{Sn^{2+}}^{\beta'} f_{Sn^{2+}}^{\beta'}\right)^{1/2}}\right)\right]$$

Assuming $f_i = 1$:

$$\boxed{U_{cell} = \underbrace{\left[\left(\frac{RT}{F}\ln\left(\frac{a_{Fe^{2+}}^{\beta}}{a_{Fe^{3+}}^{\beta}}\cdot\left(\frac{a_{Sn^{4+}}^{\beta'}}{a_{Sn^{2+}}^{\beta'}}\right)^{1/2}\right)\right)}_{U^{\circ}_{cell}} + \underbrace{\frac{RT}{F}\ln\left(\frac{C_{Fe^{2+}}^{\beta}\left(C_{Sn^{4+}}^{\beta'}\right)^{1/2}}{C_{Fe^{3+}}^{\beta}\left(C_{Sn^{2+}}^{\beta'}\right)^{1/2}}\right)}_{\text{Nernst correction}}\right]}$$

Ferric → 3+
Ferrous → 2+

(d) $$\left(U_{cell} - U^{\circ}_{cell}\right) = \frac{RT}{F}\ln\left(\frac{1\times10^{-3}}{0.5\times10^{-3}}\left(\frac{0.2\times10^{-3}}{0.7\times10^{-3}}\right)^{1/2}\right)$$

$$\boxed{U_{cell} = 1.72\ mV \text{ vs. standard cell potential}}$$

Problem 4.7

Ni electrode ; V vs. pH diagram

(a) As shown in class:

Nernst correction for HER: $H^+ + e^- \rightleftharpoons \frac{1}{2}H_2$

$$\left(U_{H_2/H^+} - U^{\circ}_{H_2/H^+}\right) = \left(U^{\circ}_{H_2/H^+} - U^{\circ}_{H_2/H^+}\right) - \frac{RT}{F}\ln\left(\frac{P_H^{1/2}}{C_{H^+}}\right)$$

$$= -0.0591\,(pH) \quad V \text{ vs. SHE}$$

Nernst correction for OER: $2H_2O \rightleftharpoons O_2 + 4H^+ + 4e^-$ ($2H_2O - O_2 - 4H^+ = 4e^-$)

$$\left(U_{O_2/H_2O} - U^{\circ}_{H_2/H^+}\right) = \left(U^{\circ}_{O_2/H_2O} - U^{\circ}_{H_2/H^+}\right) - \frac{RT}{4F}\ln\left(\frac{1}{P_{O_2}\cdot C_{H^+}^4}\right)$$

$$= 1.23 + \frac{RT}{F}\ln C_{H^+} \quad ; \quad pH = -\log H^+ \quad ; \quad \ln(a) = \frac{\log(a)}{\log(e)}$$

$$= 1.23 - 0.0591\,(pH) \quad V \text{ vs. SHE} \qquad \ln(C_{H^+}) = (2.302)_{pH}$$

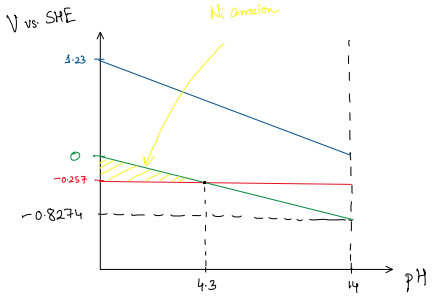
$Ni \rightleftharpoons Ni^{2+}_{(aq)} + 2e^-$

Nernst correction for Ni electrode : $\left(U_{Ni/Ni^{2+}} - U_{H_2/H^+}\right) = \left(U^{\circ}_{Ni/Ni^{2+}} - U_{H_2/H^+}\right) + \frac{RT}{2F}\ln\left(\frac{C_{Ni^{2+}}^{1}}{C_{Ni^{2+}}}\right)$

$Ni \rightleftharpoons Ni^{2+}_{(aq)} + 2e^-$

Nernst correction for Ni electrode : $(U_{Ni/Ni^{2+}} - U_{H_2/H^+}) = (U^\circ_{Ni/Ni^{2+}} - U_{H_2/H^+}) + \frac{RT}{2F} \ln\left(\frac{a_{Ni}}{C_{Ni^{2+}}}\right)$, with $a_{Ni} = 1$

$U_{Ni/Ni^{2+}} = -0.257 + \frac{RT}{F} \ln(C_{Ni^{2+}})^{1/2}$ ← Not dependent on pH



Stability: $(U_{Ni/Ni^{2+}} - U_{H_2/H^+}) < 0$, Non-spont @ pH < 4.3

↑ More negative (will be oxidized)

$(U_{Ni/Ni} - U_{H_2/H^+}) > 0$, Spontaneous , @ pH > 4.3

↑ More negative (will be oxidized)

∴ Ni will be oxidized / corrode below pH 4.3 in acidic solution

Problem 4.9

Cu electrode in $CuSO_4$ electrolyte ($Cu \rightleftharpoons Cu^{2+} + 2e^-$) ; 0.05M $[Cu^{2+}]$

(a) Calomel electrode ($Hg - Hg_2Cl_2$ RE)

$(U_{Cu/Cu^{2+}} - U_{Hg/HgCl}) = ?$

Nernst correction for Cu/Cu^{2+} ⇒ $U_{Cu/Cu^{2+}} - U_{H_2/H^+} = (U^\circ_{Cu/Cu^{2+}} - U_{H_2/H^+}) - \frac{RT}{2F} \ln C_{Cu^{2+}}^{-1}$

$= 0.337 + \frac{RT}{F} \ln(C_{Cu^{2+}})^{+1/2}$

$U_{Cu/Cu^{2+}} = 0.299$ vs. SHE @ 298 K and 0.05M $[Cu^{2+}]$

Since no $[Cl^-]$ variation : $U_{Hg/Hg_2Cl_2} = (U^\circ_{Hg/Hg_2Cl_2} - U^\circ_{H_2/H^+}) - \frac{0.0591}{2} \log C_{Cl^-}^2$ → The RE has its own electrolyte and therefore needs to corrected for depending on the electrolyte used

$= 0.244$ V vs. SHE @ 25°C and saturated solution [ChemLibreText]

∴ $(U_{Cu/Cu^{2+}} - U_{Hg/Hg_2Cl_2}) = 0.131$ V

$U_{Cu/Cu^{2+}} = 0.05$ V vs. calomel

(b) Doing a similar analysis :

$U_{Cu/Cu^{2+}} = 0.375$ vs. SHE

$U_{Ag/AgCl} = (U^\circ_{Ag/AgCl} - U_{H_2/H^+}) - 0.0591 \log C_{Cl^-}$ [ChemLibreText]

$U_{Ag/AgCl} = 0.197$ V vs. SHE @ 25°C and saturated solution

∴ $U_{Cu/Cu^{2+}} = 0.102$ V vs. Ag/AgCl RE

(c) Iron electrode. $Fe(CN)_6^{4-}_{(aq)} \rightleftharpoons Fe(CN)_6^{3-}_{(aq)} + e^-$

[10x] [x]

Nernst correction for iron electrode: $U_{Fe^{3+}/Fe^{2+}} - U_{H_2/H^+} = \left[U^\circ_{Fe^{3+}/Fe^{2+}} - U_{H_2/H^+}\right) - \frac{RT}{F} \ln\left(\frac{C_{Fe(CN)_6^{4-}}}{C_{Fe(CN)_6^{3-}}}\right)$

$= 0.358 - 0.0591 \log(10)$

$= 0.298$ V vs. SHE

∴ $U_{Fe^{3+}/Fe^{2+}} = 0.055$ V vs. Calomel

$U_{Fe^{3+}/Fe^{2+}} = 0.101$ V vs. Ag/AgCl RE

Problem 5.2

i for reduction of protons (H^+) → HER $H_2 - 2H^+ \rightleftharpoons 2e^-$; $n=2$,

+ Rate controlled by MT of H^+ in diffusion layer ($\delta = 40\times10^{-4}$ m) → i.e. we are in the limiting current density regime

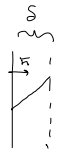
+ $D_{H^+} = 9.310 \times 10^{-9}$ m^2/s (@ www.agion.be)

+ $C_{H^+,\infty} = 0.01$ M

$\bar{N}_{H^+} = -z_j u_j F c_j \nabla\phi - D_j \nabla c_j + c_j \bar{v}$; 1D transport w/ diffusion (No migration)

(No mtg) → 0

$\frac{i}{F} = \sum z_j \bar{N}_j$ → 0 (does not cont. to i)



$\bar{n} \cdot \bar{N}_{H^+} = \frac{-s_j}{nF} i_n$

$i_n = \frac{nF}{-s_j}(\bar{n}\cdot\bar{N}_{H^+})$

$= \frac{2F}{(-2)}\left(-D_{H^+}\frac{\partial c}{\partial x}\right)$

$\frac{\partial c}{\partial x} = \frac{C_{bulk}}{\delta}$ (Linear profile assumption)

$i = +F \cdot D_{H^+} \cdot \frac{C_{bulk}}{\delta}$

$= (96486)\cdot(9.310\times10^{-9})\cdot\frac{(0.01/0.001)}{40\times10^{-4}}$

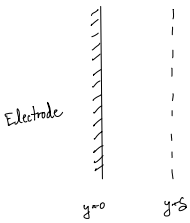
$\frac{C}{mol}\cdot\frac{z^+}{e^-}\cdot\frac{mol}{m^3}\cdot\frac{1}{m}$

$i \approx 224.6\ A\,m^{-2}$

Problem 8

$PbO_{2(s)} + SO_{4(aq)}^{2-} + 4H^+_{(aq)} + 2e^- \rightleftharpoons PbSO_{4(s)} + 2H_2O$

Binary electrolyte: H_2SO_4 → dissociating into $2H^+$ and $1\,SO_4^{2-}$



$z_+ = +1$ $z_- = -2$ $s_+ = -1$

$\nu_+ = 2$ $\nu_- = 1$ $s_- = -1$

$c = \frac{c_+}{\nu_+} = \frac{c_-}{\nu_-}$

Following class notes:

* Flux of each species:

1) $\nabla\cdot N_+ = \nabla\cdot(-z_+u_+Fc\nu_+ - D_+u_+\nabla c) + \nu_+\nabla c\cdot\bar{v}$

2) $\nabla\cdot N_- = \nabla\cdot(-z_-u_-Fc\nu_- - D_-u_-\nabla c) + \nu_-\nabla c\cdot\bar{v}$

* Material balance:

3) $\nu_+\frac{\partial c}{\partial t} = -\nabla\cdot N_+ \Rightarrow \frac{\partial c}{\partial t} = -\frac{1}{\nu_+}(\nabla\cdot N_+)$

4) $\nu_-\frac{\partial c}{\partial t} = -\nabla\cdot N_- \Rightarrow \frac{\partial c}{\partial t} = -\frac{1}{\nu_-}(\nabla\cdot N_-)$

* Subtracting ③ and ④ & rearranging: $\nabla\cdot(c\nabla\phi) = -\frac{(D_+ - D_-)}{F(z_+u_+ - z_-u_-)}\nabla^2 c$

* Plugging into ③:

$$\frac{\partial c}{\partial t} = -\frac{1}{\nu_+}\left(\nabla\cdot\left[-z_j u_j F c_j \nabla\phi - D_j\nabla c_j + c_j\bar{v}\right]\right)$$

Divergence of vector = ?

$$= -\frac{1}{\nu_+}\left(-z_j u_j F c_j \nabla\cdot(\nabla\phi) - D_j\nabla\cdot(\nabla c_j)\right)$$

$$= z_j u_j \frac{c_j}{\nu_+}\nabla\cdot\nabla\phi - \frac{D_j}{\nu_+}\nabla\cdot(\nabla c_j)$$

$$= z_j u_j c\,\nabla\cdot\nabla\phi$$

Nernst Einstein

$$\frac{\partial c}{\partial t} = D\nabla^2 c - \bar{v}\cdot\nabla c \quad ; \quad D = \frac{(z_+u_+D_- - z_-u_-D_+)}{z_+D_+ - z_-D_-} = \frac{(z_+ - z_-)D_+D_-}{z_+D_+ - z_-D_-} = \frac{3D_+D_-}{D_+ + 2D_-}$$

$i = \frac{nF}{-s_j}N_j$

@ $x=0$: $i = \frac{2F}{4}N_+$, $i = 2FN_-$; $N_+ = N_-\cdot 4$

$N_- \neq 0$ → cannot solve for $\frac{d\phi}{dy}$

Need to use this relationship!

↓

Solving this ODE for $\frac{\partial c}{\partial t} = 0$ (SS) & BC: 1) @ $y=0$, $c=0$

2) @ $y=\delta$, $c = c_{bulk}$

∝ Both species are reacting at the electrode

$$c = \frac{c_{bulk}}{\delta}y$$

& $N_- = 0$ we can get $N_+ = -\nu_+\frac{(z_-u_-D_+ - z_+u_+D_-)}{z_-u_-}\frac{dc}{dy}$

For H^+ : $z_+ = +1$, $\nu_+ = 2$

Nernst Einstein $\left(u_j = \frac{D_j}{RT}\right)$

$$N_+ = -\nu_+\left(1 - \frac{z_+}{z_-}\right)D_{H^+}\frac{dc}{dz}$$

$-2\cdot\frac{3}{2}$

$$\boxed{N_{H^+} = -3D_{H^+}\frac{dc}{dy}}$$

For binary electrolytes: $t_+ = \frac{z_+u_+}{z_+u_+ - z_-u_-} = \frac{z_+D_+}{z_+D_+ - z_-D_-}$; $D = \frac{z_+u_+D_- - z_-u_-D_+}{z_+u_+ - z_-u_-}$ or $D = \frac{(z_+u_+D_- - z_-u_-D_+)t_-}{(-z_-u_-)}$

$t_- = 1 - t_+$

$= \frac{z_+u_+ - z_-u_- - z_+u_+}{z_+u_+ - z_-u_-}$

$t_- = \frac{(-z_-u_-)}{(z_+u_+ - z_-u_-)}$

$D = \frac{(z_+u_+D_- - z_-u_-D_+)}{(z_+u_+)}\cdot t_+$

$D = \left(D_- - \frac{z_-}{z_+}D_-\right)t_+$

$D = D_-\left(1 - \frac{z_-}{z_+}\right)t_+$

$= D_-\left(1 - \frac{(-2)}{1}\right)t_+$

$D = 3D_-t_+$

$D = \left(-\frac{z_+}{z_-}D_+ + D_+\right)t_-$

$D = D_+\left(-\frac{z_+}{z_-} + 1\right)t_- = D_+\left(\frac{-1}{-2} + 1\right)t_- = D_+\left(\frac{3}{2}\right)t_-$

$\therefore D = 3D_-t_+ = \frac{3}{2}D_+t_-$

$D_+ = \frac{2D}{3t_-}$

$$N_{H^+} = -3\cdot\frac{2}{3t_-}\frac{\partial c}{\partial y}$$

$$= -\frac{2}{t_-}\frac{\partial c}{\partial y} \quad ; \quad t_- = 1 - t_+$$

$i_y = F\sum z_j\bar{N}_j$

$\frac{i_y}{F} = \sum z_j\bar{N}_j = +1N_{H^+} - 2\cancel{N_{SO_4^{2-}}}^{\,0}$ @ $y=0$

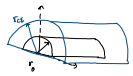
$$\boxed{\frac{i_y}{F} = \frac{-2D}{(1-t_+)}\frac{\partial c}{\partial y}} \quad @\ y=0$$

HW4

Wednesday, 19 April 2023 10:11 AM

Question 1

Hemicylinder electrode [WE's]



CE placed at a finite distance away from WE

(a) Assuming uniform composition:

$$\nabla^2\phi = 0$$

In cylindrical geometry: $\nabla^2\phi = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\phi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2\phi}{\partial\theta^2} + \frac{\partial^2\phi}{\partial z^2} = 0$

↳ Considering only the radial direction we get:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\phi}{\partial r}\right) = 0$$

BC's: $\phi = V$ @ $r = r_o$

$\phi = 0$ @ $r = r_c$

(b) Solve for ohmic resistance (R_Ω):

Obtaining ϕ profile: $\frac{\partial}{\partial r}\left(r\frac{\partial\phi}{\partial r}\right) = 0$

$$r\frac{\partial\phi}{\partial r} = c_1$$

$$\frac{\partial\phi}{\partial r} = \frac{c_1}{r}$$

$$\phi = c_1 \ln r + c_2$$

@ $r = r_o$, $\phi = V$ $\Rightarrow$ $V = c_1 \ln r_o + c_2$ ✓

@ $r = r_{CE}$, $\phi = 0$ $\Rightarrow$ (-) $0 = c_1 \ln r_{CE} + c_2$

$$V = c_1 \ln\frac{r_o}{r_{CE}}$$

$$c_1 = \frac{V}{\ln\left(\frac{r_o}{r_{CE}}\right)} \quad \Rightarrow \quad \text{Solving for } c_1: \quad 0 = \frac{V}{\ln\left(\frac{r_o}{r_{CE}}\right)}\ln r_{CE} + c_2$$

$$c_2 = -\frac{V}{\left(\ln\frac{r_o}{r_{CE}}\right)}\ln r_{CE}$$

$$\therefore \quad \phi = \frac{V}{\ln\left(\frac{r_o}{r_{CE}}\right)}\ln r - \frac{V}{\ln\left(\frac{r_o}{r_{CE}}\right)}\ln r_{CE}$$

$$\phi = \frac{V}{\ln\left(\frac{r_o}{r_{CE}}\right)}\cdot\ln\left(\frac{r}{r_{CE}}\right)$$

$R_\Omega = \frac{V}{i_\Omega}$ (Assuming no charge transfer resistances) $\Rightarrow$ $i_\Omega = -\kappa\left.\frac{\partial\phi}{\partial r}\right|_{r=r_o}$

$$\left.\frac{\partial\phi}{\partial r}\right|_{r=r_o} = \frac{c_1}{r_o} = \frac{V}{\ln\left(\frac{r_o}{r_{CE}}\right)}\cdot\frac{1}{r_o}$$

$$\therefore \quad R_\Omega = \frac{V}{-\kappa\frac{V}{r_o}\cdot\frac{1}{\ln\left(\frac{r_o}{r_{CE}}\right)}} = \boxed{\frac{r_o}{\kappa}\ln\left(\frac{r_{CE}}{r_o}\right) = R_\Omega}$$ ← In Ω

WE S.A.: $\pi r_o l$; l = length of cylinder

$$R'_\Omega = \frac{R_\Omega}{\pi r_o l} = \boxed{\frac{1}{\kappa\pi l}\ln\left(\frac{r_{CE}}{r_o}\right) = R'_\Omega}$$ ← In $\Omega\,cm^2$

As $r_{CE}\to\infty$, $R_\Omega = \frac{r_o}{\kappa}\ln\left(\frac{r_{CE}}{r_o}\right)$ ($\ln\left(\frac{r_{CE}}{r_o}\right)\to 0$)

$\lim_{r_{CE}\to\infty} R_\Omega = \infty$ ← If the CE is at an infinite distance away there is infinite ohmic resistance.

(c) For 1D: $R_\Omega = \frac{L}{\kappa}$ ⟶ This assumes that CE is at distance L

R_Ω for hemicylindrical assumes r_{CE} at finite distance away.

For hemispherical: $R_\Omega = \frac{r_o}{\kappa}$ ⟶ This assumes CE is at ∞

⟶ Comparing hemi-sphere to - cylinder

$\ln\left(\frac{r_{CE}}{r_o}\right)\to 0$ as $r_o\to\infty$

So cylindrical approaches spherical behaviour for large r_o

Question 2 [W 6.2]

Linear & Anodic Tafel kinetics

(a) $R_{ct} = \left.\frac{\partial \eta_s}{\partial i}\right|_{i=i_{avg}}$; $i = -i_o \exp\left(\frac{-\alpha_c F \eta_s}{RT}\right)$ → Cathodic Tafel ;

$$\frac{\partial i}{\partial \eta_s} = -i_o \left(\frac{-\alpha_c F}{RT}\right) \exp\left(\frac{-\alpha_c F \eta_s}{RT}\right)$$

$$= \frac{\alpha_c F \; i_o}{RT} \exp\left(-\frac{\alpha_c F \eta_s}{RT}\right)$$

@ $i = i_{avg}$

$$i_{avg} = -i_o \exp\left(-\frac{\alpha_c F \eta_s}{RT}\right)$$

$$\exp\left(-\frac{\alpha_c F}{RT}\right) = -\frac{i_{avg}}{i_o}$$

$$\therefore \left.\frac{\partial i}{\partial \eta_s}\right|_{i_{avg}} = \frac{\alpha_c F \; i_o}{RT} \cdot \left(-\frac{i_{avg}}{i_o}\right) = -\frac{\alpha_c F}{RT} i_{avg}$$

$$R_{ct} = \left(\left.\frac{\partial i}{\partial \eta_s}\right|_{i=i_{avg}}\right)^{-1} = \boxed{\frac{RT\, i_{avg}}{-\alpha_c F} = R_{ct}}$$ ← R_{ct} for Cathodic Tafel

For B-V kinetics:

$$i = i_o \left[\exp\left(\frac{\alpha_a F \eta_s}{RT}\right) - \exp\left(\frac{-\alpha_c F \eta_s}{RT}\right)\right] \Rightarrow \frac{i_{avg}}{i_o} = \exp\left(\frac{\alpha_a F \eta}{RT}\right) - \exp\left(\frac{\alpha_c F \eta}{RT}\right)$$

$$\frac{\partial i}{\partial \eta_s} = i_o \left(\frac{\alpha_a F}{RT}\right) \exp\left(\frac{\alpha_a F \eta_s}{RT}\right) - i_o \left(\frac{-\alpha_c F}{RT}\right) \exp\left(\frac{-\alpha_c F \eta_s}{RT}\right)$$

$$= i_o \left(\frac{F}{RT}\right) \left[\alpha_a \exp\left(\frac{\alpha_a F \eta_s}{RT}\right) + \alpha_c \exp\left(\frac{-\alpha_c F \eta_s}{RT}\right)\right]$$

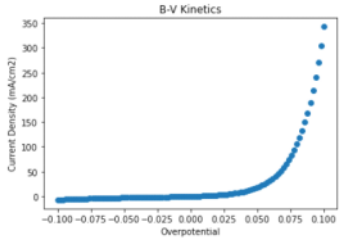
$$= \frac{F i_o}{RT} \left[\alpha_a \left(\frac{i_{avg}}{i_o} + \exp\left(\frac{\alpha_c F \eta_s}{RT}\right)\right) + \alpha_c \exp\left(\frac{-\alpha_c F \eta_s}{RT}\right)\right]$$

$$= \frac{F i_o}{RT} \left[\alpha_a \frac{i_{avg}}{i_o} + (\alpha_a + \alpha_c) \exp\left(\frac{-\alpha_c F \eta_s}{RT}\right)\right]$$

$$\boxed{R_{ct} = \left(\left.\frac{\partial i}{\partial \eta_s}\right|_{i=i_{avg}}\right)^{-1} = \frac{RT}{F i_o \left[\alpha_a \exp\left(\frac{\alpha_a F \eta}{RT}\right) + \alpha_c \exp\left(\frac{-\alpha_c F \eta}{RT}\right)\right]}}$$ ↙ BV kinetics R_{ct}

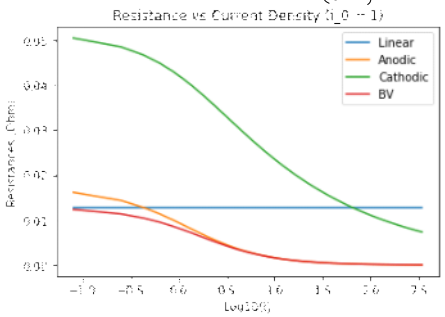
(b) Graphs shown

Plotting BV (i vs. η)

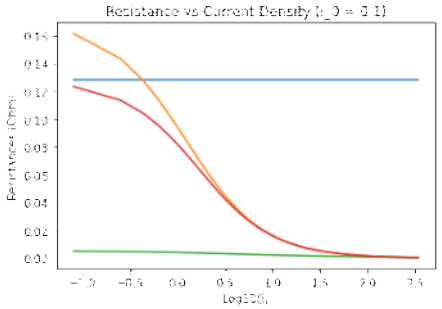


(c) Resistance vs $\log(i)$

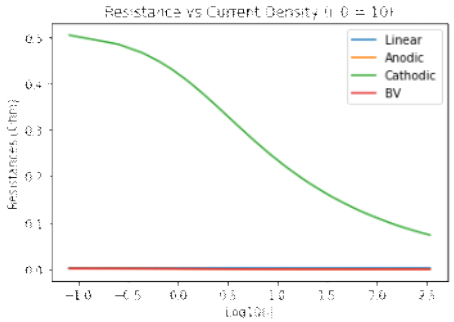
($i_0 = 1$)



($i_0 = 0.1$)



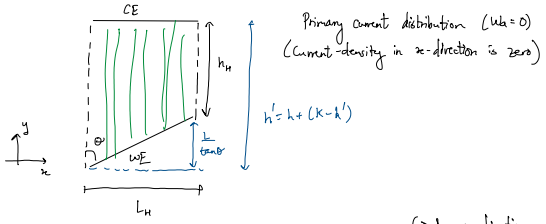
$i_0 = 10$



From the above graphs, it seems that Anodic Tafel approximation matches B-V the closest; Linear may be accurate only in small current densities but is independent of i
Cathodic diverges at first but then approaches B-V / Anodic over higher i_0 values
At higher i_0, the differences reduce (apart from cathodic)

Question 3 (W - 6.3)

Hull cell



Primary current distribution ($\mu_a = 0$)
(Current-density in x-direction is zero)

Current density only in y-direction : $\nabla^2\phi = 0$ (Only y-direction needed)

$i = -\kappa \nabla\phi$

only y-dependence

$\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} = 0$

$\frac{\partial^2\phi}{\partial y^2} = 0$

$\frac{\partial\phi}{\partial y} = c_1$

$\phi = c_1 y + c_2$

BC's :
(i) @ $y = h'$, $\phi = 0$
@ $y = K - k'$, $\phi = V$

$\Rightarrow$ $V = c_1(K - k') + c_2$
(-) $0 = c_1 h' + c_2$
$V = c_1[K - k' - h']$

$c_1 = \frac{V}{[K - k' - h']} = \frac{V}{[K - k' - K + k']} = \frac{V}{-h}$

$i = -\kappa \nabla\phi = -\kappa \frac{\partial\phi}{\partial y}$

$i = +\kappa \frac{V}{h}$

$i = \frac{\kappa V}{h}$

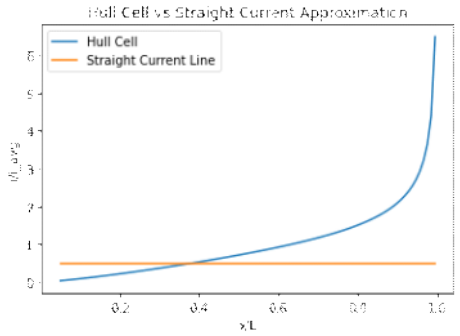
$i_{avg} = \int_0^L i \cdot dx = \frac{\kappa V}{h} \cdot L$

$\therefore$ $\boxed{\frac{i}{i_{avg}} = \frac{\kappa V}{h} \cdot \frac{h}{\kappa V L} = \frac{1}{L}}$

; $i_{avg} = \frac{\kappa V}{(h + \frac{L}{\tan\theta})}$ } Total "y" length

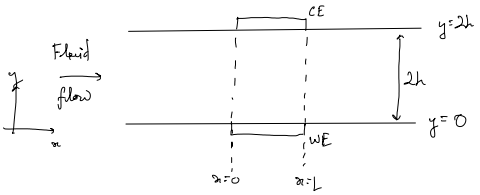
$\frac{i}{i_{avg}} = \frac{1}{h} \cdot \left(h + \frac{L}{\tan\theta}\right) = 1 + \frac{L}{h}\frac{1}{\tan\theta} = \frac{i}{i_{avg}}$

Plotting $\frac{i}{i_{avg}}$ for $\frac{L}{L} = 0.7385$ & $\theta = 0.22\pi$ we get the following graphical representation:



Question 4

Microfluidic eclsen cell; 2 line electrodes



Equation 45: $\frac{3U}{h} y \frac{\partial c_{lim}}{\partial x} = D_{lim} \frac{\partial^2 c_{lim}}{\partial y^2}$

Similarity transformation: $\xi = y \left(\frac{U}{2hD_{lim}x} \right)^{1/3}$

$$\xi^3 = y \left(\frac{U}{2hD_{lim}x} \right) \Rightarrow \frac{U}{h} y = \xi^3 \cdot 3 D_{lim} x$$

As shown in class,

$$\frac{\partial c}{\partial x} = \frac{\partial c}{\partial \xi} \cdot \frac{\partial \xi}{\partial x} \quad ; \quad \frac{\partial \xi}{\partial x} = y \left(\frac{U}{2hD} \right)^{1/3} \left(-\frac{1}{3} \right) x^{-4/3} = \frac{-\xi}{3x}$$

$$\rightarrow \frac{\partial c}{\partial x} = \frac{\partial c}{\partial \xi} \cdot \left(\frac{-\xi}{3x} \right)$$

Similarly,

$$\frac{\partial c}{\partial y} = \frac{\partial c}{\partial \xi} \cdot \frac{\partial \xi}{\partial y} \quad ; \quad \frac{\partial \xi}{\partial y} = \left(\frac{U}{3hDx} \right)^{1/3} = \frac{\xi}{y}$$

$$\frac{\partial c}{\partial y} = \left(\frac{\xi}{y} \right) \frac{\partial c}{\partial \xi}$$

We need: $\frac{\partial^2 c}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial c}{\partial y} \right) = \frac{\partial}{\partial y} \left(\frac{\xi}{y} \frac{\partial c}{\partial \xi} \right) = \frac{\xi}{y} \frac{\partial}{\partial y} \left(\frac{\partial c}{\partial \xi} \right) + \frac{\partial c}{\partial \xi} \frac{\partial}{\partial y} \left(\left[\frac{U}{2hDx} \right]^{1/3} \right)$ (the last term $= 0$)

$$\frac{\partial^2 c}{\partial y^2} = \frac{\xi}{y} \left(\frac{\partial}{\partial \xi} \frac{\partial \xi}{\partial y} \frac{\partial c}{\partial \xi} \right)$$

$$= \frac{\partial^2 c}{\partial \xi^2} \cdot \left(\frac{\partial \xi}{\partial y} \right)^2$$

$$\rightarrow \frac{\partial^2 c}{\partial y^2} = \frac{\partial^2 c}{\partial \xi^2} \cdot \left(\frac{U}{3hDx} \right)^{2/3}$$

Plugging into original PDE:

$$\frac{3U}{h} y \frac{\partial c}{\partial \xi} \left(\frac{-\xi}{3x} \right) = D \frac{\partial^2 c}{\partial \xi^2} \left(\frac{U}{3hDx} \right)^{2/3} \Big) \div \left(\frac{U}{3hDx} \right)^{2/3}$$

$$3y \left(\frac{U}{3hDx} \right)^{1/3} \frac{\partial c}{\partial \xi} (-\xi) = \frac{\partial^2 c}{\partial \xi^2}$$

$$\therefore \quad \frac{\partial^2 c}{\partial \xi^2} + 3\xi^3 \frac{\partial c}{\partial \xi} = 0 \xrightarrow{\text{Since only changes w.r.t. 1 var}} \boxed{\frac{d^2c}{d\xi^2} + 3\xi^3 \frac{dc}{d\xi} = 0}$$

BCs: @ $x=0$, $c_{lim} = c_\infty \Rightarrow \xi \to \infty$ $\quad c_{lim} = c_\infty$

@ $y=\infty$, $c_{lim} = c_\infty \Rightarrow \xi \to \infty$ $\quad c_{lim} = c_\infty$

@ $y=0$, $c_{lim} = 0 \Rightarrow @\xi = 0$ $\quad c_{lim} = 0$

Using Forbenius solution: $\quad \frac{c_{lim}}{c_\infty} = \frac{1}{\Gamma(4/3)} \int_0^\xi e^{-x^3} \cdot dx$

$$N_y = -D \frac{\partial c}{\partial y}\bigg|_{y=0} = -D \frac{\partial c}{\partial \xi}\bigg|_{\xi=0} \frac{\partial \xi}{\partial y}\bigg|_{y=0}$$

$$N_y = \frac{-D c_\infty}{\Gamma(4/3)} \underbrace{\frac{\partial}{\partial \xi}\left(\int_0^\xi e^{-x^3} dx \right)\bigg|_{\xi=0}}_{\text{F.T.C}} \left(\frac{U}{3hDx}\right)^{1/3}$$

$$N_y = \frac{-D c_\infty}{\Gamma(4/3)} e^{-\xi^3}\bigg|_{\xi=0} = \frac{-c_\infty D}{\Gamma(4/3)} \left(\frac{U}{3hDx}\right)^{1/3}$$

Find Current $\longrightarrow$ $\quad \frac{si}{nF} = -N_y$

$$\frac{si}{nF} = \frac{D c_\infty}{\Gamma(4/3)} \left(\frac{U}{3hDx}\right)^{1/3}$$

$$i_{avg} = \left(\frac{nF}{s} \frac{D c_\infty}{\Gamma(4/3)} \left(\frac{U}{3hD}\right)^{1/3} \int_0^L x^{-1/3} \cdot dx \right) \frac{1}{L}$$

$$i_{avg} = \frac{nF}{s} \left(\frac{3}{2} \frac{D^{2/3}}{\Gamma(4/3)} \right) c_\infty \left(\frac{U}{3h}\right)^{1/3} L^{-1/3}$$

$\frac{-1}{3} + \frac{1}{1/3} = \frac{2}{3}$

$\frac{L^{2/3}}{L} = L^{-1/3}$

$$\frac{i}{i_{avg}} = \frac{\frac{nF}{s} \frac{D^{1/3} c_\infty}{\Gamma(4/3)} \left(\frac{U}{3hD}\right)^{1/3} (x)^{-1/3}}{\frac{nF}{s} \frac{D^{1/3} c_\infty}{\Gamma(4/3)} \left(\frac{3}{2}\right) \left(\frac{U}{3hD}\right)^{1/3} \cdot L^{-1/3}}$$

$$\therefore \quad \boxed{\frac{i}{i_{avg}} = \frac{2}{3} \left(\frac{L}{x}\right)^{1/3}}$$

Question 5

$$S = \frac{-D c_\infty}{N} \quad ; \quad \frac{S}{S_{avg}} = \frac{\frac{-D c_\infty}{N}}{\frac{-D c_\infty}{N_{avg}}} = \frac{N_{avg}}{N}$$

From previous part, $\quad N_y = \frac{-c_\infty D}{\Gamma(4/3)} \left(\frac{U}{3hDx}\right)^{1/3}$

$$\therefore \quad S = \frac{-D c_\infty}{-c_\infty D} \Gamma(4/3) \left(\frac{U}{3hD}\right)^{-1/3} \cdot x^{1/3}$$

$$S_{avg}(x) = \left(\int_0^L \Gamma(4/3) \left(\frac{U}{3hD}\right)^{-1/3} x^{1/3} \right) \frac{1}{L}$$

$$= \Gamma(4/3) \left(\frac{U}{3hD}\right)^{-1/3} \cdot \left(\frac{3}{4}\right) x^{4/3}\Big|_0^L \cdot \frac{1}{L} \qquad \frac{4}{3} - \frac{1}{1/3} = \frac{1}{3}$$

$$= \Gamma(4/3) \left(\frac{U}{3hD}\right)^{-1/3} \left(\frac{3}{4}\right) L^{1/3}$$

$$N_y = -\frac{D}{\Gamma(4/3)} \left(\frac{U}{3hD}\right)^{1/3} \frac{c_\infty}{(x^{1/3})} = -\underbrace{\left(\frac{U^{1/3}}{(3h)^{1/3}} \cdot \frac{1}{\Gamma(4/3)}\right)}_{A} \frac{D^{2/3} c_\infty}{x^{1/3}}$$

$$N_{y,avg} = \left(-\frac{D}{\Gamma(4/3)} \left(\frac{U}{3hD}\right)^{1/3} c_\infty \int_0^L x^{-1/3} \cdot dx \right) \frac{1}{L}$$

$$N_{y,avg} = -\frac{D^{2/3}}{\Gamma(4/3)} \left(\frac{U}{3h}\right)^{1/3} c_\infty L^{-1/3} \left(\frac{3}{2}\right)$$

$$= -\underbrace{\left(\frac{U^{1/3}}{(3h)^{1/3}} \frac{1}{\Gamma(4/3)}\right)}_{A} \frac{D^{2/3} c_\infty}{L^{1/3}} \left(\frac{3}{2}\right) = \frac{3}{2} \frac{A D^{2/3} c_\infty}{S_{avg}}$$

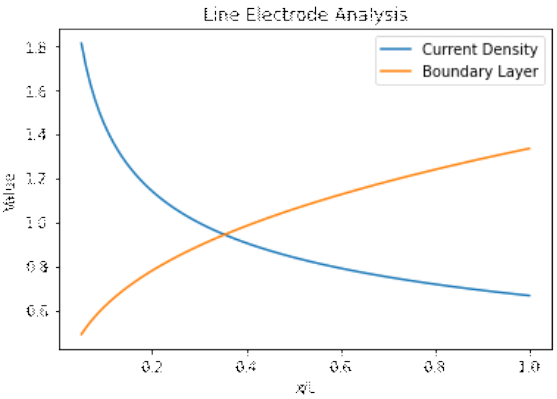
$$\therefore \quad \frac{S(x)}{S_{avg}} = \frac{N_{y,avg}}{N_y} = \frac{-A D^{2/3} c_\infty \left(\frac{3}{2}\right) / L^{1/3}}{-A D^{2/3} c_\infty} = \boxed{\frac{3}{2} \left(\frac{x}{L}\right)^{1/3} = \frac{S(x)}{S_{avg}}}$$

$$= \Gamma(4/3)\left(\frac{U}{3hD}\right)^{-1/3}\left(\frac{x}{L}\right)^{1/3}$$

$$\therefore \frac{S}{S_{avg}} = \frac{\Gamma(4/3)\left(\frac{U}{3hD}\right)^{-1/3} x^{1/3}}{\Gamma(4/3)\left(\frac{U}{3hD}\right)^{-1/3}\left(\frac{x}{L}\right)^{1/3} L^{1/3}} = \boxed{\frac{4}{3}\left(\frac{x}{L}\right)^{1/3} = \frac{S}{S_{avg}}}$$

$$\therefore \frac{S(x)}{S_{avg}} = \frac{N_{y,avg}}{N_y} = \frac{-A D^{2/3} C_\infty \left(\frac{3}{2}\right)}{\frac{L^{1/3}}{-A D^{2/3} C_\infty}} = \boxed{\frac{3}{2}\left(\frac{x}{L}\right)^{1/3} = \frac{S(x)}{S_{avg}}}$$

Plotting $\frac{i}{i_{avg}}$ & $\frac{S}{S_{avg}}$



Question 6

Levich plot @ specific overpotential → Find diffusion coefficient

$$m = 0.62\left(\frac{nFC_\infty}{S_{lim}}\right)\nu^{-1/6} D^{2/3}$$

$$\left[\frac{m}{0.62}\nu^{1/6}\left(\frac{S_{lim}}{nFC_\infty}\right)\right]^{\frac{3}{2}} = D$$

$\frac{mA}{cm^2} = \frac{10^{-3}}{(10^{-2})^2}\frac{A}{m^2}$

C_∞ → conc. of H_2 in the bulk solution

$$\frac{0.00016\ g\ H_2}{100\ g\ H_2O} \approx \frac{0.00016\ g\ H_2}{100\ mL\ H_2O} = \frac{0.00016 \times \frac{1}{2.02}}{100 \times 10^{-6}} = 0.792\ \frac{mol}{m^3}$$

ν → kinematic viscosity (can be approx. close to water since 0.1M K_2SO_4 used)

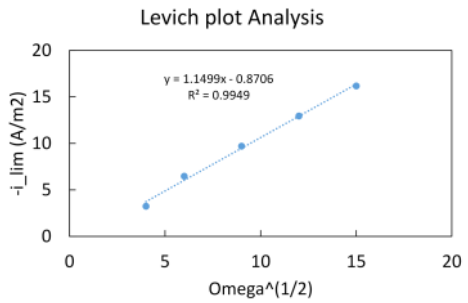
From Engineering toolbox: $\nu = 8.92 \times 10^{-7}\ m^2/s$

$n=2$, $F = 96485$, $S_{lim} = 2$

From the levich plot: slope = 1.1499

$$\therefore \boxed{D = 3.67 \times 10^{-5}\ cm^2/s}$$

$D_{lit} = 4.58 \times 10^{-6}\ cm^2/s$ } within a factor of 0.8 !



Question 7

Koutecky-Levich Analysis ; Magnitude of charge transfer resistance vs. MT resistance in system.

From the plot of $\frac{1}{i_{lim}}$ vs. $\frac{1}{\Omega^{1/2}}$ we get: $m = 0.7306$, $c = 0.0325$

$$m = \frac{1}{(0.62) D^{2/3} \nu^{-1/6} C_\infty \cdot F \cdot n}$$

$$\therefore \frac{1}{i_{lim}} = \frac{1}{i_K} + \frac{1}{i_D} \quad ; \quad \frac{1}{i_K} = 0.0325 \Rightarrow i_K = 30.77\ \frac{A}{m^2}$$

$$D^{2/3} = \frac{1}{(0.62) m \cdot \nu^{-1/6} C_\infty \cdot F \cdot n}$$

$$D = \left(\frac{1}{(0.62) m \nu^{-1/6} C_\infty \cdot F \cdot n}\right)^{3/2}$$

For 100 rpm: $\frac{1}{\ } = 0.0325$

i_{lim} i_K i_D i_K $\frac{A}{m^2}$

$D = \left(\frac{1}{(0.62)\, n^{1/6}\, \nu^{-1/6}\, c_0 \cdot F \cdot n}\right)^{3/2}$

For 100 rpm: $\frac{1}{i_K} = 0.0325$

$\frac{1}{i_D} = 0.2175$

2500 rpm: $\frac{1}{i_K} = 0.0325$

$\frac{1}{i_D} = 0.03417$

Similar to summing up resistors:

→ As rpm ↑, mass transfer resistance decreases (which charge transfer resistance stays the same)

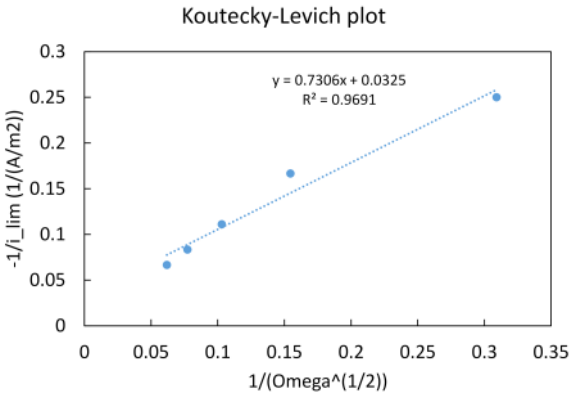
→ Charge transfer resistance always smaller than mass transfer resistance (i.e. always in diffusion limitations for the range of rpms in the demo)

$D = 1.68 \times 10^{-5}\ cm^2/s$

$D_{Lit} = 4.58 a10^{-5}\ cm^2/s$

These are off by a factor of 2.73

↳ possibly due to contributions for ORR occuring simultaneously



Question 8

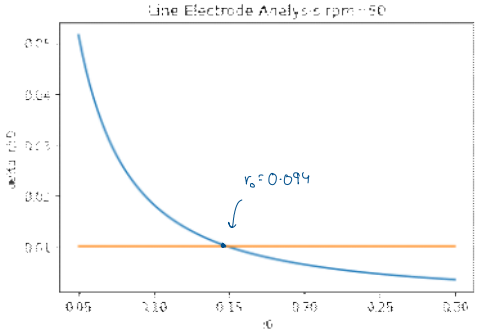
(a) $\Delta r = 0\left(\sqrt{\frac{D\nu^{1/2}}{\Omega^{3/2} r_0}}\right)$; $D = 10^{-5}\ cm^2 s^{-1}$, $\nu = 10^{-2}\ cm^2 s^{-1}$

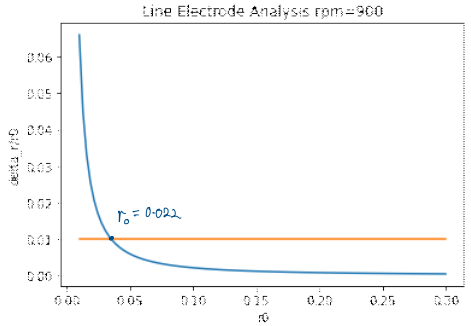
rpm → rad/s

$1\ rpm = \frac{1\ rotation}{1\ min} = \frac{2\pi}{60}\ \frac{rad}{s}$

Plotted graphs allow to calculate the required r_0 value

rpm ↑ → r_0 ↓ (Smaller r_0 needed to go below 0.01 ratio)





(b) Solving for Δr order given r_0, Ω, & ν gives:

(b) solving for Δr order given r_0, Ω, & v gives:

$$\Delta r^{100\ rpm} = 7.67 \times 10^{-4}$$

$$\Delta r^{3600\ rpm} = 5.21 \times 10^{-5}$$

↓ Higher rpm reduces the edge effect region